

The block below carries no module guard, so docstrip copies it into EVERY generated file. That is what we want for the TeX files and wrong for the latexmkrc, which is Perl and does not understand lines opening with two percent signs – hence the negated guard around it.

The guard lines themselves are hidden behind a false conditional. docstrip reads a guard line either way, but ltxdoc, which typesets this file, would otherwise set the literal text of the guard as a stray paragraph in the body of the manual.

Nothing in this area may name a TeX conditional, not even in prose: outside the meta-comment at the top of the file, every documentation line here IS LaTeX source and is executed, and a conditional named in passing opens or closes a real one. That is not hypothetical – an earlier draft of this very comment swallowed the rest of the manual.

Three conditional primitives are deliberately absent from the lists below: the ones that close or branch a conditional. `\DoNotIndex` scans its argument with TeX’s own conditional machinery live, so a bare one of them closes the wrong conditional. The manual ended its run with “Too many }’s” and exited with status 1, which failed the docs step of the harness even though the PDF had been written. Leaving them out of the lists means they get indexed, which is harmless. Note that they cannot be named here either, not even inside `...` shorthand split across a line.

The coppe document class

Version 4.0

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Abstract

In this work, it is described the `coppe` document class as well as other files distributed by the COPPETEX project. This class is suitable for writing academic dissertations, thesis and qualifying exams according to the formatting rules of the Alberto Luiz Coimbra Institute for Graduate Studies and Research in Engineering. The minimalist set of macro commands allows its users to concentrate most of their efforts on text composition rather than on the document layout.

1 Introduction

Writing documents in L^AT_EX may be a laborious task when the authors have to prepare their manuscripts rigorously respecting formatting rules imposed by publishers. Regardless of difficulty, a lot of thesis presented to the Coordination of Graduate Studies and Research in Engineering of the Federal University of Rio de Janeiro (COPPE/UFRJ) is typesetted in L^AT_EX. This demand motivated the creation of the COPPETEX project, which tries to facilitate and encourage the use of L^AT_EX within the COPPE/UFRJ scope.

The `coppe` document class is the main product of COPPETEX. It was designed to be clear and succinct. It enables the creation of dissertations, qualifying exams and thesis in a simple and automatic way. The main goal of the `coppe` class is to maintain authors strictly focused on text composition without worrying about margins sizes, line spacing, paper size, vertical and horizontal alignment, etc. The COPPETEX project comprehends also BibTEX and MakeIndex style files for creating lists of references, symbols and abbreviations. Although there aren't official guidelines to write qualifying exams, we provide this option just for convenience, as this exam is a requisite to obtain the DSc degree and, for some of the programs, the MSc degree.

In which follows, it is described the user interface of the `coppe` class. Some details about using the style files cited above are also given. We use the term *thesis* to generally refer to dissertation, qualifying exam, and thesis itself.

2 License

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copying, distributing or modifying the source code, among other acts covered by this license.

To see the full text of the GNU GPL license, go to the `COPYING` file attached to this package.

3 Support

We maintain a WhatsApp group where users can send questions, comments, and bugs to. More details can be found [here](#).

However, as the project maintenance was transferred to a new repository, bug reports, as well as new feature requests, should be directed to <https://github.com/COPPE-UFRJ/CoppeTeX>.

4 User interface

`\frontmatter` A thesis to be approved by the Academic Registry at COPPE/UFRJ must contain three-parts: *front*, *main* and *back* matters (COPPE/UFRJ Academic Registry guidelines). Each one of these parts is started by calling its corresponding macro `\frontmatter`, `\mainmatter` or `\backmatter`. The front matter of a thesis consists of front cover and face, cataloging page, dedication, acknowledgments, abstracts, table of contents, and lists of tables, algorithms, symbols and abbreviations. The main matter is just composed by chapters, while the back matter usually consists of bibliographic references, appendices and index.

You must invoke the `\frontmatter` macro immediately after the `\maketitle` one. The `\mainmatter` command comes right before the first chapter, and `\backmatter` must be typed before the list of references.

Cover (capa)

The ABNT cover suggested in the manual's Annex A. The `\maketitle` command builds it automatically, ahead of the title page: the university, the full COPPE name, and the graduate program (taken from `\department`), followed by the author, the title, and the city and year, all centred and in uppercase. The UFRJ and COPPE logos head the page. The capa is not counted.

Front face

The front face is unnumbered. There, it is not allowed to use hyphenation (COPPE/UFRJ Academic Registry guidelines). It is constructed by calling `\maketitle`. Next, it is described the commands used to enter the information required to create it.

`\author` The `\author` command was redefined. Here, it takes two arguments: the author's first names and surname, e.g., `\author{First Names}{Surname}`. The words should be typed with only first letters in uppercase.

`\title` The macros `\title` and `\foreigntitle` are used to enter the titles of your monograph in Brazilian Portuguese and English respectively (the two built-in languages of CoppeTeX 4.x). For documents whose main language is neither

brazilian nor english – e.g. spanish, french, italian, or a user-supplied language –, also call `\titlein{<lang>}{<text>}` to register the title in that language; the cover and folha de rosto print whichever title matches the main language declared in the class option. See the section “Multilingual support” below for the complete picture. The `babel` package is loaded automatically by `coppe.cls`, so you do not need to load it yourself.

`\advisor` Every COPPE student is coordinated by at least one advisor. M.Sc. and D.Sc. students can have at most 2 and 3 advisors, respectively. Their names must be provided by issuing the command `\advisor` as below:

```
\advisor{Title}{Advisor's Name}{Surname}{Degree}
\advisor{Title}{Second Advisor's Name}{Surname}{Degree}
\advisor{Title}{Third Advisor's Name}{Surname}{Degree}
```

The advisors are not necessarily members of the thesis examination board. Thus, it is required to enter the names of all examiners using the `\examiner` macro. The examiners' names are entered differently:

```
\examiner{Title}{First Examiner's Name Surname}{Degree}
\examiner{Title}{Second Examiner's Name Surname}{Degree}
...
\examiner{Title}{N-th Examiner's Name Surname}{Degree}
```

Remember that all names must be given before calling `\maketitle`.

`\department` The Alberto Luiz Coimbra institute is divided into 12 academic units: Biomedical Engineering (PEB), Civil Engineering (PEC), Electrical Engineering (PEE), Mechanical Engineering (PEM), Metallurgical and Materials Science Engineering (PEMM), Nuclear Engineering (PEN), Ocean Engineering (PENO), Energy Planning (PPE), Production Engineering (PEP), Chemical Engineering (PEQ), Systems Engineering and Computer Science (PESC), and Transportation Engineering (PET). You must specify your department using one of the above abbreviations, e.g., `\department{PEC}`.

`\date` This macro is used to set the month and year of defense. This information is required to create the front face, cataloging details page and abstracts. For example, October 2007 should be entered as `\date{10}{2007}`.

`\keyword` The keywords should describe the concentration areas of your work. You must provide them as follows:

```
\keyword{First Keyword}
\keyword{Second Keyword}
...
\keyword{N-th Keyword}
```

Usually, six words are enough.

Folha adicional: Coleta CAPES

Since August 2026 every thesis and dissertation must carry a *folha adicional* immediately after the folha de rosto, holding the fields the CAPES/MEC collection requires plus the ficha catalogafica (SiBI manual, 9th rev. ed. 2026, 3.1.2.1.2

and Annex H). The sheet is filled in by the graduate programme, and is neither counted nor numbered.

The macros below feed that sheet. They are institutional data, so the sheet is always typeset in Portuguese regardless of the main language of the work, as the COPPE norm requires of institutional identity.

`\areaconcentracao` The area of concentration, e.g. `\areaconcentracao{Engenharia de Sistemas e Computacao}`. It appears on the folha de rosto, on the folha de aprovacao and on the folha adicional.

`\linhapesquisa` The research line, required by the CAPES model of the folha de rosto.

`\tipoproducao` The kind of intellectual output, one of `bibliografica`, `artistica`, `tecnologica` or `tecnica`. Anything else raises a warning.

`\projetovinculado` Whether the work belongs to a research project (`sim` or `nao`) and, if so, its

`\nomeprojeto` name.

`\agenciafomento` A funding agency, given as full name and acronym; repeat once per agency:

```

\agenciafomento{Conselho Nacional de Desenvolvimento
                  Cientifico e Tecnologico}{CNPq}
\agenciafomento{Fundacao Carlos Chagas Filho de Amparo a
                  Pesquisa do Estado do Rio de Janeiro}{FAPERJ}

```

Cataloging details

This page contains cataloging information useful for librarians. Fortunately, it is automatically generated from the data you entered at the time you call `\maketitle`. It is not needed in qualifying exams, though.

Dedication (optional)

`\dedication` This macro was added for convenience. The input text is placed at the right bottom of a blank page. It is emphasized and in normal size.

Abstracts

`abstract (env.)` As stated by the COPPE/UFRJ Academic Registry, abstracts must be in one

`foreignabstract (env.)` page each, with at most 250 words. We recommended that they should be only one paragraph long. They must be defined inside the environments `abstract` and `foreignabstract`.

Lists of symbols and abbreviations (optional)

`\abbrev` The lists of symbols and abbreviations are optional, although highly recommended.

`\syml` It is a good practice to define a symbol/abbreviation in its first occurrence in the text. To define a symbol use `\syml[alphabetic symbol]{Symbol}{Symbol Definition}`, and for abbreviations `\abbrev[alphabetic symbol]{Abbreviation}{Abbreviation Definition}`. These commands are called *dummy*, since they don't output anything at the place they are executed, just an entry in the correspondent list.

`\makeloabbreviations` These lists are lexicographically sorted by using the MakeIndex program, which

`\makelosymbols` is part of any L^AT_EX implementation. For `\syml`, if the optional parameter is

`\printloabbreviations` provided, it will used as sort key. This was later, in 2024, implemented also for

`\printlosymbols`

`\abbrev`, otherwise `Symbol`, or `Abreviation` will be used as sort key, what can result in an undesirable order if it contains $\text{\LaTeX}$ commands, mathematical symbols, or mix of uppercase and lowercase. `MakeIndex` needs two commands to create a final sorted list: one which generates a list of entries and the other that indicates the position where the list will be printed out. To generate the lists of symbols and abbreviations, the `coppe` class provides the commands `\makeloabbreviations` and `\makelosymbols`, respectively. They must be called in the document preamble. The commands `\printlosymbols` and `\printloabbreviations` have to be invoked at the point where you want these lists appear, e.g., following the list of tables as showed in the example. Once you call `latex`, it will be created two files with extensions `abx` and `syx`, which contain `MakeIndex` input data. They must be processed with `makeindex` in order to get the lists correctly produced, redirecting the output to files with extension `lab` and `los` respectively:

```
makeindex -s coppe.ist -o example.lab example.abx
makeindex -s coppe.ist -o example.los example.syx
```

Note the `-s` option for specifying the style `coppe.ist`. Now, rerun `latex` twice to get the references solved and you are done.

References

`COPPE $\text{\TeX}$` produces citations and the reference list with `biblatex` and the `biber` backend, through its own bibliography style implementing the post-2020 ABNT NBR 6023 rules. The class loads this style automatically: there is no `\bibliographystyle` to set and no `BIB $\text{\TeX}$` `.bst` file to choose (the former `coppe-plain` and `coppe-unsrt` `BIB $\text{\TeX}$` styles have been retired).

By default, citations follow the author–date style, e.g., “Chartier (2002)”. Passing the `numbers` class option switches to the numeric style, in which citations are bracketed numbers and the reference list is numbered in order of citation (unsorted).

Declare your bibliography database in the preamble with `\addbibresource` and print the list where it should appear with `\printbibliography`:

```
\addbibresource{example.bib} % in the preamble
...
\printbibliography           % where the list goes
```

Entry types and fields may be written in English (`@book`, `title`, `author`) or with their unaccented Portuguese synonyms (`@livro`, `titulo`, `autor`).

Run in sequence `L $\text{\TeX}$` , `biber`, and twice again `L $\text{\TeX}$` to resolve the citations and references. The style is `natbib` compatible (via the `biblatex natbib=true` option), so you may freely issue `\citet` and `\citep`, as well as the other `natbib` commands.

Reference entry types

Every ABNT document category maps to a `biblatex` entry type, each with an unaccented Portuguese synonym (a worked example of every type is in the sample manuscript’s reference list):

Academic @book (@livro); @article (@artigo); @artigodejornal (newspaper); @incollection (@partedelivro); @inbook (@emlivro); @inreference (@verbete); @proceedings (@anais); @inproceedings (@trabalhodeevento); @thesis (@tese/@dissertacao, with the presets @dissertacaomestrado, @tesedoutorado, @tesephD, @tcc); @report (@relatorio); @periodical (@periodico); @manual; @misc (@diverso); @online.

Special (NBR 6023 §4.2.5–4.2.14) @patent (@patente); @legislation (@legislacao/@atonormat @jurisdiction (@jurisprudencia, with a relator field); @legal (@documentojuridico); @standard (@norma); @music (@partitura); @map (@mapa); @image (@iconografico); @artwork (@obradearte); @video (@audiovisual); @movie (@filme); @letter (@correspondencia); @dataset (@conjuntodedados).

Games and TV @game (@jogo); @boardgame (@jogodetabuleiro); @videogame (@jogoeletronico); @tvshow (@programadetv).

When an entry has no author or editor, it is entered by title, with the first significant word (and a leading article) in capitals.

Appendix and Annex (Optional)

`\appendix` Appendices and annexes are optional chapters that are part of the back matter.

`\annex` The `\appendix` command is a standard L^AT_EX command used before all appendices. COPPET_{EX} introduces the `\annex` command, which should be used only in the back matter (i.e., after the `\backmatter` command) and only after all existing `\appendix` chapters. This restriction is due to implementation constraints.

Therefore, the order for backmatter is:

```
\backmatter
\printbibliography
\appendix
\chapter{An Appendix}
\chapter{Another Appendix}
\annex
\chapter{First Annex}
\chapter{Second Annex}
```

Annex was introduced in v3.5

Final colophon page (`\coppetexfinalpage`)

`\coppetexfinalpage` Optional closing page modelled on the printer's *colophon*: it records, in Portuguese, the tooling and edition used to typeset the thesis. Call it once, after every other piece of content, immediately before `\end{document}`. The macro

- forces a verso (even) page, so in a two-sided print the colophon sits opposite the rear cover;
- types the body at the foot of the page (`\vfill`);
- renders the text in Portuguese unconditionally (Brazilian babel), even when the thesis main language is English, Spanish, French or Italian — the colophon is institutional metadata, not body content.

Three slot macros are user-overridable. Redefine them with `\renewcommand` before `\coppetexfinalpage` if the document deviates from the defaults:

`\coppefinalengine` the T_EX engine identifier. Default: auto-detected via `iftex` (`\hologo{pdfLaTeX}`, `\hologo{LuaLaTeX}` or `\hologo{XeLaTeX}`).

`\coppefinalfont` the text body font. Default: *Computer Modern (padrão L^AT_EX)*. Redefine if you loaded `lmodern`, `mathptmx`, an OpenType math font, etc.

`\coppefinalmanual` the reference manual title (typeset in italics). Default: the current COPPE/UFRJ norm. Redefine if your work follows an earlier edition.

The macro also prints the COPPE_TE_X class version (captured into `\coppe@version` at class-load time, so it survives any later package that may redefine the global `\fileversion`) and the GitHub repository URL, <https://github.com/COPPE-UFRJ/CoppeTeX>.

5 Class options

There are some options users can specify in order to customize the appearance of the output produced by the `coppe` class. These options can be passed to `coppe` as follows: `\documentclass[option1, option2]{coppe}`. In which follows, we give a brief description of all supported options.

dsc, msc, dsceexam, msceexam The `coppe` class is able to produce thesis, dissertations, and qualifying exams, which are enabled by the `dsc`, `msc`, `msceexam`, and `dsceexam` options, respectively.

doublespacing The default line spacing is one-and-a-half. For enabling double spacing between lines, use the `doublespacing` option.

numbers The default citation style is the author–date scheme. For numbered citations, specify the `numbers` option to the `coppe` class; the reference list is then numbered in order of citation (unsorted). The matching `biblatex` style is selected automatically in either case.

brazilian, english, spanish, french, italian Pick the document’s *main* language. The class loads Babel with that language as the main one (the last in the Babel option list); inside the document `\selectlanguage{...}` and `\begin{otherlanguage}{...}` keep working as usual. The default, when no language option is given, is **brazilian**.

- **brazilian** and **english** are *built-in*: their strings ship inside `coppe.cls` and need no extra file.
- **spanish, french, italian** are shipped as *language packs*: two files per language (`coppe-lang-<lang>.def` and `<lang>-coppe.lbx`) that must sit next to `coppe.cls` (or in your project directory). They are auto-loaded by the class. Picking one of them as the main language without the matching pack files installed raises a clear class error.

- Any other Babel-known language can be plugged in by writing the same two files and either passing the language name as a class option or calling `\usecoppelanguage{<lang>}` in the preamble.

See the section “Multilingual support” below for the three-slot architecture, the `\titlein / \brazilianabstract` API, and how to create a new language pack.

pdfa Produce the deposited file as **PDF/A-2b**, which 2.2(d) of the manual requires of the digital version. The class loads `pdfx` at level **a-2b** and writes the XMP metadata from the document’s own `\title`, `\author`, `\keyword` and `\department`. Level a-2b rather than a-1b: a-1b forbids transparency, which `tcolorbox` and several graphics packages produce. The metadata reaches the PDF on the SECOND run, because `pdfx` reads it from a file the class writes at the end of the first one – which is no burden, since three runs are needed for cross-references anyway.

assinaturas Give every member of the board a signature rule of its own on the folha de aprovacao, as Annex D of the manual shows, instead of the compact list. The spacing adapts to the size of the board.

coorientador List the coadvisors on the three abstract pages, under the advisors, in the same shape the folha de rosto uses. Off by default: the abstract page listing only the advisors is a COPPE tradition, and 3.1.2.1.4 does not require either form.

rascunhoficha While writing, fill the folha adicional with a ficha catalografica typeset by the class itself, so the page is not an empty frame. **Never valid for deposit:** the real ficha comes from the SiBI generator or the Programa’s library, and goes in with `\fichacatalografica`.

twoside, oneside The 2026 edition dropped the verso margins and the deposit is digital, so the class sets one-sided pages with a 3 cm left margin on every folha. **twoside** restores the mirrored layout for anyone who is going to PRINT and bind the work; **oneside** is the default and is there to be explicit.

numeraisromanos Number the pre-textual folhas in lowercase roman numerals. **Contrary to 2.7**, which has the count running from the folha de rosto with nothing printed until the textual part. Kept only so that documents written under the older norm still compile; do not use it for a new work.

5.1 Changing document identification

\freeconfig The user could *optionally* use the command `freeconfig` to modify the parameters that print the document identification. The command `freeconfig` needs all those parameters, which are degree initials, degree name, title, foreign title, local doctype, and foreign doctype as in the following example:

```
\freeconfig{Dr.}{Philosophiae Doctor}{PhD}{Doutor}{Dissertation}{Tese}
```

5.2 Multilingual support

CoppeTeX 4.x reserves three fixed language slots:

main the language of the body, captions, TOC labels and the **abstract** environment. Picked by the class option (**brazilian** by default; one of **english**, **spanish**, **french**, **italian** also accepted out of the box; any other Babel language possible with a user-supplied pack).

foreign the language of the **foreignabstract** environment. **english** by default for every main language; switches automatically to **brazilian** when **main = english**.

third optional. Loaded on demand with `\usecoppelanguage{<lang>}` in the preamble, useful when the body quotes or cites in a third language (e.g. a Spanish-main thesis citing a French source). Does not have a class option of its own.

`\titlein`
brazilianabstract (*env.*)
`\usecoppelanguage`

Two pieces of user-facing API come with the multilingual model:

- `\titlein{<lang>}{<text>}` records the title in any language. `\title{...}` still writes the Brazilian-Portuguese title and `\foreigntitle{...}` still writes the English one (back-compat); for a main language outside the **brazilian/english** pair, also call `\titlein{<lang>}{...}`. The cover and folha de rosto print whichever title matches `\coppe@mainlang`.
- The **brazilianabstract** environment is the optional third abstract used when the main language is neither **brazilian** nor **english** – typically a Spanish/French/Italian thesis that also wants a Portuguese resumo for UFRJ examiners. It wraps its body in `\begin{otherlanguage}{brazilian}` and uses the Portuguese template strings. A document with **main = brazilian** can simply keep using **abstract**; **brazilianabstract** is redundant then.
- `\usecoppelanguage{<lang>}` loads an extra language pack in the preamble without making it the main or foreign language.

Institutional names stay Portuguese

The cover, folha de rosto and ficha catalográfica are a Brazilian institutional template (NBR 14724 + COPPE manual). The strings **universityname**, **cityname**, **statename**, **countryname**, the departments declared with `\department` and the surrounding “apresentada/submetida ao...” wording are deliberately kept in Portuguese for every main language. Only the title (printed via `\coppe@selecttitle`) follows the main language.

Writing a new language pack

A language pack is just two files dropped next to **coppe.cls** (or anywhere on the TEXINPUTS path):

coppe-lang-<lang>.def class-level strings. Populate the dispatcher table with `\copperdefstring{<lang>}{<key>}{<value>}` for every key already populated for **brazilian/english** in **coppe.cls** (appendix, annex, frame,

listframe, source, program, listprogram, references, in:, art, platform, directedby, producedby, listabbreviation, listsymbol, glossary, advisor, advisors, dept, approvedmale, approvedfemale, universityname, cityname, statename, countryname, monthname1...monthname12, algorithm2eopt, babelname). Finish with

```
\AtBeginDocument{\DeclareLanguageMapping{<lang>}{<lang>-coppe}}
```

so the biblatex mapping is registered after the bibliography engine is loaded by coppe.cls (which happens *after* the language pack is read).

<lang>-coppe.lbx biblatex localization strings. \InheritBibliographyExtras{<lang>} then \DeclareBibliographyStrings for the keys availablefrom, urlseen, in, editor, editors, depositor, coppescale, msdiss, dscthis, phdthesis, tcc. The shipped spanish-coppe.lbx, french-coppe.lbx and italian-coppe.lbx are the reference templates to copy from.

Tip: the algorithm2eopt key holds whatever algorithm2e calls the language. Mind the spelling: algorithm2e ships portuguese, english, spanish, french and italiano – not italian.

Example: a Spanish-main thesis

```
\documentclass[spanish,dsc]{coppe}
\addbibresource{example.bib}
\begin{document}
  \title{T\`i tulo em portugu\`es} % Brazilian abstract slot
  \foreigntitle{English title} % English abstract slot
  \titlein{spanish}{T\`i tulo en espa\~nol} % cover/folha
  \author{Juan}{P\`erez}
  \advisor{Prof.}{Mar\`i a}{Garc\`i a}{D.Sc.}
  \department{PESC}\date{01}{2026}
  \keyword{multilenguaje}
  \maketitle
  \frontmatter
  \begin{abstract} Resumen principal en espa\~nol. \end{abstract}
  \begin{foreignabstract} English foreign abstract. \end{foreignabstract}
  \begin{brazilianabstract} Resumo em portugu\`es para a banca. \end{brazilianabstract}
  \tableofcontents
  \mainmatter
  \chapter{Introducci\`on} Cuerpo en espa\~nol\dots
\end{document}
```

6 Files in your project folder

In a normal install where CoppeTeX is part of your TEXINPUTS path (e.g. inside the local texmf tree of MiKTeX or TeX Live), you do not need to think about this list at all — the files travel with the class. This section is for two situations where you do:

- shipping the class together with your manuscript (a self-contained submission package); or
- installing CoppeTeX by hand, dropping the files next to `coppe.cls` in the project directory.

Always required

These files must be findable at every compilation, regardless of the main language or any class option:

`coppe.cls` the document class itself.

`coppe.dbx` custom `biblatex` datamodel (declares the ABNT-specific entry types `@game`, `@videogame`, `@boardgame`, `@tvshow` and `@map`, plus a few custom fields).

`coppe.bbx` and `coppe.cbx` `biblatex` bibliography and citation styles for the author–date scheme (the default).

`brazilian-coppe.lbx` and `english-coppe.lbx` `biblatex` localisation tables for Portuguese and English. Both are required *even when neither language is the main one*, because `coppe.bbx` registers

```
\DeclareLanguageMapping{brazilian}{brazilian-coppe}
\DeclareLanguageMapping{english}{english-coppe}
```

unconditionally, so `biblatex` tries to read both LBX files at every compilation.

Conditional on class options

`coppe-numeric.bbx` and `coppe-numeric.cbx` only when the `numbers` class option is given (`\documentclass[numbers]{coppe}`). Without it, `biblatex` never tries to read them.

`coppe.ist` only when the glossary, list-of-symbols, or list-of-abbreviations features are used. It is the `makeindex` style file invoked by the `makeindex -s coppe.ist ...` step described in subsection *Lists of symbols and abbreviations* above.

Conditional on the main language

The main language is chosen by the matching class option (`brazilian`, `english`, `spanish`, `french`, `italian`). The two built-in languages ship their strings inside `coppe.cls` and so need no extra file. The three language packs add two files each — one class-level (`coppe-lang-<lang>.def`) and one `biblatex`-level (`<lang>-coppe.lbx`):

Main language	Extra files needed
<code>brazilian</code> (default)	— (built into <code>coppe.cls</code>)
<code>english</code>	— (built into <code>coppe.cls</code>)
<code>spanish</code>	<code>coppe-lang-spanish.def</code> , <code>spanish-coppe.lbx</code>
<code>french</code>	<code>coppe-lang-french.def</code> , <code>french-coppe.lbx</code>
<code>italian</code>	<code>coppe-lang-italian.def</code> , <code>italian-coppe.lbx</code>

Loading a third language at preamble time via `\usecoppelanguage{<lang>}` brings in the same pair of files for that language. A user-supplied language pack follows the same scheme: drop `coppe-lang-<lang>.def` and `<lang>-coppe.lbx` next to `coppe.cls`.

Cover assets

The institutional title page produced by `\maketitle` needs two PDF logo files at compile time:

`ufrj-logo.pdf` and `coppe-logo.pdf` the UFRJ shield and the COPPE/UFRJ wordmark. Both ship with the CoppeTeX distribution; they must be findable from the project directory (or on `TEXINPUTS`).

7 Required L^AT_EX packages

The class is built on the standard `book` class (`12pt`, `a4paper`, `oneside`) and brings in the stock CTAN packages listed below with `\RequirePackage`. On a full T_EX Live or MiK_TE_X install they are all present already, so this list matters only for a minimal install — or when you simply want to know which commands are already in scope. You do *not* need to `\usepackage` any of these yourself; every command and environment they provide is available in your manuscript. Options in brackets are the ones the class passes.

Engine and encoding

`iftex` detects the running engine, so the class can branch its setup between pdfT_EX and the Unicode engines (LuaT_EX/XeT_EX).

`inputenc` (`utf8`) UTF-8 input; loaded *only under pdfT_EX* (the Unicode engines read UTF-8 natively).

`fontenc` (`T1`) T1 font encoding, for correctly composed accented glyphs and hyphenation.

Mathematics and symbols

`amsmath` the AMS mathematics environments.

`amssymb` the AMS symbol fonts; loaded *only under pdfT_EX*. Under a Unicode engine load `unicode-math` (or `amssymb`) yourself — the class warns at end of document if neither is present.

Page layout and spacing

`geometry` A4 one-sided margins (3/3/2/2 cm) and header height.

`fancyhdr` running headers and footers.

`titlesec` chapter and section title formatting.

`setspace` one-and-a-half line spacing (double with the `doublespacing` option).

Lists, tables and rules

`enumitem` customisable list spacing and labels.

`tabularx` tables with automatic-width columns.

`booktabs` professional horizontal rules (`\toprule`, `\midrule`, `\bottomrule`).

`longtable` tables that break across pages.

Floats, graphics and captions

`graphicx` image inclusion.

`float` float placement control (e.g. the `[H]` specifier).

`caption` caption typography (the ABNT caption-on-top layout).

`subcaption` subfigures and subtables.

Code, verbatim and algorithms

`listings` source-code listings.

`fancyvrb` extended verbatim.

`algorithm2e` (`linesnumbered`, `lined`, `algochapter`, `ruled`) typeset algorithms;
the natural-language option tracks the main language.

Bibliography, quotations and language

`babel` multilingual typesetting; the language options are computed from the main
and foreign language slots.

`csquotes` context-sensitive quotation marks (the recommended companion of
`biblatex`).

`biblatex` (`backend=biber`) the bibliography engine and the ABNT `coppe` styles
(`coppe` or, with the `numbers` option, `coppe-numeric`). The backend is
`biber`, not `bibTeX`.

Hyperlinks

`hyperref` clickable cross-references, PDF bookmarks and document metadata.

Low-level utilities

`etoolbox` e-TeX programming toolkit.

`ltxcmds` low-level command helpers (`\ltx@ifpackageloaded`, ...).

`ifthen` the `\ifthenelse` conditional.

`hyphenat` hyphenation control (`\nohyphens`, used on the cover).

`lastpage` `\pageref{LastPage}` for the cataloging page count.

hologo T_EX-family logos (`\hologo{LaTeX}, \dots`).

xcolor colour support (used by `listings` and `hyperref`).

8 Quick, useful tips

Pictures. The default picture format of L^AT_EX is the Encapsulated PostScript (EPS). If you use pdfL^AT_EX, the default format becomes the PDF, but you can equally load PNG files. For such, you must enter the name of your image file without extension, e.g., `\includegraphics{filename}`, and pdf_latex will firstly look for a file called `filename.pdf` and after for file `filename.png`. For producing high quality pictures with embedded fonts we recommend the Ipe drawing software available [here](#).

Fonts. The default font in L^AT_EX is the Computer Modern. If you would like to try its enhanced version, consider using the `lmodern` package. To use Times, it is recommended to load the package `mathptmx`, rather than the deprecated `times`. There is also an enhanced Times version available with the `tgtermes` package. You can still use the Arial font face with the package `uarial`.

Hyperref. When working with PDF's, there is the possibility to add extra information to the file as the author's name, document title, subject, keywords, etc. This is easily done with the `hyperref` package, which is loaded by the `coppe` class itself (since v4.0): the user must *not* load it again in the manuscript preamble. Author, title, subject and keyword metadata are filled in automatically by `\maketitle`.

Math (amssymb vs. unicode-math). The class always loads `amsmath`. The companion AMS symbol package `amssymb`, however, is loaded *only* under pdfL^AT_EX: it is a legacy 8-bit package built around the `msam/msbm` Type 1 fonts and clashes with `unicode-math`, the modern OpenType-math package used on Unicode engines (LuaL^AT_EX, XeL^AT_EX). On those engines the class skips `amssymb` and schedules an *end-of-document* check: if by then the author has loaded neither `unicode-math` nor `amssymb` (which they could have loaded by hand, though it is discouraged), a `ClassWarning` reminds them to add

```
\usepackage{unicode-math}
\setmathfont{Latin Modern Math} % or STIX Two Math, ...
```

to the manuscript preamble. The class deliberately does *not* auto-load `unicode-math`, because picking an OpenType math font is a typographic choice that belongs to the author. **Never load `amssymb` together with `unicode-math`:** it would re-establish the 8-bit AMS fonts on top of the OpenType math font and split symbols across two incompatible encodings.

Printing. To get your work correctly printed, you must ensure that any page scaling option (e.g., fit or shrink to printable area) isn't enabled. This kind of option often comes in print dialogs of document visualization softwares.

`longquote` **Quotation** A quotation longer than three lines is INDENTED BY 4 cm from the left margin – not set with a 4 cm left margin, which is a different and much narrower thing. 2.2(b) of the manual says “recuo de 4 cm da margem esquerda”, so on an A4 page with a 3 cm margin the quoted text starts 7 cm from the edge of the paper. It carries no quotation marks and is set in the smaller font, single spaced. The `coppe` class provides the `longquote` environment, which does all of that; the indentation is `\recuolongquote`, should a Programa ever need to change it.

9 A simple example

```

1 \*example>
2 \documentclass[dsc]{coppe}
3
4 \usepackage{rotating}          % rodando coisas, como tabelas
5 \usepackage[most]{tcolorbox} % caixas de texto
6 \usepackage{tikz}             % gráficos vetoriais (o tcolorbox já o traz)
7 % Sob LuaLaTeX/XeLaTeX, descomente as duas linhas a seguir para obter os
8 % símbolos AMS (\mathbb, \hbar, \varnothing, ...) via unicode-math. Sob
9 % pdfLaTeX, a classe já carrega amssymb automaticamente; deixe comentado.
10 % \usepackage{unicode-math}
11 % \setmathfont{Latin Modern Math} % ou STIX Two Math, TeX Gyre Termes Math...
12 \DeclareMathOperator{\RMSE}{RMSE} % operadores próprios (cap. de matemática)
13 \DeclareMathOperator{\diag}{diag}
14 % A classe coppe (v4.0) já carrega: amsmath, booktabs, longtable, subcaption,
15 % hyperref, biblatex, listings, fancyvrb, algorithm2e -- não recarregue esses
16 % pacotes aqui (em particular, hyperref deve ficar a cargo da classe para que
17 % a ordem com setspace e biblatex permaneça correta).
18
19 \addbibresource{example.bib}% base de referências (biblatex/biber)
20
21 \makelosymbols
22 \makeloabbreviations
23 % Algumas coisas opcionais que não cabem no coppetex.sty
24 % \renewcommand{\familydefault}{\sfdefault} % para usar fontes SEM serifa no default
25 % \renewcommand{\familydefault}{\rmdefault} % para usar fontes COM serifa no default
26
27
28 \begin{document}
29   \title{Título da Tese}
30   \foreigntitle{Thesis Title}
31   \author{Nome do Autor}{Sobrenome}
32   \advisor[UFRJ]{Prof.}{Nome do Primeiro Orientador}{Sobrenome}{D.Sc.}
33   \advisor[UFRJ]{Prof.}{Nome do Segundo Orientador}{Sobrenome}{Ph.D.}
34   % A classe aceita de um a tres \advisor; se houver apenas um, o rotulo na
35   % capa e na folha de rosto sai como "Orientador" (singular), com dois ou
36   % mais sai como "Orientadores" (plural). Para o caso 1/3 orientadores,
37   % veja tests/test_brazilian_one_advisor.tex e tests/test_brazilian_three_advisors.tex.
38
39   \examiner[UFRJ]{Prof.}{Nome do Primeiro Examinador Sobrenome}{D.Sc.}
40   \examiner[UFRJ]{Prof.}{Nome do Segundo Examinador Sobrenome}{Ph.D.}
41   \examiner[UFF]{Prof.}{Nome do Terceiro Examinador Sobrenome}{D.Sc.}
42   \examiner[UNICAMP]{Prof.}{Nome do Quarto Examinador Sobrenome}{Ph.D.}
43   \examiner[USP]{Prof.}{Nome do Quinto Examinador Sobrenome}{Ph.D.}

```



```

44 \department{PESC}
45 \date{06}{2026}
46
47 \keyword{Primeira palavra-chave}
48 \keyword{Segunda palavra-chave}
49 \keyword{Terceira palavra-chave}
50
51 % Dados da folha adicional (Coleta CAPES), obrigatoria desde agosto de 2026.
52 % A folha e preenchida pelo Programa; estes campos apenas a compoem.
53 \areaconcentracao{Engenharia de Sistemas e Computa\c c\~ao}
54 \linhapesquisa{Engenharia de Dados e Conhecimento}
55 \tipoproducao{bibliografica} % bibliografica | artistica | tecnologica | tecnica
56 \projetovinculado{sim} % sim | nao
57 \nomeprojeto{Nome do Projeto de Pesquisa ao qual o trabalho se vincula}
58 \agenciafomento{Conselho Nacional de Desenvolvimento Cientifico e
59 Tecnologico}{CNPq}
60 \agenciafomento{Fundacao Carlos Chagas Filho de Amparo a Pesquisa do
61 Estado do Rio de Janeiro}{FAPERJ}
62 % A ficha oficial vem de fichacatalografica.sibi.ufrj.br; tendo o arquivo,
63 % descomente a linha abaixo e a moldura vazia da folha adicional some.
64 % \fichacatalografica{ficha.pdf}
65
66 \maketitle
67
68 \frontmatter
69 \dedication{A alguém cujo valor é digno desta dedicatória.}
70
71 \chapter*{Agradecimentos}
72
73 Gostaria de agradecer a todos os que contribu\’iram para este trabalho:
74 orientadores, colegas, familiares e \‘as ag\~encias de fomento. Lorem ipsum
75 dolor sit amet, consectetur adipiscing elit, sed do eiusmod tempor incididunt
76 ut labore et dolore magna aliqua.
77
78 \begin{abstract}
79
80 Apresenta-se, nesta tese, um exemplo de resumo. Lorem ipsum dolor sit amet,
81 consectetur adipiscing elit, sed do eiusmod tempor incididunt ut labore et
82 dolore magna aliqua. Ut enim ad minim veniam, quis nostrud exercitation
83 ullamco laboris nisi ut aliquip ex ea commodo consequat. Duis aute irure dolor
84 in reprehenderit in voluptate velit esse cillum dolore eu fugiat nulla
85 pariatur. Excepteur sint occaecat cupidatat non proident, sunt in culpa qui
86 officia deserunt mollit anim id est laborum. Sed ut perspiciatis unde omnis
87 iste natus error sit voluptatem accusantium doloremque laudantium, totam rem
88 aperiam, eaque ipsa quae ab illo inventore veritatis et quasi architecto beatae
89 vitae dicta sunt explicabo, nemo enim ipsam voluptatem quia voluptas sit
90 aspernatur aut odit aut fugit.
91
92 \end{abstract}
93
94 \begin{foreignabstract}
95
96 In this work, we present an example abstract. Lorem ipsum dolor sit amet,
97 consectetur adipiscing elit, sed do eiusmod tempor incididunt ut labore et

```

```

98 dolore magna aliqua. Ut enim ad minim veniam, quis nostrud exercitation
99 ullamco laboris nisi ut aliquip ex ea commodo consequat. Duis aute irure dolor
100 in reprehenderit in voluptate velit esse cillum dolore eu fugiat nulla
101 pariat. Excepteur sint occaecat cupidatat non proident, sunt in culpa qui
102 officia deserunt mollit anim id est laborum. Sed ut perspiciatis unde omnis
103 iste natus error sit voluptatem accusantium doloremque laudantium, totam rem
104 aperiam, eaque ipsa quae ab illo inventore veritatis et quasi architecto beatae
105 vitae dicta sunt explicabo, nemo enim ipsam voluptatem quia voluptas sit
106 aspernatur aut odit aut fugit.
107
108 \end{foreignabstract}
109
110 % Ordem pre-textual (NBR 14724/6027, R55): listas (ilustracoes, tabelas,
111 % abreviaturas e siglas, simbolos) ANTES do sumario; sumario por ultimo.
112 \listoffigures
113 \listoftables
114 \listofquadtros
115 \listofprogramas
116 \listofalgorithms
117 \printloabbreviations
118 \printlosymbols
119 \tableofcontents
120
121 \mainmatter
122 \chapter{Introdução}
123
124 Este é um documento exemplo para o uso da classe CoppeTeX, destinado a ajudar os alunos da
125
126 A classe \verb|coppe| foi criada originalmente por Vicente Helano e George Ainsworth, poré
127 Em 2026, Geraldo Xexéo, com forte ajuda das ferramentas Claude 4.7 Opus e Max e 4.8 Opus M
128
129 Provavelmente Geraldo Xexéo, professor do Programa de Engenharia de Sistemas e Computação
130
131 A versão mais atual dessa classe é mantida no GitHub, no repositório \url{https://github.c
132
133 Esse documento segue a norma de formatação de teses e dissertações da COPPE. Ele também po
134
135 Há um grupo de WhatsApp para perguntas e discussões do estilo: \href{https://github.com/CO
136
137 Bugs devem ser reportados na \href{https://github.com/COPPE-UFRJ/CoppeTeX/issues}{página d
138
139 Esse documento é usado como exemplo de coisas que podem ser feitas. Ele está configurado p
140
141 É importante de notar que essa classe não foi construída sobre a classe \LaTeX \ para a A
142
143 Essa classe implementa citações ABNT modernas e vários tipos de entrada descritos na norma
144
145 Este documento não substitui, mas complementa, o documento que descreve a classe.
146
147
148
149 \chapter{Configurações Iniciais}
150
151 A primeira coisa a fazer é escolher o tipo de documento. Isso é feito como uma opção no c

```

152
153 Como pode ser visto nesse documento, muita coisa pode ser configurada, o que gerará o tra
154
155 Recomendo ler o documento “‘The \verb*|coppe| document class’” para entender melhor todas
156
157 \section{Linguagem principal do texto}
158
159 Essa classe considera que o texto principal está em português e algumas partes específicas
160 \begin{itemize}
161 \item A opção \verb|english| deve ser usada no comando \verb|\documentclass|.
162 \item A bibliografia segue automaticamente o idioma do Babel (português ou inglês); não é p
163 \end{itemize}
164
165 A variação de linguagem, em inglês ou português apenas, já é suportada pela classe \CopeT
166
167
168 \section{Por que usar o \LaTeX}
169
170 Há uma grande discussão entre usuário de Word e \LaTeX, principalmente, quanto ao uso dess
171
172 Nós escolhemos o \LaTeX por alguns motivos: grande facilidade de seguir um estilo sem se p
173
174 As principais desvantagens são: idiossincrasias que podem gastar tempo, pouco controle sob
175
176 \section{Como e onde usar o \LaTeX}
177
178 Existem muitos tutoriais de \LaTeX, mas basicamente, em 2024, ele é usado em dois ambiente
179 \begin{enumerate}
180 \item Na sua máquina, instalando uma versão completa como o \hologo{MiKTeX}, típico do Win
181 \item Usar um ambiente na rede, como o Overleaf.
182 \end{enumerate}
183
184 Em todo caso, recomendo fortemente que, ao mesmo tempo, mantenha versões no Git e faça o b
185
186
187 \chapter{Algumas Regras da COPPE}
188
189 Todas abreviaturas e símbolos devem ser definida antes de utilizada. Isso é facilmente fei
190
191 É imprescindível definir os símbolos, tal como o
192 conjunto dos números reais $\mathbb{R}$ e o conjunto vazio $\emptyset$.
193 \syml{\mathbb{R}}{Conjunto dos números reais}
194 \syml{\emptyset}{Conjunto vazio}. Usamos esse exemplo aqui justamente para mostrar como
195
196 Para as listas de abreviaturas e símbolos funcionarem no Overleaf é necessário rodar o \ve
197
198 Como as listas de símbolos e de abreviaturas usamo o mesmo comando usado para criar índice
199
200 \section{Citações}
201 \label{sec:citacoes}
202
203 Citações curtas podem ser feitas \enquote{o comando quote} ou direto com “‘duas crases e do
204
205 \begin{longquote}

206 Um exemplo de citação longa nas regras da ABNT (4cm de recuo e fonte menor)
207 feita com o ambiente `\verb*|longquote|` The primary objective of this
208 investigation was to determine the feasibility of detecting corrosion in
209 aluminum Naval aircraft components with neutron radiographic interrogation
210 and the use of standard corrosion penetrameters. Secondary objectives
211 included the determination of the effect of object thickness on image quality,
212 the defining of minimum levels of detectability and a preliminary investigation
213 of a means whereby the degree of corrosion could be quantified with neutron
214 radiographic data. `\citep{article-example}`
215 `\end{longquote}`
216
217 Citações devem apontar as referências. Para isso, está disponível o ótimo pacote `\verb*|na`
218
219 Em todo caso, `\textbf{deve}` se tomar enorme atenção com as citações, para evitar ocorrer em
220
221 Quando se cita uma obra por meio de outra (`\textit{apud}`), pela ABNT entra na lista de ref
222
223 `\chapter{Floats}`
224
225 Grande parte dos problemas de iniciantes, e veteranos, em `\LaTeX` é da localização dos `\texti`
226
227 A regra geral de posicionamento é que uma figura ou quadro só pode aparecer a partir da mesm
228
229 `\textbf{Pela norma atual (manual2024), a legenda}` `\verb|\caption|` `\textbf{de toda ilustração}`
230
231 Quadros são ilustrações cujos dados são delimitados por linhas em `\emph{todas}` as bordas (ao
232
233
234 `\section{Tabelas e Figuras Padrão}`
235
236 Vamos ver uma tabela padrão, como a `\autoref{tab:exemplo_numeros}`.
237
238 `\begin{table}[ht]`
239 `\centering % Centraliza a tabela`
240 `\caption{Exemplo de Tabela de Números}`
241 `\label{tab:exemplo_numeros}`
242 `\begin{tabular}{ccc} % Define a quantidade de colunas`
243 `\hline % Linha superior`
244 `\textbf{Col. 1} & \textbf{Col. 2} & \textbf{Col. 3} \\ % Cabeçalhos`
245 `\hline % Linha média`
246 `1 & 2 & 3 \\ % Primeira linha de dados`
247 `\hline`
248 `4 & 5 & 6 \\ % Segunda linha de dados`
249 `\hline`
250 `7 & 8 & 9 \\ % Terceira linha de dados`
251 `\hline`
252 `10 & 11 & 12 \\ % Quarta linha de dados`
253 `\hline % Linha inferior`
254 `\end{tabular}`
255 `\fonte{Elabora\c{c}\~ao pr'opria.} % fonte obrigatoria, abaixo do float`
256 `\end{table}`
257
258
259

```

260 Já a \autoref{fig:exemplo_figura} é uma figura padrão, com controle da largura.
261
262 \begin{figure}[ht]
263 \centering % Centraliza a figura
264 \caption{Exemplo de Figura com Legenda Acima} % Legenda em cima (norma atual)
265 \label{fig:exemplo_figura} % Etiqueta para referência cruzada
266 \includegraphics[width=0.5\textwidth]{coppe-logo.pdf} % Inclui a imagem com metade da largura
267 \source{COPPE/UFRJ.} % fonte obrigatória, abaixo da figura
268 \end{figure}
269
270 Quadros têm a mesma regra (legenda acima, fonte abaixo), mas com linhas em todas as bordas.
271
272 \begin{quadro}[ht]
273 \centering
274 \caption{Exemplo de Quadro}
275 \label{qua:exemplo}
276 \begin{tabular}{|l|l|}
277 \hline
278 \textbf{Ilustração} & \textbf{Delimitação} \\
279 \hline
280 Tabela & Linhas apenas no topo e na base. \\
281 \hline
282 Quadro & Linhas em todas as bordas. \\
283 \hline
284 \end{tabular}
285 \fonte{Elabora\c{c}\~ao pr\'opria.}
286 \end{quadro}
287
288
289
290
291
292 \section{Tabelas mais elegantes}
293
294 Atualmente a tendência é usar tabelas mais leves, como \autoref{tab:exemplo_numerosbom}. Iss
295
296 \begin{table}[ht]
297 \centering % Centraliza a tabela
298 \caption{Exemplo de Tabela de Números mais elegantes}
299 \label{tab:exemplo_numerosbom}
300 \begin{tabular}{ccc} % Define a quantidade de colunas
301 \toprule % Linha superior
302 \textbf{Col. 1} & \textbf{Col. 2} & \textbf{Col. 3} \\
303 \midrule % Linha média
304 1 & 2 & 3 \\
305 4 & 5 & 6 \\
306 7 & 8 & 9 \\
307 10 & 11 & 12 \\
308 \bottomrule % Linha inferior
309 \end{tabular}
310 \fonte{Elabora\c{c}\~ao pr\'opria.}
311 \end{table}
312
313 \section{Tabelas Longas ou Largas}

```

```

314
315 Se sua tabela é muito longa ou larga, existem várias opções.
316 \begin{itemize}
317     \item alterar o tamanho da letra
318     \item Usar o longtable
319     \item rodar a tabela, fazendo ela em \textit{landscape}
320     \item fazer a tabela dentro de um minibox
321 \end{itemize}
322
323
324 \subsection{Tabelas largas demais}
325
326 É comum em teses que as tabelas sejam largas demais. Há várias formas de resolver isso.
327
328 A \autoref{tab:tabela_largafns} é larga demais, e nela isso é resolvido diminuindo a fonte p
329
330 \begin{table}[ht]
331 \centering % Centraliza a tabela
332 \caption{Exemplo de Tabela Larga com Fonte Menor}
333 \label{tab:tabela_largafns}
334 \footnotesize % Aplica uma fonte menor para a tabela
335 \begin{tabular}{ccccccc} % Aumente o número de colunas conforme necessário
336 \toprule
337 \textbf{Col. 1} & \textbf{Col. 2} & \textbf{Col. 3} & \textbf{Col. 4} & \textbf{Col. 5} & \textbf{Col. 6} & \textbf{Col. 7} \\
338 \midrule
339 Dado 1.1 & Dado 1.2 & Dado 1.3 & Dado 1.4 & Dado 1.5 & Dado 1.6 & Dado 1.7 & Dado 1.8 \\
340 Dado 2.1 & Dado 2.2 & Dado 2.3 & Dado 2.4 & Dado 2.5 & Dado 2.6 & Dado 2.7 & Dado 2.8 \\
341 Dado 3.1 & Dado 3.2 & Dado 3.3 & Dado 3.4 & Dado 3.5 & Dado 3.6 & Dado 3.7 & Dado 3.8 \\
342 \bottomrule
343 \end{tabular}
344 \fonte{Elabora\c{c}\~ao pr'opria.}
345 \end{table}
346
347 O comando \verb|\resizebox{width}{height}{content}| permite ajustar o tamanho de qualquer co
348
349 \begin{table}[ht]
350 \centering
351 \caption{Exemplo de Tabela Redimensionada}
352 \label{tab:examplerb}
353 \resizebox{\textwidth}{!}{%
354 \begin{tabular}{llll}
355 \toprule
356 Coluna 1 & Coluna 2 & Coluna 3 & Coluna 4 \\
357 \midrule
358 Dados 1 & Dados 2 & Dados 3 & Dados 4 \\
359 Dados 5 & Dados 6 & Dados 7 & Dados 8 \\
360 \bottomrule
361 \end{tabular}}%
362 }
363 \fonte{Elabora\c{c}\~ao pr'opria.}
364 \end{table}
365
366
367 Para rodar uma tabela muito larga em 90 graus no LaTeX, você pode usar o pacote \verb*|rotat

```

```

368
369 Aqui está um exemplo de como usar o ambiente \verb*|sidewaystable| para girar uma tabela. Pr
370
371 \begin{sidewaystable}
372 \centering
373 \caption{Sua Legenda Aqui}
374 \label{tab:sua_tabela}
375 \begin{tabular}{lll}
376 \toprule
377 Coluna 1 & Coluna 2 & Coluna 3 \\
378 \midrule
379 Item 1 & Item 2 & Item 3 \\
380 Item 4 & Item 5 & Item 6 \\
381 \bottomrule
382 \end{tabular}
383 \fonte{Elabora\c{c}\~ao pr\'opria.}
384 \end{sidewaystable}
385
386 Se a tabela for muito longa, o ambiente \verb|longtable| é o ideal. Ele fornece comandos par
387
388 % Exemplo de tabela longa que se estende por várias páginas
389 \begin{longtable}{|c|c|c|}
390 % primeiro cabeçalho (é o caption)
391 \caption{Exemplo de Tabela Longa}\label{tab:longa} \\
392 \hline \textbf{Col. 1} & \textbf{Col. 2} & \textbf{Col. 3} \\
393 \endfirsthead
394 % cabeçalho normal
395 \multicolumn{3}{c}{%
396 {\tablename\ \thetable} -- continuação da página anterior}} \\
397 \hline \textbf{Col. 1} & \textbf{Col. 2} & \textbf{Col. 3} \\
398 \endhead
399 % pé normal
400 \hline \multicolumn{3}{|r|}{\Continua na próxima página} \\
401 \endfoot
402 \hline
403 % último pé
404 \multicolumn{3}{|r|}{\Continua na próxima página}} \\
405 \hline \hline
406 \endlastfoot
407
408 % Conteúdo da tabela
409 1 & 2 & 3 \\
410 4 & 5 & 6 \\
411 1 & 2 & 3 \\
412 4 & 5 & 6 \\
413 1 & 2 & 3 \\
414 4 & 5 & 6 \\
415 1 & 2 & 3 \\
416 4 & 5 & 6 \\
417 1 & 2 & 3 \\
418 4 & 5 & 6 \\
419 1 & 2 & 3 \\
420 4 & 5 & 6 \\
421 1 & 2 & 3

```

422 4 & 5 & 6 \\
 423 1 & 2 & 3 \\
 424 4 & 5 & 6 \\
 425 1 & 2 & 3 \\
 426 1 & 2 & 3 \\
 427 4 & 5 & 6 \\
 428 1 & 2 & 3 \\
 429 4 & 5 & 6 \\
 430 1 & 2 & 3 \\
 431 4 & 5 & 6 \\
 432 1 & 2 & 3 \\
 433 4 & 5 & 6 \\
 434 1 & 2 & 3 \\
 435 4 & 5 & 6 \\
 436 1 & 2 & 3 \\
 437 4 & 5 & 6 \\
 438 1 & 2 & 3 \\
 439 4 & 5 & 6 \\
 440 1 & 2 & 3 \\
 441 4 & 5 & 6 \\
 442 1 & 2 & 3 \\
 443 4 & 5 & 6 \\
 444 1 & 2 & 3 \\
 445 4 & 5 & 6 \\
 446 1 & 2 & 3 \\
 447 4 & 5 & 6 \\
 448 1 & 2 & 3 \\
 449 4 & 5 & 6 \\1 & 2 & 3 \\
 450 4 & 5 & 6 \\
 451 1 & 2 & 3 \\
 452 4 & 5 & 6 \\
 453 1 & 2 & 3 \\
 454 1 & 2 & 3 \\
 455 4 & 5 & 6 \\
 456 1 & 2 & 3 \\
 457 4 & 5 & 6 \\
 458 1 & 2 & 3 \\
 459 4 & 5 & 6 \\
 460 1 & 2 & 3 \\
 461 4 & 5 & 6 \\
 462 1 & 2 & 3 \\
 463 4 & 5 & 6 \\
 464 1 & 2 & 3 \\
 465 4 & 5 & 6 \\
 466 1 & 2 & 3 \\
 467 4 & 5 & 6 \\
 468 1 & 2 & 3 \\
 469 4 & 5 & 6 \\
 470 1 & 2 & 3 \\
 471 4 & 5 & 6 \\
 472 1 & 2 & 3 \\
 473 4 & 5 & 6 \\
 474 1 & 2 & 3 \\
 475 4 & 5 & 6 \\


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476 1 & 2 & 3 \\
477 4 & 5 & 6 \\
478 1 & 2 & 3 \\
479 4 & 5 & 6 \\
480 1 & 2 & 3 \\
481 4 & 5 & 6 \\
482 1 & 2 & 3 \\
483 4 & 5 & 6 \\
484 1 & 2 & 3 \\
485 4 & 5 & 6 \\
486 1 & 2 & 3 \\
487 4 & 5 & 6 \\
488 1 & 2 & 3 \\
489 4 & 5 & 6 \\
490 1 & 2 & 3 \\
491 4 & 5 & 6 \\
492 1 & 2 & 3 \\
493 4 & 5 & 6 \\
494 1 & 2 & 3 \\
495 4 & 5 & 6 \\
496 1 & 2 & 3 \\
497 4 & 5 & 6 \\
498 1 & 2 & 3 \\
499 4 & 5 & 6 \\
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501 4 & 5 & 6 \\
502 1 & 2 & 3 \\
503 4 & 5 & 6 \\
504 1 & 2 & 3 \\
505 4 & 5 & 6 \\
506 1 & 2 & 3 \\
507 4 & 5 & 6 \\
508 1 & 2 & 3 \\
509 4 & 5 & 6 \\
510 1 & 2 & 3 \\
511 4 & 5 & 6 \\
512 1 & 2 & 3 \\
513 4 & 5 & 6 \\
514 1 & 2 & 3 \\
515 4 & 5 & 6 \\
516
517 % Repetir linhas semelhantes conforme necessário para estender a tabela por 3 páginas
518 \end{longtable}
519 \fonte{Elabora\c{c}\~ao pr\'opria.}% a longtable não é float: a fonte vem após o ambiente
520
521 \section{Subfiguras (subcaption)}
522
523 Para colocar várias imagens (ou tabelas) lado a lado, cada uma com sua própria legenda, use
524
525 \begin{figure}[ht]
526   \centering
527   \begin{subfigure}[b]{0.42\textwidth}
528     \centering\includegraphics[width=\linewidth]{example-image-a}
529     \caption{Primeira vista}\label{fig:sub-a}

```

```

530 \end{subfigure}
531 \hfill
532 \begin{subfigure}[b]{0.42\textwidth}
533   \centering\includegraphics[width=\linewidth]{example-image-b}
534   \caption{Segunda vista}\label{fig:sub-b}
535 \end{subfigure}
536 \caption{Figura com duas subfiguras}
537 \label{fig:subs}
538 \fonte{Elabora\c{c}\~ao pr\`opria.}
539 \end{figure}
540
541 Referencie a figura inteira com \verb|\autoref{fig:subs}|. Cada parte tem
542 rótulo próprio: \verb|\autoref{fig:sub-a}| leva à \verb|\autoref{fig:sub-a}|.
543
544 \section{Imagens (graphicx)}
545
546 A classe já carrega o \verb|graphicx|. Insira imagens com \verb|\includegraphics|; as opções
547 \begin{itemize}
548   \item \verb|[width=0.8\textwidth]| ou \verb|[scale=0.5]| --- tamanho;
549   \item \verb|[angle=90]| --- rotação;
550   \item \verb|[trim=l b r t, clip]| --- recorte (corta as bordas \emph{esquerda, baixo, direi}
551 \end{itemize}
552 Formatos aceitos com \hologo{pdfLaTeX}/\hologo{LuaLaTeX}: \texttt{pdf}, \texttt{png}, \texttt{t}
553
554 \section{Introdução ao tikz}
555
556 0 \verb|tikz| (disponível também via \verb|tcolorbox|) desenha gráficos vetoriais no próprio
557
558 \begin{figure}[ht]
559   \centering
560   \begin{tikzpicture}
561     \node[draw, rounded corners] (a) at (0,0) {Início};
562     \node[draw, rounded corners] (b) at (4,0) {Fim};
563     \draw[->, thick] (a) -- (b) node[midway, above] {processo};
564   \end{tikzpicture}
565   \caption{Diagrama simples em tikz}
566   \label{fig:tikz}
567   \fonte{Elabora\c{c}\~ao pr\`opria.}
568 \end{figure}
569
570 Para gráficos de dados (curvas, barras), veja o \verb|pgfplots|, construído sobre o \verb|ti
571
572 \section{Float ou direto no texto?}
573
574 Colocar a tabela ou imagem dentro de \verb|figure|/\verb|table| (um \emph{float}) deixa o \L
575
576 \chapter{Using BibLaTeX}
577 \label{cap:biblatex}
578
579 Esta classe gera as referências com BibLaTeX e o \emph{backend} \texttt{biber}, no padrão
580
581 \textbf{É} obrigatório compilar com \texttt{biber} (e não com \hologo{BibTeX}). A sequênci
582
583 Os comandos do \texttt{natbib} (\verb|\citet|, \verb|\citep|) continuam disponíveis porque

```

```

584
585 \begin{table}[h]
586 \caption{Exemplos de cita{\c c}\~oes utilizando o comando padr\~ao
587 \texttt{\textbackslash cite} do \LaTeX\ e
588 o comando \texttt{\textbackslash citep},
589 disponível pela opção \texttt{\natbib} do BibLaTeX.}
590 \label{tab:citation}
591 \centering
592 {\footnotesize
593 \begin{tabular}{|c|c|c|}
594 \hline
595 Tipo da Publicação & \verb|\cite| & \verb|\citep|\\
596 \hline
597 Livro & \cite{book-example} & \citep{book-example}\\
598 Artigo & \cite{article-example} & \citep{article-example}\\
599 Relatório & \cite{techreport-example} & \citep{techreport-example}\\
600 Relatório & \cite{techreport-exampleIn} & \citep{techreport-exampleIn}\\
601 Anais de Congresso & \cite{inproceedings-example} &
602 \citep{inproceedings-example}\\
603 Séries & \cite{incollection-example} & \citep{incollection-example}\\
604 Em Livro & \cite{inbook-example} & \citep{inbook-example}\\
605 Dissertação de mestrado & \cite{mastersthesis-example} &
606 \citep{mastersthesis-example}\\
607 Tese de doutorado & \cite{phdthesis-example} & \citep{phdthesis-example}\\
608 \hline
609 \end{tabular}}
610 \fonte{Elabora\c c}\~ao pr\~o'pria.}
611 \end{table}
612
613 \begin{table}[h]
614 \caption{Exemplos de cita{\c c}\~oes utilizando o comando padr\~ao
615 \texttt{\textbackslash cite} do \LaTeX\ e
616 o comando \texttt{\textbackslash citet},
617 disponível pela opção \texttt{\natbib} do BibLaTeX. Além disso, usando o booktabs.}
618 \label{tab:citation1}
619 \centering
620 {\footnotesize
621 \begin{tabular}{ccc}
622 \toprule
623 Tipo da Publicação & \verb|\cite| & \verb|\citet|\\
624 \midrule
625 Livro & \cite{book-example} & \citet{book-example}\\
626 Artigo & \cite{article-example} & \citet{article-example}\\
627 Relatório & \cite{techreport-example} & \citet{techreport-example}\\
628 Relatório & \cite{techreport-exampleIn} & \citet{techreport-exampleIn}\\
629 Anais de Congresso & \cite{inproceedings-example} &
630 \citet{inproceedings-example}\\
631 Séries & \cite{incollection-example} & \citet{incollection-example}\\
632 Em Livro & \cite{inbook-example} & \citet{inbook-example}\\
633 Dissertação de mestrado & \cite{mastersthesis-example} &
634 \citet{mastersthesis-example}\\
635 Tese de doutorado & \cite{phdthesis-example} & \citet{phdthesis-example}\\
636 \bottomrule
637 \end{tabular}}

```

```

638 \fonte{Elabora\c{c}\~ao pr\'opria.}
639 \end{table}
640
641 \section{Comandos de citação}
642
643 Os principais comandos são: \verb|\cite| (referência simples), \verb|\citet| (autor por ex
644
645 \subsection{Parâmetro opcional: página e “tradução nossa”}
646
647 Todos os comandos aceitam um parâmetro opcional para a localização (página, seção) e obser
648
649 \subsection{Citação de citação (\emph{apud})}
650
651 Para citar uma obra através de outra, use \verb|\citetapud| e \verb|\citepapud| (demonstra
652
653 \subsection{Múltiplas bibliografias}
654
655 Para separar as referências (por tipo, capítulo ou tema), use o ambiente \verb|refsection|
656
657 \section{Tipos de entrada bibliográfica}
658
659 Cada tipo de documento da ABNT corresponde a um tipo de entrada do BibLaTeX, com um sinôni
660
661 \subsection{Tipos acadêmicos principais}
662 \begin{itemize}
663 \item Livro / monografia --- \verb|@book| (\verb|@livro|)
664 \item Artigo de periódico --- \verb|@article| (\verb|@artigo|); matéria de jornal --- \verb|@journal|
665 \item Capítulo de livro --- \verb|@incollection| (\verb|@partedelivro|), \verb|@inbook|
666 \item Evento no todo (anais) --- \verb|@proceedings| (\verb|@anais|); trabalho em evento
667 \item Tese/dissertação --- \verb|@thesis| (\verb|@tese|, \verb|@dissertacao|); com tipo
668 \item Relatório --- \verb|@report| (\verb|@relatorio|, \verb|@relatoriotecnico|); periód
669 \item Manual --- \verb|@manual|; diversos --- \verb|@misc| (\verb|@diverso|); página/sit
670 \end{itemize}
671
672 \subsection{Tipos especiais (ABNT §4.2.5--4.2.14)}
673 \begin{itemize}
674 \item Patente --- \verb|@patent| (\verb|@patente|)
675 \item Legislação / ato normativo --- \verb|@legislation| (\verb|@legislacao|, \verb|@ato|
676 \item Norma técnica --- \verb|@standard| (\verb|@norma|); partitura --- \verb|@music| (\verb|@musica|)
677 \item Documento cartográfico (mapa) --- \verb|@map| (\verb|@mapa|, \verb|@cartografico|)
678 \item Audiovisual --- \verb|@video| (\verb|@audiovisual|); filme --- \verb|@movie| (\verb|@filme|)
679 \end{itemize}
680
681 \subsection{Jogos e TV (tipos próprios da CoppeTeX)}
682 \begin{itemize}
683 \item Jogo --- \verb|@game| (\verb|@jogo|); de tabuleiro --- \verb|@boardgame| (\verb|@jogo|)
684 \item Programa de TV / série --- \verb|@tvshow| (\verb|@programadetv|, \verb|@serietv|)
685 \end{itemize}
686
687 Os campos também aceitam sinônimos em português (sem acentos): \verb|autor|, \verb|titulo|
688
689 \chapter{Alguns outros exemplo úteis}
690
691 \begin{tcolorbox}[title=Meu Textbox]

```

```

692 Este é o conteúdo do meu textbox. Você pode adicionar qualquer texto aqui, bem como incluir
693 \end{tcolorbox}
694
695 \begin{tcolorbox}
696 Este é o conteúdo do meu textbox sem título. Você pode adicionar qualquer texto aqui, bem como
697 \end{tcolorbox}
698
699 \begin{figure}[ht]
700     \centering
701     \begin{tikzpicture}
702         \node[anchor=south west,inner sep=0] (image) at (0,0) {\includegraphics[width=0.5\textwidth]{image.png}}
703         \begin{scope}[x={(image.south east)},y={(image.north west)}]
704             % Definindo o textbox dentro da figura
705             \node[anchor=north west, text width=0.3\textwidth, fill=white, opacity=0.7, text=Este textbox fala sobre como inserir um textbox dentro de uma figura usando tcolorbox]
706                 \begin{tcolorbox}[colback=red!5!white,colframe=red!75!black,title=Textbox de exemplo]
707                     Este textbox fala sobre como inserir um textbox dentro de uma figura usando tcolorbox
708                 \end{tcolorbox}
709             };
710         \end{scope}
711     \end{tikzpicture}
712     \caption{Figura com Textbox}
713     \label{fig:figura_com_textbox1}
714     \fonte{Elabora\c{c}\~ao pr\'opria.}
715 \end{figure}
716
717
718 \begin{figure}[ht]
719     \centering
720     \begin{tcolorbox}
721 Este é o conteúdo do meu textbox sem título. Você pode adicionar qualquer texto aqui, bem como
722 \end{tcolorbox}
723     \caption{Figura com Textbox simples}
724     \label{fig:figura_com_textbox}
725     \fonte{Elabora\c{c}\~ao pr\'opria.}
726 \end{figure}
727
728 \section{Listagens de código e algoritmos}
729
730 O CoppeTeX traz embutido o suporte a listagens de programas (pacote \verb|listings|) e a alg
731
732 \begin{python}[caption={Exemplo de programa em Python},label=prog:exemplo]
733 def saudacao(nome):
734     # imprime uma saudação acentuada
735     print(f"Olá, {nome}!")
736
737 saudacao("COPPE")
738 \end{python}
739 \source{Elaboração própria.}
740
741 Para inserir código no meio do texto use \verb|\pythoninline|, como em \pythoninline{saudacao("COPPE")}
742
743 \begin{Verbatim}[frame=lines,numbers=left,numbersep=0.5em]
744 Olá, COPPE!
745 \end{Verbatim}

```

```

746
747 Há estilos prontos também para \verb|java|, \verb|xml|, \verb|html| e \verb|prolog|.
748 Há também \verb|\pythoninline| e \verb|\javainline| para código no meio do
749 texto, e o ambiente \verb|Verbatim| (com as opções acima) para a saída de programas. Para al
750
751 \begin{algorithm}[H]
752 \caption{Soma dos elementos de um vetor}
753 \label{alg:soma}
754 \Dados{vetor  $v$  com  $n$  elementos}
755 \Resultado{soma  $s$  dos elementos de  $v$ }
756  $s \leftarrow 0$ ;
757 \Para{ $i \leftarrow 1$  \Ate{ }  $n$ }{
758    $s \leftarrow s + v[i]$ ;
759 }
760 \Retorna  $s$ ;
761 \end{algorithm}
762
763 \chapter{Escrevendo matemática}
764
765 A classe carrega \texttt{amsmath} em qualquer motor. Sob \hologo{pdfLaTeX} ela também carr
766
767 \section{Alinhamento (align)}
768 O ambiente \verb|align| alinha equações pelo \verb|&| e numera cada linha (\verb|\\| quebr
769 \begin{align}
770   f(x) &= (x+1)^2 \\
771         &= x^2 + 2x + 1 .
772 \end{align}
773
774 \section{Matrizes (matrix)}
775 O \texttt{amsmath} traz \verb|pmatrix|, \verb|bmatrix|, \verb|vmatrix|, entre outros:
776 \begin{equation}
777   A = \begin{bmatrix} a_{11} & a_{12} \\ a_{21} & a_{22} \end{bmatrix}
778   \quad
779   \det A = \begin{vmatrix} a_{11} & a_{12} \\ a_{21} & a_{22} \end{vmatrix}
780 \end{equation}
781
782 \section{Subequações (subequations)}
783 Para numerar um grupo de equações sob o mesmo número, com letras (a), (b):
784 \begin{subequations}
785 \begin{align}
786   a &= b + c , \\
787   d &= e - f .
788 \end{align}
789 \end{subequations}
790
791 \section{Estruturas alinhadas (array)}
792 O \verb|array| monta estruturas dentro do modo matemático, como definições por partes:
793 \begin{equation}
794   |x| = \left\{ \begin{array}{l}
795     x, \quad \text{se } x \geq 0 , \\
796     -x, \quad \text{caso contrário.}
797   \end{array} \right.
798 \end{equation}
799

```

```

800 \section{Operadores (\textbackslash DeclareMathOperator)}
801 Para operadores com a tipografia correta (romano, espaçamento adequado), declare-os no pre
802 \begin{verbatim}
803 \usepackage{amsmath}
804 \DeclareMathOperator{\RMSE}{RMSE}
805 \DeclareMathOperator{\diag}{diag}
806 \end{verbatim}
807 Assim:
808 \begin{itemize}
809   \item \verb|$\RMSE(y,\hat{y})$| produz  $\RMSE(y,\hat{y})$ ;
810   \item \verb|$\diag(\lambda_1,\lambda_2)$| produz
811          $\diag(\lambda_1,\lambda_2)$ .
812 \end{itemize}
813
814 \section{Unicode e LuaLaTeX (unicode-math)}
815 \label{sec:unicode-math}
816 Ao migrar para \hologo{LuaLaTeX} (ou \hologo{XeLaTeX}), \textbf{não carregue} \texttt{amss
817 \begin{verbatim}
818 \usepackage{amsmath}      % a classe já carrega
819 \usepackage{unicode-math} % NÃO use \usepackage{amssymb} junto
820 \setmathfont{Latin Modern Math} % ou STIX Two Math
821 \end{verbatim}
822 O código clássico continua valendo ( $\mathbb{R}$ ,  $\int_a^b f(x)dx$ ,  $\alpha$ ) e você ai
823
824 \chapter{Referências cruzadas, notas e índices}
825 \label{cap:refs}
826
827 Este capítulo reúne os recursos para \emph{apontar} para coisas: rótulos, notas de rodapé,
828
829 \section{Rótulos e referências cruzadas}
830
831 Marque um elemento com \verb|\label{tipo:nome}| e aponte para ele com \verb|\ref| (só o nú
832
833 \section{Notas de rodapé}
834
835 Use \verb|\footnote{texto}| \footnote{Como esta nota, que a classe imprime em espaçamento s
836
837 \section{hyperref e marcadores}
838
839 A classe carrega o \verb|hyperref|: o sumário, as citações e os \verb|\ref|/\verb|\autoref|
840
841 \section{URLs}
842
843 Para endereços, use \verb|\url{https://exemplo.org}|: o \verb|hyperref| cria o link e queb
844 use \verb|\href{https://exemplo.org}{texto}|. Evite digitar URLs como texto comum --- elas n
845
846 \section{Índice remissivo}
847
848 Marque os termos com \verb|\index{termo}| e imprima o índice com \verb|\printindex| (carre
849
850 \section{Traduções (NBR 10520)}
851
852 Ao citar um trecho que você mesmo traduziu, coloque a tradução no corpo do texto e escreva
853

```

```

854 \chapter{Truques de \LaTeX}
855
856 \section{Comentários mágicos (\texttt{\% !TeX})}
857 Linhas \verb|\% !TeX| no topo do arquivo dizem ao editor (\hologo{TeX}studio, Overleaf, \ho
858 \begin{verbatim}
859 % !TeX program = lualatex
860 % !TeX encoding = UTF-8
861 % !TeX spellcheck = pt_BR
862 % !BIB program = biber
863 \end{verbatim}
864 Com isso o documento compila igual em qualquer ambiente. O \texttt{latexmkrc} cumpre papel
865
866 \section{Espaço engolido}
867 Um comando sem argumento engole o espaço seguinte: \verb|\LaTeX é bom| sai como “\LaTeX é
868
869 \section{Escopo (grupos)}
870 Chaves \verb|\{...\}| (ou \verb|\begingroup|\dots\verb|\endgroup|) criam um escopo: mudanças
871
872 \section{Descarregar figuras: \texttt{\textbackslash clearpage} * \texttt{\textbackslash m
873 \verb|\newpage| apenas quebra a página e leva os floats pendentes adiante, acumulando-os.
874
875 \section{\texttt{latexmkrc}, perl e Overleaf}
876 O \verb|latexmk| automatiza a compilação (roda \LaTeX, \verb|biber| e \verb|makeindex| qua
877
878 \chapter{Método Proposto}
879 \chapter{Resultados e Discuss\~oes}
880
881 \section{Algumas Demonstra\c c\~oes}
882
883 A Lista de Símbolos precisa usar comandos específicos. Aqui vamos usar os símbolos $\alpha
884 \syml[beta]{Beta}{A palavra Beta more e corrigida}
885 \syml[zbeta]{\beta}{A letra $\beta$ corrigida}
886 \syml{beta}{A palavra beta}
887 \syml{alpha}{A palavra alpha}
888 \syml[alpha]{Alpha}{A palavra Alpha}
889 \syml[zzalpha]{\alpha}{A letra $\alpha$ corrigida}
890 \syml[marco]{Marco}{A palavra Marco corrigida}
891
892 A Lista de Abreviações segue, a partir de 2024, a mesma regra, e aqui seguem alguns exempl
893 \abbrev{GoT}{Game of Thrones}
894 \abbrev[GOT]{GoT}{Game of Thrones ordenado como GOT}
895 \abbrev[iot]{IoT}{IoT ordenado como iot}
896 \abbrev[IoT]{IoT}{IoT ordenado como IoT}
897 \abbrev[IOT]{IoT}{IoT ordenado como IOT}
898 \abbrev{IoT}{IoT com ordenação default}
899 \abbrev[ITU]{ITU}{ITU mesmo}
900
901 \newsigla{ufrj}{UFRJ}{Universidade Federal do Rio de Janeiro}
902 \newsigla{cpgp}{CPGP}{Comissão de Pós-Graduação e Pesquisa de Engenharia}
903
904 As siglas (manual2024, seção 2.8) são expandidas na primeira ocorrência e registradas auto
905
906
907

```



```

908
909 \chapter{Conclusões}
910
911 \backmatter
912
913 % Tipos exóticos da ABNT (NBR 6023): patente, legislação, norma, mapa,
914 % partitura --- declarados via sinônimos em português em example.bib.
915 \nocite{patente-example,legislacao-example,norma-example,mapa-example,partitura-example}
916 \nocite{jurisprudencia-example}
917 \nocite{videogame-example,filme-example,semautor-example,semautorart-example}
918 \nocite{jornalcaderno-example,jornal-example}
919 \printbibliography
920
921
922
923 \appendix
924
925 \chapter{Um apêndice}
926
927 Segundo a norma da ABNT (Associação Brasileira de Normas Técnicas), a definição e utilização
928
929 Apêndice: O apêndice é um texto ou documento elaborado pelo autor do trabalho com o objetivo
930
931
932 \chapter{Outro apêndice}
933
934 \annex
935
936
937 \chapter{Um Anexo}
938 Segundo a norma da ABNT (Associação Brasileira de Normas Técnicas), a definição e utilização
939
940
941
942 Anexo: O anexo, por sua vez, consiste em um texto ou documento não elaborado pelo autor, que
943
944 No modelo \CoppeTeX os anexos devem obrigatoriamente vir depois dos apêndices e usam o comando
945
946
947 \chapter{Outro Anexo}
948
949
950 \coppetexfinalpage% colofão institucional (página par, face do contracapa)
951
952 \end{document}
953
954 \example

```

10 Implementation

10.1 The ‘coppe.cls’ file

```

955 \class
956 \def\filename{coppe.dtx}

```

```

957 \def\fileversion{v4.0}
958 \def\filedate{2026/05/28}
959 \NeedsTeXFormat{LaTeX2e}[1995/12/01]
960 \ProvidesClass{coppe}[\filedate\ \fileversion\ COPPE Dissertations and Thesis]
961 % Save the class version and date into private macros immediately, before any
962 % required package can redefine the very common \fileversion / \filedate
963 % global names (LaTeX has no convention reserving them for the host class).
964 % \coppe@version / \coppe@date are what user-facing macros (e.g.
965 % \coppetexfinalpage) should reference.
966 \let\coppe@version\fileversion
967 \let\coppe@date\filedate
968 % Document date stored in private slots so \date can set them without ever
969 % touching TeX's \month/\year primitives (which would break \today). They
970 % default to the current system month/year, so a document without \date
971 % still renders sensibly on the cover and folha de rosto.
972 \edef\coppe@docmonth{\the\month}
973 \edef\coppe@docyear{\the\year}
974 \LoadClass[12pt,a4paper,oneside]{book}
975 % setspace, hyperref and the bibliography engine (biblatex+biber) are loaded at
976 % the END of the class (after every class hook is defined), in that order:
977 % setspace -> hyperref -> biblatex. hyperref must come late so all the macros
978 % it patches (TOC, refs, lists) are already defined, and biblatex must come
979 % after hyperref so its citation links and back-references work.
980 \RequirePackage{hyphenat}
981 \RequirePackage{lastpage}
982 \RequirePackage{ifthen}
983 \RequirePackage{graphicx}
984 \RequirePackage{tabularx}
985 \RequirePackage{booktabs} % regras de tabela profissionais (toprule/midrule/bottomrule)
986 \RequirePackage{longtable} % tabelas que quebram entre páginas
987 \RequirePackage{etoolbox}
988 \RequirePackage{ltxcmds}
989 \RequirePackage{hologo} % TeX-family logos: \hologo{TeX}, \hologo{LaTeX}, ...
990 % Render \TeX, \LaTeX and friends via hologo (\CoppTeX is a custom logo, kept).
991 \def\TeX{\hologo{TeX}}
992 \def\LaTeX{\hologo{LaTeX}}
993 \def\LaTeXe{\hologo{LaTeXe}}
994 \def\BibTeX{\hologo{BibTeX}}
995 \def\pdfLaTeX{\hologo{pdfLaTeX}}
996 \def\pdfTeX{\hologo{pdfTeX}}
997 \def\LuaLaTeX{\hologo{LuaLaTeX}}
998 \def\XeLaTeX{\hologo{XeLaTeX}}
999 \def\MiKTeX{\hologo{MiKTeX}}
1000 % Engine-aware UTF-8 input: pdfLaTeX needs inputenc(utf8); LuaLaTeX (and XeTeX)
1001 % are UTF-8 native, so inputenc must NOT be loaded there (it would clash). Source
1002 % files may use accented Latin characters (áéíóú) directly. Under pdfLaTeX we also
1003 % suggest LuaLaTeX, which handles Unicode/fonts better (at the cost of speed).
1004 \RequirePackage{iftex}
1005 \ifPDFTeX
1006 \RequirePackage[utf8]{inputenc}
1007 \AtEndOfClass{\ClassWarningNoLine{coppe}{%
1008 Although slower, you should try compiling with LuaLaTeX}}
1009 \fi
1010 % lmodern MUST come with the T1 encoding. Without it, T1 falls back to the EC

```

```

1011 % fonts, which exist only as METAFONT bitmaps in most installations, and the
1012 % whole document is typeset in Type3 bitmap fonts. Measured before this change:
1013 % example.pdf carried 87 Type3 fonts against 6 Type1, and coppe.pdf 133 against
1014 % 2. The consequences are not cosmetic:
1015 %
1016 % * the text renders blurred at any zoom and prints poorly;
1017 % * copying and searching text in the PDF is unreliable;
1018 % * PDF/A -- required for deposit by 2.2(d) of the 2026 manual -- is not
1019 %   attainable with bitmap text.
1020 %
1021 % Latin Modern is the vector successor of Computer Modern, metrically and
1022 % visually near-identical, and ships with every current distribution, so the
1023 % page layout does not move.
1024 \RequirePackage{lmodern}
1025 \RequirePackage[T1]{fontenc}
1026 % Math support. amsmath is universal and works on every engine, so the class
1027 % brings it in unconditionally (it must precede hyperref, which is loaded at
1028 % the end of the class). amssymb is the legacy AMS symbol package, built
1029 % around the 8-bit Type 1 fonts msam/msbm; on a Unicode engine (LuaLaTeX or
1030 % XeLaTeX) it clashes with unicode-math, which already provides all of its
1031 % symbols through Unicode math codes. So amssymb is loaded ONLY under
1032 % pdfLaTeX; Unicode-engine users get a ClassWarning pointing them at
1033 % unicode-math (the class does not auto-load unicode-math because choosing
1034 % an OpenType math font is a typographic decision that belongs to the author).
1035 \RequirePackage{amsmath}
1036 \ifPDFTeX
1037   \RequirePackage{amssymb}
1038 \else
1039 % Unicode engine: defer the check to end of document. Only warn if the
1040 % user did NOT add either unicode-math (the modern OpenType-math route)
1041 % or amssymb (the legacy route, which they would have loaded manually).
1042 % This avoids nagging authors who already followed the documented
1043 % \usepackage{unicode-math} workflow.
1044 \AtEndDocument{%
1045   \@ifpackageloaded{unicode-math}{}{%
1046     \@ifpackageloaded{amssymb}{}{%
1047       \ClassWarningNoLine{coppe}{%
1048         You are compiling with a Unicode engine (LuaLaTeX or XeLaTeX) and\MessageBreak
1049         neither unicode-math nor amssymb is loaded. AMS-style symbols\MessageBreak
1050         (\string\mathbb, \string\hbar, \string\varnothing, ...) will be undefined.\MessageBreak
1051         Add to your preamble:\MessageBreak
1052         \space\space\string\usepackage{unicode-math}\MessageBreak
1053         \space\space\string\setmathfont{Latin Modern Math}\MessageBreak
1054         Do NOT load amssymb together with unicode-math -- it would clash\MessageBreak
1055         with the OpenType math font setup}}}}
1056 \fi
1057 % Margins per the SiBI manual, 9th rev. ed. (2026), 2.3: left 3cm, top 3cm,
1058 % right 2cm, bottom 2cm -- ONE-SIDED. The 2025 edition also specified verso
1059 % margins (right 3cm / left 2cm) and recommended printing the body on both
1060 % sides; the 2026 revision DELETED the verso margins entirely, following CEPG
1061 % Resolution 246/2023, which ended the printed deposit. A digital-only document
1062 % has no gutter, so the mirrored layout (previously inner=outer=2cm plus a 1cm
1063 % bindingoffset) no longer has any normative basis and is now opt-in via the
1064 % \texttt{twoside} class option, handled after \ProcessOptions below.

```

```

1065 %
1066 % headheight=15pt is given here (not patched later) so geometry's own page
1067 % arithmetic stays self-consistent: fancyhdr needs >= 14.49998pt of header
1068 % height to host the running head, and the standard 12pt default triggers
1069 % its "headheight is too small" warning. Letting geometry know up front
1070 % keeps \topmargin a derived value -- we never touch it by hand.
1071 % (geometry has no \texttt{oneside} key -- only \texttt{twoside}; with the key
1072 % absent it follows the class, which \LoadClass set to one-sided above.)
1073 % headsep is given explicitly to place the folio. 2.7 measures it the way it
1074 % measures the right margin -- to the digits themselves: "a 2cm da borda
1075 % superior, ficando o ultimo algarismo a 2cm da borda direita". The right edge
1076 % already landed at 2.02cm; the top of the digits sat at 1.87cm, too high.
1077 % geometry derives \topmargin from top - 1in - headheight - headsep, so
1078 % shortening headsep lowers the header and leaves the body top untouched:
1079 % \topmargin + headheight + headsep is 13.088pt either way, and 3cm is where the
1080 % body still starts. The value is CALIBRATED, not computed -- the folio sits a
1081 % little lower than the arithmetic alone predicts, because fancyhdr's box does
1082 % not put the digits flush with the top of \headheight. Two measured points,
1083 % taken from the rendered page rather than from the PDF's font metrics:
1084 %   headsep 18.07pt -> digits at 1.8669cm   headsep 14.29pt -> 2.0616cm
1085 % which put 2.0000cm at 15.49pt while the folio was still set at the body size.
1086 % It is \footnotesize now (2.2(b) puts pagination in the smaller font), the
1087 % digits are shorter, and their top therefore sits lower for the same baseline:
1088 % re-measured, 15.49pt gave 2.0659cm and the value moved to 16.87pt. That is
1089 % precisely why this is a measured constant and not a computed one -- change
1090 % the folio's size or headheight and it has to be measured again, not scaled.
1091 % Verified on the rendered page: the top of the digits and their right edge come
1092 % out at the same 2.019cm, and the right margin is 2cm by construction, so that
1093 % shared 0.019cm is the ruler's own bias and not a residual error.
1094 \RequirePackage[a4paper,headheight=15pt,headsep=16.87pt,%
1095 top=3cm,bottom=2cm,left=3cm,right=2cm]{geometry}
1096 % The first paragraph of every section must be indented like any other. LaTeX's
1097 % default is the Anglo-Saxon convention -- no indent after a heading -- which is
1098 % not what Brazilian typography (nor the SiBI manual's own examples) uses.
1099 % indentfirst was mentioned only as a commented suggestion in the example
1100 % preamble, so every document built with this class came out with the first
1101 % paragraph of each section flush left. Loading it in the class fixes it for
1102 % everybody instead of asking each author to remember.
1103 %
1104 % Loaded BEFORE titlesec so titlesec's \@afterindent handling sees the patched
1105 % \@afterindentfalse.
1106 % Long unbreakable tokens -- \verb|...| in running prose, command names, URLs --
1107 % cannot be hyphenated, so TeX overfills the line rather than stretching the
1108 % spaces. example.tex alone produced 17 overfull boxes this way, the worst 86pt,
1109 % which is the "text does not always respect the margins" a recent review
1110 % reported. \emergencystretch gives TeX a final pass in which it may stretch
1111 % interword spaces instead of running into the margin. It cannot save a single
1112 % token longer than the measure, but it clears the ordinary cases.
1113 \setlength\emergencystretch{3em}
1114 \RequirePackage{indentfirst}
1115 \RequirePackage{titlesec}
1116 % --- Section heading formats (manual2024 2.6 / proposta P22) -----
1117 % chapter (numbered): "N TITLE", ALL CAPS, bold, flush left (no "Capitulo N").
1118 % APENDICE/ANEXO chapters print "APENDICE A -- Title" (R75/R76); normal chapters

```

```

1119 % keep "N Title". \appendix and \annex raise the flag (see below).
1120 \newif\if@coppeappendix \@coppeappendixfalse
1121 \titleformat{\chapter}[hang]
1122   {\normalfont\Large\bfseries}
1123   {\if@coppeappendix\MakeUppercase{\appendixname}\ \thechapter\ \textendash\else\thechapter\
1124   {1ex}\{\MakeUppercase}
1125 \titlespacing*{\chapter}{0pt}{10pt}{1.5\baselineskip}
1126 % 2.6 says the numeric indicative of a section precedes its title flush left,
1127 % and that titles WITHOUT a numeric indicative are centred -- and then lists
1128 % them, "apendice(s)" and "anexo(s)" among them. A letter is not a numeric
1129 % indicative, and the manual sets its own Annex A centred. The chapter format is
1130 % therefore re-issued when \appendix or \annex is called: same label, same
1131 % travessao, but a centred block instead of a flush-left hang. Numbered chapters
1132 % are untouched -- they DO carry a numeric indicative and stay flush left.
1133 \newcommand\coppe@appendixheadings{%
1134   \titleformat{\chapter}[block]
1135     {\normalfont\Large\bfseries\filcenter}
1136     {\MakeUppercase{\appendixname}\ \thechapter\ \textendash}
1137     {1ex}\{\MakeUppercase}%
1138   \titlespacing*{\chapter}{0pt}{10pt}{1.5\baselineskip}}
1139 \let\coppe@orig@appendix\appendix
1140 \renewcommand\appendix{%
1141   \coppe@orig@appendix\@coppeappendixtrue\coppe@appendixheadings}
1142 % Sumario (TOC) entries for APENDICE/ANEXO chapters read "APENDICE A -- Title"
1143 % / "ANEXO A -- Title" (R75/R76), not the bare "A". The label word is chosen by
1144 % which macro is written (so it is correct at .toc-render time even though the
1145 % appendixname has changed back); the chapter letter is baked in at write time.
1146 \newif\if@coppeannex \@coppeannexfalse
1147 \newcommand\coppe@tocapp[1]{%
1148   \MakeUppercase{\coppestring{appendix}}%
1149   \nobreakspace#1\nobreakspace\textendash\nobreakspace}
1150 \newcommand\coppe@tocanx[1]{%
1151   \MakeUppercase{\coppestring{annex}}%
1152   \nobreakspace#1\nobreakspace\textendash\nobreakspace}
1153 \def\@chapter[#1]#2{%
1154   \ifnum \c@secnumdepth >\m@ne
1155     \if@mainmatter
1156       \refstepcounter{chapter}%
1157       \typeout{\@chapapp\space\thechapter.}%
1158       \if@coppeappendix
1159         \if@coppeannex
1160           \addcontentsline{toc}{chapter}{\protect\coppe@tocanx{\thechapter}#1}%
1161         \else
1162           \addcontentsline{toc}{chapter}{\protect\coppe@tocapp{\thechapter}#1}%
1163         \fi
1164       \else
1165         \addcontentsline{toc}{chapter}{\protect\numberline{\thechapter}#1}%
1166       \fi
1167     \else
1168       \addcontentsline{toc}{chapter}{#1}%
1169     \fi
1170   \else
1171     \addcontentsline{toc}{chapter}{#1}%
1172   \fi

```

```

1173 \chaptermark{#1}%
1174 \addtocontents{lof}{\protect\addvspace{10\p@}}%
1175 \addtocontents{lot}{\protect\addvspace{10\p@}}%
1176 \if@twocolumn
1177   \topnewpage[\@makechapterhead{#2}]%
1178 \else
1179   \makechapterhead{#2}%
1180   \afterheading
1181 \fi}
1182 % chapter (unnumbered): centred ALL CAPS bold (errata, lists, resumo, sumario,
1183 % referencias, glossario, ... -- unnumbered headings are centred, P29-P30).
1184 \titleformat{name=\chapter,numberless}[hang]
1185   {\normalfont\Large\bfseries\filcenter}{0pt}{0pt}{\MakeUppercase}
1186 \titlespacing*{name=\chapter,numberless}{0pt}{10pt}{1.5\baselineskip}
1187 % section: ALL CAPS, normal weight -- the \bfseries here contradicted this very
1188 % comment until v4.1, and made the secondary level as heavy as the primary one,
1189 % defeating the "destaque gradativo" of 2.6. The gradation is now
1190 % chapter = ALL CAPS bold, section = ALL CAPS regular, subsection = Title Case
1191 % bold, subsubsection = Title Case bold italic, as read from the manual.
1192 \titleformat{section}
1193   {\normalfont\large}{\thesection}{1ex}{\MakeUppercase}
1194 % subsection: bold (author writes Title Case).
1195 \titleformat{subsection}
1196   {\normalfont\bfseries}{\thesubsection}{1ex}{}
1197 % subsubsection: bold italic.
1198 \titleformat{subsubsection}
1199   {\normalfont\bfseries\itshape}{\thesubsubsection}{1ex}{}
1200 % paragraph = quinary section: italic, the author capitalising only the first
1201 % word. 2.6 recommends going as deep as the quinary, and the annotation on the
1202 % 2025 manual (specs/ANOTACOEES.md) describes exactly this shape.
1203 %
1204 % book makes \paragraph a RUN-IN heading, so the text continued on the same
1205 % line -- 2.6 requires "o texto deve ser iniciado em outra linha". The positive
1206 % "after" spacing in \titlespacing turns it into a displayed heading.
1207 \titleformat{paragraph}[hang]
1208   {\normalfont\itshape}{\theparagraph}{1ex}{}
1209 \titlespacing*{paragraph}{0pt}{2.5ex plus 1ex minus .2ex}{1.5ex plus .2ex}
1210 % book defaults secnumdepth to 2, so \subsubsection and \paragraph came out
1211 % UNNUMBERED however carefully \titleformat named \thesubsubsection. 2.6 has
1212 % the numbering following the whole chain down to the quinary, and the manual's
1213 % own sumario is four levels deep, so tocdepth follows.
1214 \setcounter{secnumdepth}{4}
1215 \setcounter{tocdepth}{4}
1216 % 2.6: "no sumario, as secoes devem ser grafadas conforme apresentadas no corpo
1217 % do trabalho". The body sets chapter and section in ALL CAPS (via the
1218 % \MakeUppercase in their \titleformat), but that case change belongs to the
1219 % printed heading alone -- the .toc keeps the title as typed, so the sumario
1220 % came out in ordinary case and contradicted the body.
1221 %
1222 % \l@chapter and \l@section are the two TOC line formatters; wrapping their
1223 % entry argument is enough, and leaves \l@subsection and below untouched, which
1224 % is what we want since those levels are NOT uppercased in the body either.
1225 % The entry carries \numberline{N} plus the title; \MakeUppercase passes the
1226 % digits through unchanged and leaves robust commands (including hyperref's

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1227 % link wrappers) intact.
1228 \let\coppe@orig@lchapter\l@chapter
1229 \renewcommand\l@chapter[2]{\coppe@orig@lchapter{\MakeUppercase{#1}}{#2}}
1230 \let\coppe@orig@lsection\l@section
1231 \renewcommand\l@section[2]{\coppe@orig@lsection{\MakeUppercase{#1}}{#2}}
1232 % --- Captions (manual2024 2.10-2.11 / proposta P6): smaller font, ABNT
1233 % "Figura 1 -- titulo" travessao separator (sources handled by \source). ---
1234 \RequirePackage{caption}
1235 % justification=centering makes MULTI-LINE captions centre like the one-line
1236 % ones. Without it, caption's singlelinecheck centres a caption that fits on one
1237 % line and justifies one that wraps, so in a document with mixed caption lengths
1238 % a couple of tables looked left-aligned while the rest looked centred -- which
1239 % is what a recent review noticed about tables 5.1 and 5.2. 2.10 does not
1240 % prescribe the alignment; it does not tolerate two alignments either.
1241 \captionsetup{font=footnotesize,labelsep=endash,justification=centering}
1242 % subcaption (NÃO subfig/subfigure) for subfigures / subtables; load right after
1243 % caption so it picks up the ABNT captionsetup above.
1244 \RequirePackage{subcaption}
1245 % --- Floats: caption on TOP and a mandatory "Fonte:" line at the bottom.
1246 % manual2024 2.10-2.11 (R46/R51: identification on top; R48/R52: the source
1247 % at the bottom is mandatory). The float package's plaintop style forces the
1248 % caption above the body for figure, table and the new quadro float. -----
1249 \RequirePackage{float}
1250 \floatstyle{plaintop}
1251 \restylefloat{figure}
1252 \restylefloat{table}
1253 % New "quadro" float -- "Quadro" (pt) / "Frame" (en) (R53) -- with its own list
1254 % file (.loq). A quadro's data is bordered on all sides (drawn by the author),
1255 % unlike a tabela (rules on top and bottom only).
1256 \providecommand{\quadroname}{\coppestring{frame}}
1257 \providecommand{\listquadroname}{\coppestring{listframe}}
1258 \newfloat{quadro}{htbp}{loq}[chapter]
1259 \floatname{quadro}{\quadroname}
1260 \providecommand{\quadroautorefname}{\quadroname}
1261 % Format the Lista de Quadros entries exactly like the Lista de Figuras ones
1262 % (float's own \l@quadro lays the number/title out incorrectly).
1263 \let\l@quadro\l@figure
1264 \newcommand\listofquadros{%
1265   \chapter*{\listquadroname}%
1266   \coppe@listtoc{\listquadroname}%
1267   \@mkboth{\MakeUppercase\listquadroname}{\MakeUppercase\listquadroname}%
1268   \@starttoc{loq}}
1269 % \source / \fonte: the illustration source (ABNT-mandatory). Typeset below the
1270 % float as a smaller, single-spaced, centred line; it does NOT advance any
1271 % caption counter. \cpsource/\cpfonte are clash-safe aliases (the rename rule);
1272 % the label is "Fonte" (pt) / "Source" (en).
1273 \newcommand{\cpsourcenam}{\coppestring{source}}
1274 \newcommand{\cpsource}[1]{%
1275   \par\addvspace{\smallskipamount}%
1276   {\footnotesize\singlespacing\centering\cpsourcenam: #1\par}}
1277 \let\cpfonte\cpsource
1278 \@ifundefined{source}{\let\source\cpsource}{}%
1279 \@ifundefined{fonte}{\let\fonte\cpsource}{}%
1280 % \newcoppefloat{env}{caption name}{list title}: declare an extra caption-on-top

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1281 % float (e.g. "picture") that supports \source and gains a \listof<env>s command.
1282 \newcommand{\newcoppefloat}[3]{%
1283   {\floatstyle{plaintop}\newfloat{#1}{htbp}{lo#1}}%
1284   \floatname{#1}{#2}%
1285   \expandafter\newcommand\csname listof#1s\endcsname{%
1286     \chapter*{#3}\coppe@listtoc{#3}%
1287     \@mkboth{\MakeUppercase{#3}}{\MakeUppercase{#3}}%
1288     \@starttoc{lo#1}}}}
1289 % --- Footnotes (manual2024 / R16, R8, R21): single spacing and a 5 cm rule
1290 %   (filete) from the left margin; footnotesize is the book default. -----
1291 \let\coppe@orig@footnotetext\@footnotetext
1292 \long\def\@footnotetext#1{\coppe@orig@footnotetext{\singlespacing#1}}
1293 \renewcommand\footnoterule{\kern-3\p@\hrule\@width 5cm\kern 2.6\p@}
1294 % --- alineas: list keyed by lowercase letters a) b) c); subitems with "--"
1295 %   (manual2024 2.6 / proposta P24). -----
1296 \RequirePackage{enumitem}
1297 \newlist{alineas}{enumerate}{2}
1298 \setlist{alineas,1}{label=\alph*},leftmargin=1.8em,itemsep=0pt}
1299 \setlist{alineas,2}{label=--,leftmargin=1.8em,itemsep=0pt}
1300 % --- Page numbering: arabic in the top OUTER corner on textual pages
1301 %   (manual2024 2.7 / proposta P25-P26). Pre-textual numbering is set by
1302 %   \frontmatter (unnumbered by default; roman with the numeraisromanos
1303 %   class option). -----
1304 \RequirePackage{fancyhdr}
1305 % \headheight is set to 15pt by geometry up above (see the geometry call),
1306 % which is what fancyhdr needs (>= 14.49998pt) for the running head.
1307 % 2.2(b) lists what is set in "fonte menor e uniforme": long quotations,
1308 % footnotes, PAGINATION, and the captions of illustrations and tables. The
1309 % folio was the one item on that list still coming out at the body size of 12pt.
1310 % It is \footnotesize now, which is the same smaller size the other three
1311 % already use -- "uniforme" is part of the requirement, not just "menor".
1312 % Pre-textual lists reach the sumario only under the listasnosumario option
1313 % (see the option's declaration for the reason). Post-textual material calls
1314 % \addcontentsline directly and is unaffected.
1315 \newcommand\coppe@listtoc[1]{%
1316   \if@coppe@listastoc\addcontentsline{toc}{chapter}{#1}\fi}
1317 % \coppe@glosspre marks the two PRE-textual uses of the glossary environment --
1318 % the list of abbreviations and the list of symbols -- apart from the genuine
1319 % post-textual Glossario, which shares the same environment but must appear in
1320 % the sumario.
1321 \newif\if@coppeglosspre \@coppeglossprefalse
1322 \newcommand\coppe@folio{\footnotesize\thepage}
1323 \fancypagestyle{coppe}{\fancyhf{}\fancyhead[R0,LE]{\coppe@folio}%
1324   \renewcommand{\headrulewidth}{0pt}}
1325 % Blank pages inserted by \cleardoublepage (two-sided) must be empty: no page
1326 % number may leak onto them (e.g. the verso right after the cover).
1327 \renewcommand{\cleardoublepage}{%
1328   \clearpage
1329   \if@twoside
1330     \ifodd\c@page\else
1331       \hbox{}\thispagestyle{empty}\newpage
1332     \if@twocolumn\hbox{}\newpage\fi
1333   \fi
1334   \fi}

```



```

1335 \def\CoppeTeX{{\rm C\kern-.05em{\sc o\kern-.025em p\kern-.025em
1336 p\kern-.025em e}}\kern-.08em
1337 T\kern-.1667em\lower.5ex\hbox{E}\kern-.125emX\spacefactor1000}

1338 \newboolean{maledoc}
1339 \setboolean{maledoc}{false}
1340 %
1341 % Class options.
1342 % Multilingual model (CoppeTeX 4.x, branch nlinguas). The class holds three
1343 % fixed language slots:
1344 % * main -- the language of the body, captions, and the main abstract.
1345 % Set by a class option: brazilian (default), english,
1346 % spanish, french, italian, or any user-supplied language
1347 % that ships a coppe-lang-<name>.def file.
1348 % * foreign -- the language used by the foreignabstract environment and
1349 % by \foreign@... registers. Defaults to english (or to
1350 % brazilian when main=english) for back-compat.
1351 % * third -- optional, loaded on demand with \usecoppelanguage{<name>}
1352 % in the preamble (for citations or quotations in a third
1353 % language). Has no class option of its own.
1354 % Note: the legacy \if@english boolean (defined and toggled here in
1355 % v3.x for backward compatibility) was removed in v4.0 -- nothing
1356 % inside the class or in the biblatex style files (.bbx/.cbx/.dbx/.ltx)
1357 % tests it. Code that needs to branch on the main language should test
1358 % \ifx\coppe@mainlang\coppe@englishstr instead.
1359 \def\coppe@mainlang{brazilian}
1360 \def\coppe@foreignlang{english}
1361 \DeclareOption{brazilian}{%
1362 \def\coppe@mainlang{brazilian}\def\coppe@foreignlang{english}}
1363 \DeclareOption{english}{%
1364 \def\coppe@mainlang{english}\def\coppe@foreignlang{brazilian}}
1365 \DeclareOption{spanish}{%
1366 \def\coppe@mainlang{spanish}\def\coppe@foreignlang{english}}
1367 \DeclareOption{french}{%
1368 \def\coppe@mainlang{french}\def\coppe@foreignlang{english}}
1369 \DeclareOption{italian}{%
1370 \def\coppe@mainlang{italian}\def\coppe@foreignlang{english}}
1371 \DeclareOption{msc}{%
1372 \newcommand{\@degree}{M.Sc.}
1373 \newcommand{\@degreename}{Mestrado}
1374 \newcommand{\local@degname}{Mestre}
1375 \newcommand{\foreign@degname}{Master}
1376 \newcommand{\local@doctype}{Disserta{\c c}{\~ a}o}
1377 \newcommand{\foreign@doctype}{Dissertation}
1378 }
1379 \DeclareOption{dsccexam}{%
1380 \newcommand{\@degree}{D.Sc.}
1381 \newcommand{\@degreename}{Doutorado}
1382 \newcommand{\local@degname}{Doutor}
1383 \newcommand{\foreign@degname}{Doctor}
1384 \setboolean{maledoc}{true}
1385 \newcommand{\local@doctype}{Exame de Qualifica{\c c}{\~ a}o}
1386 \newcommand{\foreign@doctype}{Qualifying Exam}
1387 }
1388 \DeclareOption{msccexam}{%

```

```

1389 \newcommand{\@degree}{M.Sc.}
1390 \newcommand{\@degreename}{Mestrado}
1391 \newcommand{\local@degname}{Mestre}
1392 \newcommand{\foreign@degname}{Master}
1393 \setboolean{maledoc}{true}
1394 \newcommand\local@doctype{Exame de Qualifica{\c c}{\~ a}o}
1395 \newcommand\foreign@doctype{Qualifying Exam}
1396 }
1397 \DeclareOption{dsc}{%
1398 \newcommand{\@degree}{D.Sc.}
1399 \newcommand{\@degreename}{Doutorado}
1400 \newcommand{\local@degname}{Doutor}
1401 \newcommand{\foreign@degname}{Doctor}
1402 \newcommand\local@doctype{Tese}
1403 \newcommand\foreign@doctype{Thesis}
1404 }
1405 \newif\if@coppenumeric\@coppenumericfalse
1406 \DeclareOption{numbers}{\@coppenumerictrue}
1407 % numeraisromanos: show roman numerals on the pre-textual pages. Default: the
1408 % pre-textual pages are counted but not numbered (manual2024 2.7).
1409 \newif\if@copperoman\@copperomanfalse
1410 \DeclareOption{numeraisromanos}{\@copperomantrue}

```

Default line spacing is one-and-a-half; the doublespacing option switches to double spacing. Since `setspace` is loaded at the end of the class (after `hyperref`), the actual `\onehalfspacing`/`\doublespacing` call is deferred: the option only flips a flag, which the late-load block reads.

```

1411 \newif\if@coppedoublespace\@coppedoublespacefalse
1412 \DeclareOption{doublespacing}{\@coppedoublespacetrue}
1413 % Two-sided layout is opt-in since v4.1: the 2026 manual dropped the verso
1414 % margins and the deposit is digital (CEPG 246/2023). Given \texttt{twoside},
1415 % the mirrored 3cm inner / 2cm outer layout is restored after \ProcessOptions.
1416 % With no \fichacatalografica file given, the folha adicional leaves a framed
1417 % blank where the ficha goes. The rascunhoficha option fills that blank with the
1418 % class's own LaTeX-composed ficha instead -- useful while writing, never valid
1419 % for deposit: the official ficha comes from the SiBI generator.
1420 % Annex D of the manual draws one signature rule per board member, with the
1421 % identification underneath. The COPPE folha de aprovacao has listed the names
1422 % compactly, without rules, for years. The compact list stays the default and
1423 % the rules are opt-in, so existing theses do not change shape overnight.
1424 % PDF/A output. Opt-in for now: 2.2(d) of the 2026 manual requires the deposited
1425 % file to be PDF/A, but the constraints (embedded fonts, colour profile, no
1426 % encryption, restricted transparency) can trip up a document that pulls in
1427 % arbitrary images or packages, and breaking existing theses to enforce it would
1428 % be worse than letting the author switch it on when depositing.
1429 \newif\if@coppepdfa\@coppepdfafalse
1430 \DeclareOption{pdfa}{\@coppepdfatrue}
1431 \newif\if@coppeassin\@coppeassinfalse
1432 \DeclareOption{assinaturas}{\@coppeassintrue}
1433 % As paginas de resumo listam os ORIENTADORES abaixo do titulo. O coorientador
1434 % nao aparece ali, e isso e tradicao da COPPE, nao exigencia do manual: a
1435 % 3.1.2.1.4 descreve o resumo e nao lista quem assina a orientacao. Como ha
1436 % Programas que preferem ve-lo, a opcao acrescenta os coorientadores nas tres
1437 % paginas de resumo, no mesmo formato da folha de rosto. Sem a opcao nada muda.

```

```

1438 \newif\if@coppecoadvresumo \@coppecoadvresumofalse
1439 \DeclareOption{coorientador}{\@coppecoadvresumotrue}
1440 % 3.1.2.1.6 says the sumario enumerates the divisions "na mesma ordem e grafia
1441 % em que a materia nele se sucede" and adds "inclusive os elementos
1442 % pos-textuais" -- and then settles the question by pointing at its own sumario
1443 % as the model. That sumario opens at "1 INTRODUCAO": the lists of figures, of
1444 % tables, of abbreviations and of symbols come BEFORE the sumario and are not
1445 % listed in it, which is also the rule of NBR 6027 for pre-textual elements.
1446 % The class was adding all of them, so every thesis carried six entries the
1447 % manual's own example does not have. They come out. Post-textual elements --
1448 % Referencias, Glossario, Apendices, Anexos, Indice -- stay, as required.
1449 %
1450 % Programas that want the old behaviour ask for it by name.
1451 \newif\if@coppelistastoc \@coppelistastocfalse
1452 \DeclareOption{listasnosumario}{\@coppelistastoctrue}
1453 \newif\if@copperascunho \@copperascunhofalse
1454 \DeclareOption{rascunhoficha}{\@copperascunhotrue}
1455 \newif\if@coppetwoside \@coppetwosidefalse
1456 \DeclareOption{twoside}{\@coppetwosidetrue}
1457 \DeclareOption{oneside}{\@coppetwosidefalse}
1458 \ProcessOptions\relax
1459 % \LoadClass and geometry ran before \ProcessOptions (see above), so the
1460 % two-sided layout can only be applied here. geometry reads left/right as
1461 % inner/outer once twoside is on, so the 3cm/2cm pair set above becomes the
1462 % mirrored gutter with no further arithmetic. \@twosidetrue is what book's own
1463 % twoside option sets, and is what \cleardoublepage and fancyhdr test.
1464 % In one-sided mode fancyhdr treats every page as odd, so the \fancyhead[RO,LE]
1465 % declarations elsewhere already put the folio in the top right corner of every
1466 % folha, as 2.7 requires.
1467 \if@coppetwoside
1468   \@twosidetrue
1469   \geometry{twoside}%
1470 \fi
1471 % Built-in language names (used by the loader and the dispatcher below).
1472 \def\coppe@@brazilianstr{brazilian}
1473 \def\coppe@@englishstr{english}
1474 % Load Babel with the foreign language first and the main language last
1475 % (Babel treats the LAST option in the list as the main language). With
1476 % pt-main this produces [english,brazilian]{babel} and with en-main
1477 % [brazilian,english] -- identical to the historical \if@english calls.
1478 \edef\coppe@babelopts{\coppe@foreignlang,\coppe@mainlang}
1479 \expandafter\RequirePackage\expandafter[\coppe@babelopts]{babel}
1480 % --- Multilingual string dispatcher -----
1481 % \copperdefstring{<lang>}{<key>}{<value>} adds one entry to the table for
1482 % language <lang>. The three retrievers below look strings up either at the
1483 % current babel language (\coppestring -- follows \selectlanguage), or pinned
1484 % to the main slot (\coppemainstring) or the foreign slot (\coppeforeignstring).
1485 \providecommand{\copperdefstring}[3]{%
1486   \expandafter\gdef\csname coppe@str@#1@#2\endcsname{#3}}
1487 \providecommand{\coppestring}[1]{\@nameuse{coppe@str@language@#1}}
1488 \providecommand{\coppemainstring}[1]{\@nameuse{coppe@str@coppe@mainlang@#1}}
1489 \providecommand{\coppeforeignstring}[1]{\@nameuse{coppe@str@coppe@foreignlang@#1}}
1490 % Load an extra language pack at any point in the preamble (third-language
1491 % slot for citations or quotations beyond the main+foreign pair).

```

```

1492 \providecommand{\usecoppelanguage}[1]{%
1493   \def\coppe@@cur{#1}%
1494   \ifx\coppe@@cur\coppe@@brazilianstr\else
1495   \ifx\coppe@@cur\coppe@@englishstr\else
1496     \InputIfFileExists{coppe-lang-#1.def}{}
1497     {\ClassWarning{coppe}{Language pack coppe-lang-#1.def not found}}%
1498   \fi\fi}
1499 % \titlein{<lang>}{<text>}: record the title in any language. Will be
1500 % consulted by the cover/folha renderers once they are migrated in step 2.
1501 % In step 1 it is just a no-op store -- the historical \title / \foreigntitle
1502 % still drive every visible site, so behavior for pt/en is unchanged.
1503 \providecommand{\titlein}[2]{%
1504   \expandafter\gdef\csname coppe@title@#1\endcsname{#2}}
1505 % --- Built-in string tables: brazilian + english. The values mirror the
1506 % literals currently scattered through the class; step-1 only populates the
1507 % tables -- no site reads them yet, so the rendered output is unchanged. ----
1508 \copperdefstring{brazilian}{appendix}{Ap\~endice}
1509 \copperdefstring{english}{appendix}{Appendix}
1510 \copperdefstring{brazilian}{annex}{Anexo}
1511 \copperdefstring{english}{annex}{Annex}
1512 \copperdefstring{brazilian}{frame}{Quadro}
1513 \copperdefstring{english}{frame}{Frame}
1514 \copperdefstring{brazilian}{listframe}{Lista de Quadros}
1515 \copperdefstring{english}{listframe}{List of Frames}
1516 \copperdefstring{brazilian}{source}{Fonte}
1517 \copperdefstring{english}{source}{Source}
1518 \copperdefstring{brazilian}{program}{Programa}
1519 \copperdefstring{english}{program}{Program}
1520 \copperdefstring{brazilian}{listprogram}{Lista de Programas}
1521 \copperdefstring{english}{listprogram}{List of Programs}
1522 \copperdefstring{brazilian}{references}{Refer\~encias}
1523 \copperdefstring{english}{references}{References}
1524 \copperdefstring{brazilian}{in:}{em}
1525 \copperdefstring{english}{in:}{in}
1526 \copperdefstring{brazilian}{art}{Ilustra\c{c}\~ao}
1527 \copperdefstring{english}{art}{Art}
1528 \copperdefstring{brazilian}{platform}{Plataforma}
1529 \copperdefstring{english}{platform}{Platform}
1530 \copperdefstring{brazilian}{directedby}{Dire\c{c}\~ao}
1531 \copperdefstring{english}{directedby}{Directed by}
1532 \copperdefstring{brazilian}{producedby}{Produ\c{c}\~ao}
1533 \copperdefstring{english}{producedby}{Produced by}
1534 \copperdefstring{brazilian}{listabbreviation}{Lista de Abreviaturas e Siglas}
1535 \copperdefstring{english}{listabbreviation}{List of Abbreviations and Acronyms}
1536 \copperdefstring{brazilian}{listsymbol}{Lista de S{\~i}mbolos}
1537 \copperdefstring{english}{listsymbol}{List of Symbols}
1538 \copperdefstring{brazilian}{glossary}{Gloss{\~a}rio}
1539 \copperdefstring{english}{glossary}{Glossary}
1540 \copperdefstring{brazilian}{advisor}{Orientador}
1541 \copperdefstring{english}{advisor}{Advisor}
1542 \copperdefstring{brazilian}{advisors}{Orientadores}
1543 \copperdefstring{english}{advisors}{Advisors}
1544 \copperdefstring{brazilian}{dept}{Programa}
1545 \copperdefstring{english}{dept}{Department}

```

```

1546 \copperdefstring{brazilian}{approvedmale} {Aprovado por}
1547 \copperdefstring{brazilian}{approvedfemale}{Aprovada por}
1548 \copperdefstring{brazilian}{approvedonmale} {Aprovado em}
1549 \copperdefstring{brazilian}{approvedonfemale}{Aprovada em}
1550 \copperdefstring{brazilian}{keywordsname} {Palavras-chave}
1551 \copperdefstring{brazilian}{volumename} {Volume}
1552 \copperdefstring{brazilian}{coadvisor} {Coorientador}
1553 \copperdefstring{brazilian}{coadvisors} {Coorientadores}
1554 \copperdefstring{brazilian}{ofname} {de}
1555 \copperdefstring{english} {approvedmale} {Approved by}
1556 \copperdefstring{english} {approvedfemale}{Approved by}
1557 \copperdefstring{english} {approvedonmale} {Approved on}
1558 \copperdefstring{english} {approvedonfemale}{Approved on}
1559 \copperdefstring{english} {keywordsname} {Keywords}
1560 \copperdefstring{english} {volumename} {Volume}
1561 \copperdefstring{english} {coadvisor} {Co-advisor}
1562 \copperdefstring{english} {coadvisors} {Co-advisors}
1563 \copperdefstring{english} {ofname} {of}
1564 % Institutional names: always Portuguese on the cover/folha (UFRJ template).
1565 \copperdefstring{brazilian}{universityname}{Universidade Federal do Rio de Janeiro}
1566 \copperdefstring{english} {universityname}{Universidade Federal do Rio de Janeiro}
1567 \copperdefstring{brazilian}{cityname} {Rio de Janeiro}
1568 \copperdefstring{english} {cityname} {Rio de Janeiro}
1569 \copperdefstring{brazilian}{statename} {RJ}
1570 \copperdefstring{english} {statename} {RJ}
1571 \copperdefstring{brazilian}{countryname} {Brasil}
1572 \copperdefstring{english} {countryname} {Brasil}
1573 % Month names.
1574 \copperdefstring{brazilian}{monthname1} {Janeiro}
1575 \copperdefstring{brazilian}{monthname2} {Fevereiro}
1576 \copperdefstring{brazilian}{monthname3} {Mar{\c c}o}
1577 \copperdefstring{brazilian}{monthname4} {Abril}
1578 \copperdefstring{brazilian}{monthname5} {Maio}
1579 \copperdefstring{brazilian}{monthname6} {Junho}
1580 \copperdefstring{brazilian}{monthname7} {Julho}
1581 \copperdefstring{brazilian}{monthname8} {Agosto}
1582 \copperdefstring{brazilian}{monthname9} {Setembro}
1583 \copperdefstring{brazilian}{monthname10}{Outubro}
1584 \copperdefstring{brazilian}{monthname11}{Novembro}
1585 \copperdefstring{brazilian}{monthname12}{Dezembro}
1586 \copperdefstring{english}{monthname1} {January}
1587 \copperdefstring{english}{monthname2} {February}
1588 \copperdefstring{english}{monthname3} {March}
1589 \copperdefstring{english}{monthname4} {April}
1590 \copperdefstring{english}{monthname5} {May}
1591 \copperdefstring{english}{monthname6} {June}
1592 \copperdefstring{english}{monthname7} {July}
1593 \copperdefstring{english}{monthname8} {August}
1594 \copperdefstring{english}{monthname9} {September}
1595 \copperdefstring{english}{monthname10}{October}
1596 \copperdefstring{english}{monthname11}{November}
1597 \copperdefstring{english}{monthname12}{December}
1598 % algorithm2e package option for this language (empty -> english default).
1599 \copperdefstring{brazilian}{algorithm2eopt}{portuguese}

```

```

1600 \copperdefstring{english} {algorithm2eopt}{}
1601 % Babel package name for the foreignabstract \begin{otherlanguage}{...}.
1602 \copperdefstring{brazilian}{babelname}{brazilian}
1603 \copperdefstring{english} {babelname}{english}
1604 % Auto-load language packs for any non-built-in language listed in babelopts.
1605 % Built-ins (brazilian, english) ship their string tables inside coppe.cls
1606 % (defined just above) and need no extra file; others (spanish/french/italian
1607 % /...) ship as coppe-lang-<name>.def alongside the class. The dispatcher
1608 % must already be defined when these run, so the loader sits AFTER the tables.
1609 \@for\coppe@@lang:=\coppe@babelopts\do{%
1610   \edef\coppe@@cur{\coppe@@lang}%
1611   \ifx\coppe@@cur\coppe@@brazilianstr\else
1612   \ifx\coppe@@cur\coppe@@englishstr\else
1613     \InputIfFileExists{coppe-lang-\coppe@@cur.def}
1614     {\ClassInfo{coppe}{Loaded language pack coppe-lang-\coppe@@cur.def}}
1615     {\ClassError{coppe}
1616       {Missing language pack 'coppe-lang-\coppe@@cur.def'.}
1617       {Install it next to coppe.cls, or pick a built-in language.}}}%
1618 \fi\fi}
1619 % Quotation marks: csquotes gives the language-aware \enquote; \citacao is its
1620 % Portuguese alias for short, inline quotations (proposta P3).
1621 \RequirePackage{csquotes}
1622 \newcommand{\citacao}[1]{\enquote{#1}}
1623 % --- Code listings & algorithms (TODO3 "tratar listagens"): the listings
1624 % package gives the "Programa" listing (caption on top, ABNT R46) with
1625 % \listofprogramas; algorithm2e gives algorithms. Names are babel-aware and
1626 % the language styles below are based on the user's listagens.tex. -----
1627 \RequirePackage{xcolor}
1628 \RequirePackage{listings}
1629 \RequirePackage{fancyvrb}
1630 \DeclareFixedFont{\ttb}{T1}{ttx}{bx}{n}{10}
1631 \DeclareFixedFont{\ttm}{T1}{ttx}{m}{n}{10}
1632 \lstset{captionpos=t,extendedchars=true,basicstyle=\ttfamily,inputencoding=utf8,%
1633   literate=%
1634   {\~}{\~{a}}1 {\~}{\~{a}}1 {\~}{\~{a}}1 {\~}{\~{a}}1 {\~}{\~{e}}1%
1635   {\~}{\~{e}}1 {\~}{\~{e}}1 {\~}{\~{i}}1 {\~}{\~{i}}1 {\~}{\~{o}}1%
1636   {\~}{\~{o}}1 {\~}{\~{o}}1 {\~}{\~{u}}1 {\~}{\~{u}}1 {\~}{\~{c}}1%
1637   {\~}{\~{c}}1 {\~}{\~{A}}1 {\~}{\~{A}}1 {\~}{\~{A}}1 {\~}{\~{E}}1%
1638   {\~}{\~{E}}1 {\~}{\~{I}}1 {\~}{\~{O}}1 {\~}{\~{O}}1 {\~}{\~{O}}1%
1639   {\~}{\~{U}}1%
1640 }
1641 % Babel-aware listing names, set after listings' own babel setup.
1642 \AtBeginDocument{%
1643   \renewcommand\lstlistingname{\coppestring{program}}%
1644   \renewcommand\lstlistlistingname{\coppestring{listprogram}}%
1645 % \listofprogramas in the coppe list style (heading + ToC + running head).
1646 \newcommand\listofprogramas{%
1647   \chapter*{\lstlistlistingname}%
1648   \coppe@listtoc{\lstlistlistingname}%
1649   \mkboth{\MakeUppercase\lstlistlistingname}{\MakeUppercase\lstlistlistingname}%
1650   \@starttoc{lol}}
1651 % Per-language styles + environments (python, java, xml, html, prolog).
1652 % The red hook that marks a wrapped line is built ONCE, into a box, and the
1653 % styles below insert the box. Writing the colour inline -- postbreak=\mbox{%

```

```

1654 % \textcolor{red}{\hookrightarrow}\space} -- worked for years and broke the
1655 % moment the pdfa option was switched on: pdfx puts xcolor into a converting
1656 % mode (\selectcolormodel plus \convertcolorsU/D), and a colour command run
1657 % from inside listings' line-breaking machinery then fails with "Missing
1658 % number, treated as zero" at the first wrapped line. Every listing in the
1659 % document dies, so the class's own example could not be compiled as PDF/A at
1660 % all. A box is immune: it is built here, in ordinary horizontal mode where
1661 % xcolor works normally, and listings only ever copies it.
1662 %
1663 % The \sbox waits for \begin{document} because xcolor's colours are not
1664 % defined while the class is still loading.
1665 \newsavebox\coppe@lstarrow
1666 \AtBeginDocument{\sbox\coppe@lstarrow{\textcolor{red}{\hookrightarrow}\space}}
1667 \newcommand\pythonstyle{\lstset{language=Python,basicstyle=\ttm,numbers=left,%
1668   numberstyle=\small\ttfamily,numbersep=0.5em,commentstyle=\color{purple},%
1669   morekeywords={self,as},keywordstyle=\ttb\color{rgb}{0,0,0.5},%
1670   emph={MyClass,__init__},emphstyle=\ttb\color{rgb}{0.6,0,0},%
1671   stringstyle=\color{rgb}{0,0.5,0},frame=tb,showstringspaces=false,%
1672   extendedchars=true,firstnumber=auto,inputencoding=utf8,breaklines=true,%
1673   escapechar=',postbreak=\usebox\coppe@lstarrow}}
1674 \lstnewenvironment{python}[1][\pythonstyle\lstset{#1}%
1675   \csname\@lst @SetFirstNumber\endcsname}{\csname\@lst @SaveFirstNumber\endcsname}
1676 \newcommand\pythoninline[1][\pythonstyle\lstinline!#1!}]
1677 \newcommand\pythonexternal[2][\pythonstyle\lstinputlisting{#1}{#2}]}
1678 \newcommand\xmlstyle{\lstset{language=XML,basicstyle=\ttm,numbers=left,%
1679   numberstyle=\small\ttfamily,numbersep=0.5em,%
1680   morekeywords={xs:schema,xs:element,xs:sequence,xs:complexType},%
1681   keywordstyle=\ttb\color{rgb}{0,0,0.5},%
1682   emph={name,type,attributeFormDefault,elementFormDefault,xmlns:xs},%
1683   emphstyle=\ttb\color{rgb}{0.6,0,0},stringstyle=\color{rgb}{0,0.5,0},frame=tb,%
1684   showstringspaces=false,extendedchars=true,inputencoding=utf8,breaklines=true,%
1685   postbreak=\usebox\coppe@lstarrow}}
1686 \lstnewenvironment{xml}[1][\xmlstyle\lstset{#1}]{%
1687 \newcommand\htmlstyle{\lstset{language=HTML,basicstyle=\ttm,numbers=left,%
1688   numberstyle=\small\ttfamily,numbersep=0.5em,%
1689   keywordstyle=\ttb\color{rgb}{0,0,0.5},emphstyle=\ttb\color{rgb}{0.6,0,0},%
1690   stringstyle=\color{rgb}{0,0.5,0},frame=tb,showstringspaces=false,%
1691   extendedchars=true,inputencoding=utf8,breaklines=true,%
1692   postbreak=\usebox\coppe@lstarrow}}
1693 \lstnewenvironment{html}[1][\htmlstyle\lstset{#1}]{%
1694 \newcommand\prologstyle{\lstset{language=Prolog,basicstyle=\ttm,numbers=left,%
1695   numberstyle=\small\ttfamily,numbersep=0.5em,commentstyle=\color{purple},%
1696   keywordstyle=\ttb\color{rgb}{0,0,0.5},emph={MyClass,__init__},%
1697   emphstyle=\ttb\color{rgb}{0.6,0,0},stringstyle=\color{rgb}{0,0.5,0},frame=tb,%
1698   showstringspaces=false,extendedchars=true,firstnumber=auto,inputencoding=utf8,%
1699   breaklines=true,escapechar=',%
1700   postbreak=\usebox\coppe@lstarrow}}
1701 \lstnewenvironment{prolog}[1][\prologstyle\lstset{#1}%
1702   \csname\@lst @SetFirstNumber\endcsname}{\csname\@lst @SaveFirstNumber\endcsname}
1703 \newcommand\javastyle{\lstset{language=Java,basicstyle=\ttm,numbers=left,%
1704   numberstyle=\small\ttfamily,numbersep=0.5em,commentstyle=\color{purple},%
1705   morekeywords={self,as},keywordstyle=\ttb\color{rgb}{0,0,0.5},%
1706   emphstyle=\ttb\color{rgb}{0.6,0,0},stringstyle=\color{rgb}{0,0.5,0},frame=tb,%
1707   showstringspaces=false,extendedchars=true,firstnumber=auto,inputencoding=utf8,%

```

```

1708 breaklines=true,escapechar=',%
1709 postbreak=\usebox\coppe@lstarrow}}
1710 \lstnewenvironment{java}[1] [] {\javastyle\lstset{#1}%
1711 \csname\@lst @SetFirstNumber\endcsname}{\csname\@lst @SaveFirstNumber\endcsname}
1712 \newcommand\javainline[1]{\{\javastyle\lstinline!#1!\}}
1713 \newcommand\javaexternal[2] [] {\{\javastyle\lstinputlisting[#1]{#2}\}}
1714 % fancyvrb's Verbatim environment is available for program output; the
1715 % \DefineVerbatimEnvironment{gxoutput}/{gxoutputs} aliases that used to live
1716 % here were author-personal (gx prefix) and were removed in v4.0. Use, for
1717 % example,
1718 % \begin{Verbatim}[frame=lines,numbers=left,numbersep=0.5em]
1719 % ...
1720 % \end{Verbatim}
1721 % (or \DefineVerbatimEnvironment{...}{Verbatim}{...} in the manuscript preamble).
1722 \renewcommand{\theFancyVerbLine}{\small\ttfamily\arabic{FancyVerbLine}}
1723 % Algorithms: algorithm2e, language driven by the main-slot \coppemainstring
1724 % lookup (key "algorithm2eopt"). Empty means use algorithm2e's default
1725 % language (English) -- needed because algorithm2e errors on an empty option.
1726 \edef\coppe@@algotlang{\coppemainstring{algorithm2eopt}}
1727 \ifx\coppe@@algotlang\@empty
1728 \RequirePackage[linesnumbered,lined,algochapter,ruled]{algorithm2e}
1729 \else
1730 \expandafter\RequirePackage\expandafter[\coppe@@algotlang,linesnumbered,lined,algochapter,r
1731 \fi
1732 % algorithm2e composes its own caption and therefore ignores the class-wide
1733 % \captionsetup{font=footnotesize,labelsep=endash}. Left alone it printed
1734 % "Algoritmo 6.1:Titulo" in bold at body size, while every other illustration
1735 % printed "Figura 6.1 -- Titulo" in footnotesize -- 2.10 asks for the
1736 % designating word, the number, a TRAVESSAO and the title, uniformly.
1737 %
1738 % The separator and the caption fonts are therefore set to match. The 'ruled'
1739 % layout (rules above and below) is kept: 2.10 prescribes where the caption and
1740 % the source go, not whether the float is delimited, and the rules read here
1741 % like the top-and-bottom delimitation a table carries.
1742 \SetAlgoCaptionSeparator{\textendash}
1743 \SetAlCapFnt{\footnotesize}
1744 \SetAlCapNameFnt{\footnotesize}
1745 \SetAlCapSty{textnormal}
1746 \SetAlCapNameSty{textnormal}
1747 \SetKwFor{Para}{para}{faça}{fim do para}
1748 \SetKwFor{Enquanto}{enquanto}{faça}{fim do enquanto}
1749 % setspace + hyperref + biblatex are loaded at the very end of the class
1750 % (see the block just before %</class>), so their hooks see every macro the
1751 % class defines, and the load order setspace -> hyperref -> biblatex is
1752 % respected.

```

`\department` This macro is used to set the author's affiliation. There are twelve options which correspond to all academic units at COPPE/UFRJ. It defines the current and the foreign names of these units.

```

1753 \newcommand\coppe@deptcode{}
1754 \newcommand\meta@deptname{}
1755 \newcommand\department[1]{%
1756 % The programme CODE is kept alongside the resolved name, and each name is
1757 % declared TWICE. \local@deptname keeps the TeX accent escapes with braces

```



```

1758 % (Computa{\c c}{\~ a}o) that the cover and the folha de rosto typeset;
1759 \meta@deptname is the same name in literal UTF-8, and is the one that goes
1760 into the XMP. pdfx copies whatever it is given straight through, so feeding it
1761 % the escaped form produced "Computa{c}o" in dc:description -- which is why the
1762 % subject line used to fall back to the bare programme code. Both forms are
1763 % written out here rather than derived from one another because there is no
1764 % reliable way to strip TeX accent syntax at the point where the .xmpdata is
1765 % written, and a table of thirteen entries is cheaper than that machinery.
1766 \gdef\coppe@deptcode{#1}%
1767 \ifthenelse{\equal{#1}{PEB}}{
1768   {\global\def\local@deptname{Engenharia Biom{\` e}dica}
1769   \global\def\meta@deptname{Engenharia Biomédica}
1770   \global\def\foreign@deptname{Biomedical Engineering}}{}
1771 \ifthenelse{\equal{#1}{PEC}}{
1772   {\global\def\local@deptname{Engenharia Civil}
1773   \global\def\meta@deptname{Engenharia Civil}
1774   \global\def\foreign@deptname{Civil Engineering}}{}
1775 \ifthenelse{\equal{#1}{PEE}}{
1776   {\global\def\local@deptname{Engenharia El{\` e}trica}
1777   \global\def\meta@deptname{Engenharia Elétrica}
1778   \global\def\foreign@deptname{Electrical Engineering}}{}
1779 \ifthenelse{\equal{#1}{PEM}}{
1780   {\global\def\local@deptname{Engenharia Mec{\` a}nica}
1781   \global\def\meta@deptname{Engenharia Mecânica}
1782   \global\def\foreign@deptname{Mechanical Engineering}}{}
1783 \ifthenelse{\equal{#1}{PEMM}}{
1784   {\global\def\local@deptname{Engenharia Metal{\` u}rgica e de Materiais}
1785   \global\def\meta@deptname{Engenharia Metalúrgica e de Materiais}
1786 \global\def\foreign@deptname{Metallurgical and Materials Engineering}}{}
1787 \ifthenelse{\equal{#1}{PEN}}{
1788   {\global\def\local@deptname{Engenharia Nuclear}
1789   \global\def\meta@deptname{Engenharia Nuclear}
1790   \global\def\foreign@deptname{Nuclear Engineering}}{}
1791 \ifthenelse{\equal{#1}{PEN0}}{
1792   {\global\def\local@deptname{Engenharia Oce{\` a}nica}
1793   \global\def\meta@deptname{Engenharia Oceânica}
1794   \global\def\foreign@deptname{Ocean Engineering}}{}
1795 \ifthenelse{\equal{#1}{PPE}}{
1796   {\global\def\local@deptname{Planejamento Energ{\` e}tico}
1797   \global\def\meta@deptname{Planejamento Energético}
1798   \global\def\foreign@deptname{Energy Planning}}{}
1799 \ifthenelse{\equal{#1}{PEP}}{
1800   {\global\def\local@deptname{Engenharia de Produ{\c c}{\~ a}o}
1801   \global\def\meta@deptname{Engenharia de Produção}
1802   \global\def\foreign@deptname{Production Engineering}}{}
1803 \ifthenelse{\equal{#1}{PEQ}}{
1804   {\global\def\local@deptname{Engenharia Qu{\` i}mica}
1805   \global\def\meta@deptname{Engenharia Química}
1806   \global\def\foreign@deptname{Chemical Engineering}}{}
1807 \ifthenelse{\equal{#1}{PESC}}{
1808   {\global\def\local@deptname{Engenharia de Sistemas e Computa{\c c}{\~ a}o}
1809   \global\def\meta@deptname{Engenharia de Sistemas e Computação}
1810   \global\def\foreign@deptname{Systems Engineering and Computer Science}}{}
1811 \ifthenelse{\equal{#1}{PET}}{

```

```

1812     {\global\def\local@deptname{Engenharia de Transportes}
1813      \global\def\meta@deptname{Engenharia de Transportes}
1814      \global\def\foreign@deptname{Transportation Engineering}}{}
1815 \ifthenelse{\equal{#1}{PENT}}
1816     {\global\def\local@deptname{Engenharia de Nanotecnologia}
1817      \global\def\meta@deptname{Engenharia de Nanotecnologia}
1818      \global\def\foreign@deptname{Nanotechnology Engineering}}{}
1819 }

```

`\title` Used to enter the title in Brazilian Portuguese. Mirrors into `\coppe@title@brazilian` so that the per-language title slot used by the cover/folha renderers (`\coppe@selecttitle`) picks the right text for a pt-main thesis.

```

1820 \renewcommand\title[1]{%
1821   \global\def\local@title{#1}%
1822   \expandafter\gdef\csname coppe@title@brazilian\endcsname{#1}%
1823 }

```

`\foreigntitle` Used to enter the foreign title. Also mirrors into `\coppe@title@english` so that the cover renders the right text for an english-main thesis.

```

1824 \newcommand\foreigntitle[1]{%
1825   \global\def\foreign@title{#1}%
1826   \expandafter\gdef\csname coppe@title@english\endcsname{#1}%
1827 }

```

`\coppe@selecttitle` Expands to the title stored for the current main language. For pt-main this is `\local@title`; for en-main `\foreign@title`; for any other main language the user must call `\titlein{<lang>}{...}` to populate it.

```

1828 \newcommand\coppe@selecttitle{\@nameuse{coppe@title@\coppe@mainlang}}
1829 % Subtitle, mirroring the title machinery slot for slot: \subtitle feeds the
1830 % brazilian slot, \foreignsubtitle the english one, \subtitlein any other.
1831 % 3.1.1 and 3.1.2.1.1(c) require the subtitle on the capa and on the folha de
1832 % rosto, subordinated to the title by a colon.
1833 \newcommand\subtitle[1]{%
1834   \expandafter\gdef\csname coppe@subtitle@brazilian\endcsname{#1}}
1835 \newcommand\foreignsubtitle[1]{%
1836   \expandafter\gdef\csname coppe@subtitle@english\endcsname{#1}}
1837 \newcommand\subtitlein[2]{%
1838   \expandafter\gdef\csname coppe@subtitle@#1\endcsname{#2}}
1839 \newcommand\coppe@selectsubtitle{\@nameuse{coppe@subtitle@\coppe@mainlang}}
1840 % Title as the capa and the folha de rosto print it. Annexes A and B show the
1841 % title in caps and the subtitle in ordinary case after a colon, so the case
1842 % change is applied to the title alone.
1843 \newcommand\coppe@titleblock{%
1844   \MakeUppercase{\coppe@selecttitle}%
1845   \@ifundefined{coppe@subtitle@\coppe@mainlang}{: \coppe@selectsubtitle}}
1846 % Number of volumes (3.1.1 and 3.1.2.1.1(d)): printed only when there is more
1847 % than one, since a single-volume work says nothing about volumes.
1848 % Coorientador (3.1.2.1.1(f)). Same shape as \advisor, including the optional
1849 % institution; the label switches to the plural on the second call, as the
1850 % advisor label already did.
1851 \newcount\@coadvisor\@coadvisor0
1852 \newcommand\coadvisor[5][1]{%
1853   \global\@namedef{CoppeCoadvisorInst:\expandafter\the\@coadvisor}{#1}

```

```

1854 \global\@namedef{CoppeCoadvisorTitle:\expandafter\the\@coadvisor}{#2}
1855 \global\@namedef{CoppeCoadvisorName:\expandafter\the\@coadvisor}{#3}
1856 \global\@namedef{CoppeCoadvisorSurname:\expandafter\the\@coadvisor}{#4}
1857 \global\@namedef{CoppeCoadvisorDegree:\expandafter\the\@coadvisor}{#5}
1858 \global\advance\@coadvisor by 1
1859 }
1860 \newcommand\local@coadvisorstring{%
1861 \ifnum\@coadvisor>1
1862 \coppemainstring{coadvisors}%
1863 \else
1864 \coppemainstring{coadvisor}%
1865 \fi}
1866 % Mesma coisa no idioma estrangeiro, para a pagina de abstract.
1867 \newcommand\foreign@coadvisorstring{%
1868 \ifnum\@coadvisor>1
1869 \coppeforeignstring{coadvisors}%
1870 \else
1871 \coppeforeignstring{coadvisor}%
1872 \fi}
1873 \newcommand\coppe@nvolumes{1}
1874 \newcommand\coppe@thisvolume{1}
1875 \newcommand\volumes[1]{\gdef\coppe@nvolumes{#1}}
1876 \newcommand\volume[1]{\gdef\coppe@thisvolume{#1}}
1877 \newcommand\coppe@volumeline{%
1878 \ifnum\coppe@nvolumes>1
1879 \par\vspace*{4mm}%
1880 \coppemainstring{volumename}\space\coppe@thisvolume\space
1881 \coppemainstring{ofname}\space\coppe@nvolumes
1882 % The closing \par matters: on the capa the next thing is \vspace{\stretch{3}},
1883 % which does not end a paragraph, so without it the city ran on into the volume
1884 % line ("Volume 2 de 3 RIO DE JANEIRO").
1885 \par
1886 \fi}

```

\advisor Defines globally the title, name and academic degree of the advisors.

```

1887 \newcount\@advisor\@advisor0
1888 % The optional first argument is the member's institution, required by
1889 % 3.1.2.1.3(e). Omitting it keeps every pre-4.1 call working unchanged.
1890 \newcommand\advisor[5][\%
1891 \global\@namedef{CoppeAdvisorInst:\expandafter\the\@advisor}{#1}
1892 \global\@namedef{CoppeAdvisorTitle:\expandafter\the\@advisor}{#2}
1893 \global\@namedef{CoppeAdvisorName:\expandafter\the\@advisor}{#3}
1894 \global\@namedef{CoppeAdvisorSurname:\expandafter\the\@advisor}{#4}
1895 \global\@namedef{CoppeAdvisorDegree:\expandafter\the\@advisor}{#5}
1896 \global\advance\@advisor by 1
1897 \ifnum\@advisor>1
1898 \renewcommand\local@advisorstring{Orientadores}
1899 \renewcommand\foreign@advisorstring{Advisors}
1900 \fi
1901 }

```

\examiner

```

1902 \newcount\@examiner\@examiner0
1903 % Until v4.1 this stored only "#1 #2" and DISCARDED the third argument: the

```

```

1904 % titulacao every document passed was silently thrown away and never printed,
1905 % though 3.1.2.1.3(e) requires it. Each part is now kept separately, and the
1906 % optional first argument carries the institution.
1907 % CoppeExaminer: is retained with its old meaning for anything reading it.
1908 \newcommand\examiner[4][]{%
1909   \global\@namedef{CoppeExaminerInst:\expandafter\the\@examiner}{#1}
1910   \global\@namedef{CoppeExaminerTitle:\expandafter\the\@examiner}{#2}
1911   \global\@namedef{CoppeExaminerName:\expandafter\the\@examiner}{#3}
1912   \global\@namedef{CoppeExaminerDegree:\expandafter\the\@examiner}{#4}
1913   \global\@namedef{CoppeExaminer:\expandafter\the\@examiner}{#2\ #3}
1914   \global\advance\@examiner by 1
1915 }

```

`\author` It was redefined to allow the identification of the author's first names and surname.

```

1916 \renewcommand\author[2]{%
1917   \global\def\@authname{#1}
1918   \global\def\@authsurn{#2}
1919 }

```

`\date` Sets the *document's* nominal month and year (used by the cover, the folha de rosto, and the catalog). Stores them in private slots (`\coppe@docmonth` / `\coppe@docyear`) rather than mutating TeX's `\month` / `\year` primitives, which would clobber `\today`.

```

1920 \renewcommand\date[2]{%
1921   \def\coppe@docmonth{#1}%
1922   \def\coppe@docyear{#2}}

```

`\local@monthname`

```

1923 \newcommand\local@monthname{\ifcase\coppe@docmonth\or
1924   Janeiro\or Fevereiro\or Mar{\c c}o\or Abril\or Maio\or Junho\or
1925   Julho\or Agosto\or Setembro\or Outubro\or Novembro\or Dezembro\fi}

```

`\foreign@monthname`

```

1926 \newcommand\foreign@monthname{\ifcase\coppe@docmonth\or
1927   January\or February\or March\or April\or May\or June\or
1928   July\or August\or September\or October\or November\or December\fi}

```

`\keyword`

```

1929 \newcounter{keywords}
1930 \newcommand\keyword[1]{%
1931   \global\@namedef{CoppeKeyword:\expandafter\the\c@keywords}{#1}
1932   \global\addtocounter{keywords}{1}
1933 }
1934 % Keywords for the foreign abstract. Annex F of the manual shows the English
1935 % abstract closing with English keywords, and \keyword alone cannot serve both
1936 % pages. When no \foreignkeyword is given the foreign abstract falls back to
1937 % the \keyword list, which is what a document written before v4.1 expects.
1938 \newcounter{foreignkeywords}
1939 \newcommand\foreignkeyword[1]{%
1940   \global\@namedef{CoppeForeignKeyword:\expandafter\the\c@foreignkeywords}{#1}
1941   \global\addtocounter{foreignkeywords}{1}
1942 }
1943 % Keywords for the optional third abstract (Portuguese, for the banca). Same

```

```

1944 % shape and same fallback as \foreignkeyword: without it the Portuguese page
1945 % would close with the main-language keywords under a Portuguese label, which
1946 % is what a Spanish-, French- or Italian-main work would produce.
1947 \newcounter{braziliankeywords}
1948 \newcommand\braziliankeyword[1]{%
1949   \global\@namedef{CoppeBrazilianKeyword:\expandafter\the\c@braziliankeywords}{#1}
1950   \global\addtocounter{braziliankeywords}{1}
1951 }
1952 % 3.1.2.1.4: the keywords close the resumo, preceded by the term, a colon, and
1953 % separated from one another by semicolons.
1954 % #1 label, #2 csname prefix, #3 how many.
1955 \newcommand\coppe@printkeywords[3]{%
1956   \ifnum#3>0
1957     \par\vspace*{7mm}%
1958     \noindent{\singlespacing\textbf{#1:}\enspace
1959       \count1=0
1960       \@whilenum\count1<#3\do{%
1961         \ifnum\count1>0 ;\space\fi
1962         \csname #2\the\count1\endcsname
1963         \advance\count1 by 1}%
1964       .\par}%
1965   \fi}

\areaconcentracao Data for the folha adicional (Coleta CAPES) and for the natureza block of the
\linhapesquisa folha de rosto and folha de aprovacao. Every one of them defaults to empty;
\tipoproducao \makefolhaadicional prints an empty rule where a value is missing, so the pro-
\projetovinculado gramme can complete the sheet by hand if need be.
\nomeprojeto 1966 \newcommand\coppe@areaconc{}
\agenciafomento 1967 \newcommand\areaconcentracao[1]{\gdef\coppe@areaconc{#1}}
1968 \newcommand\coppe@dataaprov{}
1969 \newcommand\dataaprovacao[1]{\gdef\coppe@dataaprov{#1}}
1970 \newcommand\coppe@linhapesq{}
1971 \newcommand\linhapesquisa[1]{\gdef\coppe@linhapesq{#1}}
1972 \newcommand\coppe@tipoproducao{}
1973 \newcommand\tipoproducao[1]{\gdef\coppe@tipoproducao{#1}}
1974 \newcommand\coppe@projetovinc{}
1975 \newcommand\projetovinculado[1]{\gdef\coppe@projetovinc{#1}}
1976 \newcommand\coppe@nomeprojeto{}
1977 \newcommand\nomeprojeto[1]{\gdef\coppe@nomeprojeto{#1}}
1978 \newcounter{coppeagencias}
1979 \newcommand\agenciafomento[2]{%
1980   \global\@namedef{CoppeAgencia:\expandafter\the\c@coppeagencias}{#1}%
1981   \global\@namedef{CoppeAgenciaSigla:\expandafter\the\c@coppeagencias}{#2}%
1982   \global\addtocounter{coppeagencias}{1}}

\freeconfig This command allows easy changing of core class parameters.
1983 \newcommand\freeconfig[6]{%
1984   \providecommand\@degree{}
1985   \renewcommand\@degree{#1}
1986   \providecommand\@degreename{}
1987   \renewcommand\@degreename{#2}
1988   \providecommand\local@degname{}
1989   \renewcommand\local@degname{#3}
1990   \providecommand\foreign@degname{}

```

```

1991 \renewcommand\foreign@degname{#4}
1992 \providecommand\local@doctype{}
1993 \renewcommand\local@doctype{#5}
1994 \providecommand\foreign@doctype{}
1995 \renewcommand\foreign@doctype{#6}%
1996 }
1997 % \end{macrocode}
1998 % \end{macro}
1999 %
2000 % \begin{macro}{\frontmatter}
2001 % The number of pages for both frontmatter and mainmatter printed
2002 % in the cataloging details page is computed by means of simple
2003 % \LaTeX\ labels.
2004 % \begin{macrocode}
2005 \renewcommand\frontmatter{%
2006 \cleardoublepage
2007 \@mainmatterfalse
2008 % The count already started at the folha de rosto (see \maketitle) and simply
2009 % continues here: 2.7 requires every folha from the folha de rosto onwards to
2010 % be counted sequentially, printing arabic numerals only from the first textual
2011 % folha. Resetting the counter here -- as this class did until v4.1 -- made the
2012 % Introducao come out as page 1, which the norm does not allow.
2013 %
2014 % The numeraisromanos option keeps the legacy behaviour (lowercase roman in the
2015 % pre-textual part, arabic restarting at 1 in the textual part). That departs
2016 % from 2.7, which has the pre-textual folhas counted but NOT numbered, so the
2017 % option warns.
2018 \if@copperoman
2019 \ClassWarningNoLine{coppe}{Option ‘numeraisromanos’ prints roman numerals
2020 on the pre-textual folhas and restarts the arabic count at 1 in the
2021 textual part. Section 2.7 of the SiBI manual (9th rev. ed., 2026) has
2022 those folhas counted but not numbered, and the arabic numbering
2023 continuing the count. Drop the option to comply}%
2024 \pagenumbering{roman}%
2025 \fi
2026 \thispagestyle{empty}
2027 \makefrontpage
2028 \clearpage
2029 \if@copperoman
2030 \fancypagestyle{plain}{\fancyhf{}\fancyhead[R0,LE]{\coppe@folio}}%
2031 \renewcommand{\headrulewidth}{0pt}}%
2032 \pagestyle{coppe}%
2033 \else
2034 \fancypagestyle{plain}{\fancyhf{}\renewcommand{\headrulewidth}{0pt}}%
2035 \pagestyle{empty}%
2036 \fi
2037 }

\mainmatter
2038 \renewcommand\mainmatter{%
2039 \coppe@mainBegin
2040 \cleardoublepage
2041 \@mainmattertrue
2042 \fancypagestyle{plain}{\fancyhf{}\fancyhead[R0,LE]{\coppe@folio}}%

```

```

2043 \renewcommand{\headrulewidth}{0pt}}%
2044 \pagestyle{coppe}%
2045 % No \pagenumbering{arabic} in the default path: it would reset the counter to
2046 % 1 and make the Introducao folha 1, contrary to 2.7. The counter has been
2047 % arabic and running since the folha de rosto; all that changes here is that
2048 % the folio starts being PRINTED. Only the legacy numeraisromanos path, which
2049 % switched to roman in the front matter, has to switch back -- and there the
2050 % restart at 1 is intrinsic to that (non-conforming) scheme.
2051 \if@copperoman\pagenumbering{arabic}\fi

```

\backmatter

```

2052 \renewcommand\backmatter{%
2053 \if@openright
2054 \cleardoublepage
2055 \else
2056 \clearpage
2057 \fi}
2058 %

```

\coppetexfinalpage Optional colophon-style closing page intended to face the rear cover. The macro forces a verso (even) page so that, in a two-sided print, the reader sees the colophon on the left and the inside of the rear cover on the right; pushes the text to the bottom of the page with `\vfill`; and types it in Portuguese unconditionally (the colophon is institutional metadata, not body content, so it is rendered in pt-BR regardless of the thesis main language). Three slot macros are user-overridable: `\coppefinalengine` (engine name; auto-detected via `iftex`), `\coppefinalfont` (text body font; defaults to Computer Modern, the L^AT_EX default), and `\coppefinalmanual` (the reference manual title; defaults to the current COPPE/UFRJ norm). Override them before `\coppetexfinalpage` with `\renewcommand` if the document uses a non-default font or follows an earlier edition of the norm.

```

2059 \providecommand{\coppefinalengine}{%
2060 \ifPDFTeX\hologo{pdfLaTeX}\fi
2061 \ifLuaTeX\hologo{LuaLaTeX}\fi
2062 \ifXeTeX\hologo{XeLaTeX}\fi}
2063 \providecommand{\coppefinalfont}{Computer Modern (padr\~ao \LaTeX{})}
2064 \providecommand{\coppefinalmanual}{%
2065 \emph{Norma para a Elaborac\~ao Gr\~afica de Teses e Disserta\~oes
2066 da COPPE/UFRJ}}
2067 \newcommand{\coppetexfinalpage}{%
2068 \clearpage
2069 % Force the colophon onto a verso (even) page so it faces the rear cover.
2070 % If the current page is odd (recto), fill it with an empty verso first.
2071 \if@twoside
2072 \ifodd\c@page
2073 \hbox{}\thispagestyle{empty}\newpage
2074 \fi
2075 \fi
2076 \thispagestyle{empty}%
2077 \null\vfill
2078 \begin{otherlanguage}{brazilian}%
2079 \begin{center}
2080 {\footnotesize\setstretch{1.0}}%

```

```

2081 \textsc{Colof\~ao}\par
2082 \vspace{0.7em}
2083 \begin{minipage}{0.75\textwidth}\centering
2084 Este documento foi composto em \LaTeX{} com \coppefinalengine{}
2085 em \today, na fonte \coppefinalfont. A bibliografia foi processada
2086 por Bib\LaTeX{} e Biber. Foi utilizada a classe \CoppeTeX,
2087 vers\~ao\~\coppe@version, dispon\~\i vel no reposit\~\orio GitHub
2088 \mbox{\url{https://github.com/COPPE-UFRJ/CoppeTeX}}, em
2089 conformidade com a \coppefinalmanual.
2090 \end{minipage}\par}
2091 \end{center}
2092 \end{otherlanguage}%
2093 \vspace*{2cm}%
2094 \clearpage}

```

\makecover The ABNT cover (capa, NBR 14724) suggested in the manual's Anexo A. It is the institution block — the university, the full COPPE name, and the graduate program taken from `\department` — followed by the author, the title, and the city and year, all centred and uppercased. The optional UFRJ and COPPE logos head the page. The capa is not counted: it precedes the (roman) front matter and is produced before the `titlepage` (folha de rosto).

```

2095 \newcommand\makecover{%
2096 \begin{titlepage}%
2097 \centering
2098 \makebox[\linewidth]{%
2099 \includegraphics[height=2cm]{ufrj-logo}\hfill
2100 \includegraphics[height=2cm]{coppe-logo}}\par
2101 \vspace{1.5cm}
2102 {\singlespacing
2103 \MakeUppercase{\local@universityname}\par
2104 \nohyphens{\MakeUppercase{Instituto Alberto Luiz Coimbra de \local@deptname}
2105 P\~\o s-Gradua\~\c c}{\~\a o e Pesquisa de Engenharia}}\par
2106 \MakeUppercase{Programa de \local@deptname}\par}
2107 \vspace{\stretch{2}}
2108 \textbf{\MakeUppercase{\textbf{\@authname\ \@authsur}}}\par
2109 \vspace{\stretch{2}}
2110 \nohyphens{\coppe@titleblock}\par
2111 \coppe@volumeline
2112 \vspace{\stretch{3}}
2113 \MakeUppercase{\local@cityname}\par
2114 \coppe@docyear
2115 \end{titlepage}%
2116 }

```

\maketitle

```

2117 \renewcommand\maketitle{%
2118 % The capa is neither counted nor numbered (2.7), so it lives in its own
2119 % throwaway numbering scheme; \pagenumbering{arabic} right after it restarts
2120 % the counter, making the FOLHA DE ROSTO page 1 -- which is where 2.7 says the
2121 % count begins. Nothing is printed here: \frontmatter keeps \pagestyle{empty}
2122 % until the textual part starts.
2123 \pagenumbering{alph}
2124 \ltx@ifpackageloaded{hyperref}{\coppe@hypersetup}{}%

```



```

2125 \makecover
2126 \pagenumbering{arabic}
2127 \begin{titlepage}
2128 \begin{flushleft}
2129 \vspace*{1.5mm}
2130 \setlength\baselineskip{0pt}
2131 \setlength\parskip{1mm}
2132 \makebox[\linewidth]{%
2133   \includegraphics[height=2cm]{ufrj-logo}\hfill
2134   \includegraphics[height=2cm]{coppe-logo}}
2135 \end{flushleft}
2136 \vspace{1.05cm}
2137 \begin{center}
2138   \nohyphens{\@authname\ \@authsur}\par
2139   \vspace*{3cm}
2140   \nohyphens{\coppe@titleblock}\par
2141   \coppe@volumeline
2142 \end{center}
2143 \vspace*{2.1cm}
2144 \begin{flushright}
2145 % 2.4: on the folha de rosto the natureza/objetivo/instituicao block is set
2146 % SINGLE spaced, and aligned from the middle of the mancha grafica to the right
2147 % margin. 0.5\textwidth tracks the mancha, so the block follows automatically if
2148 % the margins ever change -- the old fixed 8.45cm did not.
2149 \begin{minipage}{0.5\textwidth}
2150 \singlespacing\frontcover@maintext
2151 \end{minipage}\par
2152 \end{flushright}
2153 % Only the natureza block is indented to the right (2.4). The linha de pesquisa
2154 % and the orientadores return to the LEFT MARGIN, as the CAPES model of the
2155 % folha de rosto and Annex B of the manual show. Until v4.1 they sat inside the
2156 % right-hand block, which is what a recent review flagged.
2157 \vspace*{12mm}
2158 {\singlespacing
2159 \ifthenelse{\equal{\coppe@linhapesq}{}}{\}{%
2160   \noindent Linha de pesquisa: \coppe@linhapesq\par
2161   \vspace*{5mm}}}%
2162 \nohyphens{%
2163 \noindent\begin{tabularx}{\textwidth}[b]{@{}l@{ }>{\raggedright\arraybackslash}X@{}}
2164   \local@advisorstring: &
2165   \count1=0
2166   \toks@={}
2167   \@whilenum \count1<\@advisor \do{%
2168     \ifcase\count1 % same as \ifnum0=\count1
2169       \toks@=\expandafter{\csname CoppeAdvisorName:\the\count1%
2170         \expandafter\endcsname\expandafter\space%
2171         \csname CoppeAdvisorSurname:\the\count1\endcsname\\}
2172     \else
2173       \toks@=\expandafter\expandafter\expandafter{%
2174         \expandafter\the\expandafter\toks@%
2175         \expandafter&\expandafter\space%
2176         \csname CoppeAdvisorName:\the\count1\expandafter\endcsname%
2177         \expandafter\space\csname CoppeAdvisorSurname:\the\count1\endcsname\\}
2178     }%

```

```

2179 \fi
2180 \advance\count1 by 1}
2181 \the\toks@
2182 \end{tabularx}}}%
2183 \ifnum\@coadvisor>0
2184 \par\vspace*{2mm}%
2185 \nohyphens{%
2186 \noindent\begin{tabularx}{\textwidth}[b]{\@{1}{ }>{\raggedright\arraybackslash}X@{}}
2187 \local@coadvisorstring: &
2188 \count1=0
2189 \toks@={ }
2190 \@whilenum \count1<\@coadvisor \do{%
2191 \ifcase\count1
2192 \toks@=\expandafter{\csname CoppeCoadvisorName:\the\count1%
2193 \expandafter\endcsname\expandafter\space%
2194 \csname CoppeCoadvisorSurname:\the\count1\endcsname\\}
2195 \else
2196 \toks@=\expandafter\expandafter\expandafter{%
2197 \expandafter\the\expandafter\toks@%
2198 \expandafter&\expandafter\space%
2199 \csname CoppeCoadvisorName:\the\count1\expandafter\endcsname%
2200 \expandafter\space\csname CoppeCoadvisorSurname:\the\count1\endcsname\\}
2201 }%
2202 \fi
2203 \advance\count1 by 1}
2204 \the\toks@
2205 \end{tabularx}}}%
2206 \fi
2207 \par}
2208 \vspace*{\fill}
2209 % 3.1.2.1.1(g): the folha de rosto carries the ANO DE DEPOSITO alone -- the year
2210 % the revised final version is handed to the Programa. The month was never asked
2211 % for here (the capa, Annex A, also shows only the year), and printing it
2212 % invited confusion with the defence date. \date still takes month and year:
2213 % the month is what the abstract pages and the colophon use.
2214 \begin{center}
2215 \local@cityname\par
2216 \coppe@docyear
2217 \end{center}
2218 \end{titlepage}
2219 % Ficha catalográfica on the verso of the folha de rosto (NBR 14724, R57):
2220 % the title page is a recto, so this lands on the immediately following verso,
2221 % ahead of the approval sheet produced by \frontmatter.
2222 % LaTeX's titlepage environment ends with \setcounter{page}{1} whenever the
2223 % document is one-sided (see \endtitlepage in classes.dtx). That silently ate
2224 % the folha de rosto from the count, putting the folha de aprovacao at 1 instead
2225 % of 2. The count is therefore restated here: the folha de rosto WAS folha 1, so
2226 % whatever follows it is folha 2.
2227 \setcounter{page}{2}%
2228 % The page carrying the ficha catalografica is explicitly excluded from the
2229 % count -- 2.7 of the 2026 manual ("a folha adicional que contem a ficha
2230 % catalografica nao pode ser contada ou numerada"), and the 2025 edition said
2231 % the same of the verso of the folha de rosto. Hence the counter is stepped
2232 % back after it.

```

```

2233 \ifthenelse{\boolean{maledoc}}{\%
2234 \makefolhaadicional\clearpage\addtocounter{page}{-1}}%
2235 \global\let\maketitle\relax%
2236 \global\let\and\relax}

2237 % 3.1.2.1.3 lists the elements of the folha de aprovacao in order: (a) the
2238 % AUTHOR, (b) the title in full, (c) the natureza/objetivo/instituicao/area de
2239 % concentracao, (d) the date of approval, (e) the examining board. Until v4.1
2240 % this page opened with the title in caps and the author under it, and set the
2241 % natureza as a wall of uppercase (\frontpage@maintext). Annex D shows neither.
2242 % One board member. \coppe@membroid is the identification line required by
2243 % 3.1.2.1.3(e) -- name, titulacao and institution -- with the last two omitted
2244 % when not given, so a document that never supplied them looks as it did before.
2245 \newcommand\coppe@membroid[4]{%
2246 #1\ #2%
2247 \ifthenelse{\equal{#3}{}}{\%{, #3}%
2248 \ifthenelse{\equal{#4}{}}{\%{, #4}%
2249 % The gap above each signature rule is \coppe@membrogap, set from the size of
2250 % the board just before the list is assembled (see \makefrontpage). A fixed 9mm
2251 % fitted five members and overflowed at seven -- the sheet silently continued
2252 % onto a second page, which also printed a folio in the pre-textual part, where
2253 % 2.7 forbids one. The shrink lives in the length itself -- \newlength makes a
2254 % skip register, so "9mm minus 2mm" survives \setlength and reaches \vspace as
2255 % real glue, and a title long enough to wrap an extra line is absorbed instead
2256 % of costing a page. It has to go through \vspace and not a bare \vskip: the
2257 % board list is emitted inside \nohyphens, i.e. in horizontal mode, where
2258 % \vskip is an error and TeX then typesets the "plus ... minus ..." after it as
2259 % ordinary text -- which is exactly what landed on the page the first time.
2260 \newlength\coppe@membrogap
2261 \setlength\coppe@membrogap{9mm minus 2mm}
2262 \newcount\coppe@bancasize
2263 \newcommand\coppe@membro[4]{%
2264 \if@coppeassin
2265 \vspace{\coppe@membrogap}%
2266 \begin{center}
2267 \rule{95mm}{0.4pt}\par
2268 \nopagebreak{\footnotesize\coppe@membroid{#1}{#2}{#3}{#4}}\par
2269 \end{center}%
2270 \else
2271 \noindent\hspace*{8mm}%
2272 \begin{minipage}{\dimexpr\textwidth-8mm\relax}
2273 \coppe@membroid{#1}{#2}{#3}{#4}
2274 \end{minipage}\par\vspace{2mm}%
2275 \fi}
2276 \newcommand\makefrontpage{%
2277 \begin{center}
2278 \nohyphens{\@authname\ \@authsur}\par
2279 \vspace*{7mm}
2280 \sloppy\nohyphens{\coppe@selecttitle
2281 \@ifundefined{coppe@subtitle@\coppe@mainlang}{%{:
2282 \coppe@selectsubtitle}}\par
2283 \end{center}\par
2284 \vspace*{8mm}
2285 % Same block, same indentation and same single spacing as the folha de rosto:

```

```

2286 % 2.4 names the two pages together.
2287 \begin{flushright}
2288 \begin{minipage}{0.5\textwidth}
2289 \singlespacing\frontcover@maintext
2290 \end{minipage}
2291 \end{flushright}
2292 \vspace*{12mm}
2293 % (d) date of approval. Annex D prints "Aprovada em:" followed by a blank to be
2294 % filled in; \dataaprovacao supplies it when the date is already known.
2295 \noindent\local@approvedonname:\enspace
2296 \ifthenelse{\equal{\coppe@dataaprov}{}}{\hrulefill}{\coppe@dataaprov}\par
2297 \vspace*{12mm}
2298 % (e) the examining board, with the ORIENTADOR FIRST, as president of the board
2299 % (3.1.2.1.3(e)). Until v4.1 the advisors were printed in a block of their own
2300 % above a second list headed "Aprovada por:", which split the board in two and
2301 % did not mark the advisor as its president. They are now one list.
2302 %
2303 % The list is assembled into \coppe@bancalist with \g@addto@macro instead of
2304 % being built inside a tabularx: each entry now carries four fields (treatment,
2305 % name, titulacao, institution) and the old \toks@/\expandafter accumulation
2306 % could not carry them without becoming unreadable. \protected@edef keeps
2307 % accents and other protected commands in the names intact.
2308 % Size of the board decides the spacing above. The thresholds are measured, not
2309 % guessed: with the 9mm gap a seven-member board pushed three members onto a
2310 % second page, 7mm brought six back onto one sheet and 5mm did the same for
2311 % seven and eight. Boards of five or fewer keep the original 9mm, so every
2312 % document written before this change renders exactly as it did.
2313 \coppe@bancasize=\@advisor
2314 \advance\coppe@bancasize by \@coadvisor
2315 \advance\coppe@bancasize by \@examiner
2316 \ifnum\coppe@bancasize>6
2317 \setlength\coppe@membrogap{5mm minus 2mm}%
2318 \else\ifnum\coppe@bancasize>5
2319 \setlength\coppe@membrogap{7mm minus 2mm}%
2320 \fi\fi
2321 \gdef\coppe@bancalist{}%
2322 \count1=0
2323 \@whilenum\count1<\@advisor\do{%
2324 \protected@edef\coppe@tmp{\noexpand\coppe@membro
2325 {\csname CoppeAdvisorTitle:\the\count1\endcsname}%
2326 {\csname CoppeAdvisorName:\the\count1\endcsname\space
2327 \csname CoppeAdvisorSurname:\the\count1\endcsname}%
2328 {\csname CoppeAdvisorDegree:\the\count1\endcsname}%
2329 {\csname CoppeAdvisorInst:\the\count1\endcsname}}%
2330 \expandafter\g@addto@macro\expandafter\coppe@bancalist
2331 \expandafter{\coppe@tmp}%
2332 \advance\count1 by 1}%
2333 % Coorientadores come right after the orientadores and before the rest of the
2334 % board, keeping the order 3.1.2.1.3(e) prescribes.
2335 \count1=0
2336 \@whilenum\count1<\@coadvisor\do{%
2337 \protected@edef\coppe@tmp{\noexpand\coppe@membro
2338 {\csname CoppeCoadvisorTitle:\the\count1\endcsname}%
2339 {\csname CoppeCoadvisorName:\the\count1\endcsname\space

```

```

2340     \csname CoppeCoadvisorSurname:\the\count1\endcsname}%
2341     {\csname CoppeCoadvisorDegree:\the\count1\endcsname}%
2342     {\csname CoppeCoadvisorInst:\the\count1\endcsname}}%
2343     \expandafter\g@addto@macro\expandafter\coppe@bancalist
2344     \expandafter{\coppe@tmp}%
2345     \advance\count1 by 1}%
2346 \count1=0
2347 \@whilenum\count1<\@examiner\do{%
2348     \protected@edef\coppe@tmp{\noexpand\coppe@membro
2349     {\csname CoppeExaminerTitle:\the\count1\endcsname}%
2350     {\csname CoppeExaminerName:\the\count1\endcsname}%
2351     {\csname CoppeExaminerDegree:\the\count1\endcsname}%
2352     {\csname CoppeExaminerInst:\the\count1\endcsname}}}%
2353     \expandafter\g@addto@macro\expandafter\coppe@bancalist
2354     \expandafter{\coppe@tmp}%
2355     \advance\count1 by 1}%
2356 \noindent\local@approvedname:\par
2357 \nohyphens{\singlespacing\coppe@bancalist}\par
2358 % No city/date block at the foot: 3.1.2.1.3 ends the folha de aprovacao at the
2359 % examining board, and Annex D shows nothing below the signature lines. The date
2360 % that belongs on this page is the date of APPROVAL, printed above.
2361 % \frontpage@bottomtext stays defined for anyone calling it directly.
2362 \vspace*{\fill}}

2363 \newcommand\coppe@hypersetup{%
2364 \begingroup
2365 % changes to \toks@ and \count@ are kept local;
2366 % it's not necessary for them, but it is usually the case
2367 % for \count1, because the first ten counters are written
2368 % to the DVI file, thus you got lucky because of PDF output
2369 \toks@={}% in this special case not necessary
2370 \count@=0 %
2371 \@whilenum\count@<\value{keywords}\do{%
2372     % * a keyword separator is not necessary,
2373     %     if there is just one keyword
2374     % * \csname CoppeKeyword:\the\count@\endcsname must be expanded
2375     %     at least once, to get rid of the loop depended \count@
2376     \ifcase\count@ % same as \ifnum0=\count@
2377         \toks@=\expandafter{\csname CoppeKeyword:\the\count@\endcsname}%
2378     \else
2379         \toks@=\expandafter\expandafter\expandafter{%
2380             \expandafter\the\expandafter\toks@
2381             \expandafter;\expandafter\space
2382             \csname CoppeKeyword:\the\count@\endcsname
2383         }%
2384     \fi
2385     \advance\count@ by 1 %
2386 }%
2387 \edef\x{\endgroup
2388     \noexpand\hypersetup{%
2389         pdfkeywords={\the\toks@}%
2390     }%
2391 }%
2392 \x

```

```

2393 \hypersetup{%
2394 pdfauthor={\@authname\ \@authsur},
2395 pdftitle={\local@title},
2396 pdfsubject={\local@doctype\ de \@degreename\ em \local@deptname\ da COPPE/UFRJ},
2397 pdfcreator={LaTeX with CoppeTeX toolkit},
2398 breaklinks={true},
2399 raiselinks={true},
2400 pageanchor={true},
2401 }}

```

`\makecatalog` When the document has illustrations, it is required to insert “: il.,” between the number of pages of the textual part and the page dimension. We have created a label to flag the existence of lists of figures. It is checked to be undefined using the plain `TeX` command `\isundefined` (cf. the `TeX` FAQ).

```

2402 \newcommand\makecatalog{%
2403   \vspace*{\fill}
2404   \begin{center}
2405     \setlength{\fboxsep}{5mm}
2406     \framebox[120mm][c]{\makebox[5mm][c]{}%
2407       \begin{minipage}[c]{105mm}
2408         \setlength{\parindent}{5mm}
2409         \noindent\sloppy\nohyphens\@authsur,
2410         \nohyphens\@authname\par
2411         \nohyphens{%
2412           \coppe@selecttitle/\@authname\ \@authsur. -- \local@cityname:
2413           UFRJ/COPPE, \coppe@docyear.}\par
2414         \pageref{front:pageno},
2415         \pageref{LastPage}
2416         p.\@ifundefined{r@cat:lofflag}{\pageref{cat:lofflag}}{$29,7$cm.}\par
2417         % There is an issue here. When the last entry must be split between lines,
2418         % the spacing between it and the next paragraph becomes smaller.
2419         % Should we manually introduce a fixed space? But how could we know that
2420         % a name was split? Is this happening yet?
2421         \nohyphens{%
2422           \begin{tabularx}{100mm}[b]{@{}l@{ }>{\raggedright\arraybackslash}X@{}}
2423             \local@advisorstring: &
2424             \count1=0
2425             \toks@={}
2426             \@whilenum \count1<\@advisor \do{%
2427               \ifcase\count1 % same as \ifnum0=\count1
2428                 \toks@=\expandafter{\csname CoppeAdvisorName:\the\count1%
2429                   \expandafter\endcsname\expandafter\space%
2430                   \csname CoppeAdvisorSurname:\the\count1\endcsname\\}
2431               \else
2432                 \toks@=\expandafter\expandafter\expandafter{%
2433                   \expandafter\the\expandafter\toks@
2434                   \expandafter&\expandafter\space
2435                   \csname CoppeAdvisorName:\the\count1\expandafter\endcsname%
2436                   \expandafter\space\csname CoppeAdvisorSurname:\the\count1\endcsname\\}
2437                 }%
2438               \fi
2439               \advance\count1 by 1}
2440             \the\toks@
2441           \end{tabularx}}\par

```

```

2442 \nohyphens{\local@doctype\ ({\MakeLowercase\@degreename}) --
2443 UFRJ/COPPE/Programa de \local@deptname, \coppe@docyear.}\par
2444 Refer{\^ e}ncias Bibliogr{\' a}ficas: p. \pageref{bib:begin} -- \pageref{bib:end}.\par
2445 \count1=0
2446 \count2=1
2447 \nohyphens{\@whilenum \count1<\value{keywords} \do {%
2448 \number\count2. \csname CoppeKeyword:\the\count1 \endcsname.
2449 \advance\count1 by 1
2450 \advance\count2 by 1}
2451 I. \csname CoppeAdvisorSurname:0\endcsname,%
2452 \ \csname CoppeAdvisorName:0\endcsname%
2453 \ifthenelse{\@advisor>1}{\ \emph{et~al.}}{}}.
2454 II. \local@universityname, COPPE, Programa de \local@deptname.
2455 III. T{\' i}tulo.}
2456 \end{minipage}}
2457 \end{center}
2458 \vspace*{\fill}}

```

`\fichacatalografica` The folha adicional required from August 2026 (SiBI manual, 9th rev. ed., `\makefolhaadicional` 3.1.2.1.2 and Annex H): the CAPES collection fields plus the ficha catalografica, sitting immediately after the folha de rosto and counted by nobody. `\fichacatalografica` takes the PDF produced by the SiBI generator at <http://fichacatalografica.sibi.ufrj.br/>, optionally with a page number.

The sheet is institutional, so it is always in Portuguese whatever the main language of the work – and deliberately without `\selectlanguage`, since the brazilian babel language need not be loaded in, say, a Spanish-main document.

```

2459 \newcommand\coppe@fichafila{}
2460 \newcommand\coppe@fichapage{1}
2461 \newcommand\fichacatalografica[2][1]{%
2462 \gdef\coppe@fichapage{#1}\gdef\coppe@fichafila{#2}}
2463 \newcommand\coppe@marcatipo[1]{%
2464 \ifthenelse{\equal{\coppe@tipoproducao}{#1}}{$\times$}{\phantom{$\times$}}}
2465 \newcommand\coppe@marcavinc[1]{%
2466 \ifthenelse{\equal{\coppe@projetovinc}{#1}}{$\times$}{\phantom{$\times$}}}
2467 \newcommand\coppe@campo[1]{%
2468 \ifthenelse{\equal{#1}}{\hrulefill}{#1\enspace\hrulefill}}
2469 \newcommand\coppe@checatipo{%
2470 \ifthenelse{\equal{\coppe@tipoproducao}}{}}{ }{%
2471 \ifthenelse{\equal{\coppe@tipoproducao}{bibliografica}}\OR
2472 \equal{\coppe@tipoproducao}{artistica}}\OR
2473 \equal{\coppe@tipoproducao}{tecnologica}}\OR
2474 \equal{\coppe@tipoproducao}{tecnica}}{ }{%
2475 \ClassWarning{coppe}{Unknown \string\tipoproducao{} value
2476 '\coppe@tipoproducao'. Annex H of the SiBI manual allows only
2477 bibliografica, artistica, tecnologica or tecnica; no box will be
2478 ticked}}}}
2479 \newcommand\makefolhaadicional{%
2480 \clearpage
2481 \thispagestyle{empty}%
2482 \coppe@checatipo
2483 {\singlespacing
2484 \begin{center}
2485 {\large\bfseries Informa\c c\~oes Coleta CAPES}

```

```

2486 \end{center}
2487 \vspace{7mm}
2488 \noindent Tipo de produ\c c\~ao intelectual:\par
2489 \vspace{1mm}
2490 \hspace*{10mm}(\coppe@marcatipo{bibliografica})\enspace Bibliogr\'afica\par
2491 \hspace*{10mm}(\coppe@marcatipo{artistica})\enspace Art\'istica\par
2492 \hspace*{10mm}(\coppe@marcatipo{tecnologica})\enspace Tecnol\'ogica\par
2493 \hspace*{10mm}(\coppe@marcatipo{tecnica})\enspace T\'ecnica\par
2494 \vspace{5mm}
2495 \noindent Projeto de pesquisa vinculado \'a produ\c c\~ao intelectual?\par
2496 \vspace{1mm}
2497 \hspace*{10mm}(\coppe@marcavinc{sim})\enspace Sim\par
2498 \hspace*{10mm}(\coppe@marcavinc{nao})\enspace N\~ao\par
2499 \vspace{5mm}
2500 \noindent Nome do projeto de pesquisa:\par
2501 \vspace{1mm}
2502 \noindent\coppe@campo{\coppe@nomeprojeto}\par
2503 \vspace{5mm}
2504 \noindent \'Area de concentra\c c\~ao da produ\c c\~ao intelectual:\par
2505 \vspace{1mm}
2506 \noindent\coppe@campo{\coppe@areaconc}\par
2507 \vspace{5mm}
2508 \noindent Ag\~encia(s) de fomento financiadora(s) da produ\c c\~ao
2509 intelectual:\par
2510 \vspace{1mm}
2511 \ifthenelse{\value{coppeagencias}=0}{%
2512 \noindent\hrulefill\par}{%
2513 \count1=0
2514 \@whilenum\count1<\value{coppeagencias}\do{%
2515 \noindent\hspace*{10mm}\csname CoppeAgencia:\the\count1\endcsname
2516 \ (\csname CoppeAgenciaSigla:\the\count1\endcsname)\par
2517 \advance\count1 by 1}}%
2518 \vspace{9mm}
2519 \begin{center}
2520 {\large\bfseries Ficha catalogr\'afica}\par
2521 \vspace{4mm}
2522 \ifthenelse{\equal{\coppe@fichafile}{}}{%
2523 \if@copperascunho
2524 \makecatalog
2525 \else
2526 \framebox[125mm][c]{\parbox[c][62mm][c]{115mm}{\centering\footnotesize
2527 Espa\c co reservado \'a ficha catalogr\'afica.\\[1ex]
2528 Gere-a em \texttt{fichacatalografica.sibi.ufrj.br} e inclua o
2529 arquivo com \texttt{\string\fichacatalografica\{arquivo.pdf\}},
2530 ou solicite-a \'a biblioteca do seu Programa.}}%
2531 \fi}{%
2532 \includegraphics[page=\coppe@fichapage,width=125mm]{\coppe@fichafile}}%
2533 \end{center}
2534 \par}}

```

\dedication

```

2535 \newcommand\dedication[1]{
2536 \gdef\@dedic{#1}
2537 \cleardoublepage

```



```

2538 \vspace*{\fill}
2539 \begin{flushright}
2540 \begin{minipage}{60mm}
2541 \raggedleft \it \normalsize \@dedic
2542 \end{minipage}
2543 \end{flushright}}

```

`\coppe@coadvisorrows` The coadvisor block of the abstract pages, in the shape the folha de rosto already uses. It prints nothing at all unless the `coorientador` class option is given AND the document actually declared a `\coadvisor`, so the default output is byte for byte what it was. The argument is the label macro, which differs per page: the main-language pages pass `\local@coadvisorstring` and the foreign one passes `\foreign@coadvisorstring`.

```

2544 \newcommand\coppe@coadvisorrows[1]{%
2545 \if@coppecoadvresumo
2546 \ifnum\@coadvisor>0
2547 \par\vspace*{2mm}%
2548 \noindent\begin{tabularx}{\textwidth}[b]{@{}l@{ }>{\raggedright\arraybackslash}X@{}}
2549 #1: &
2550 \count1=0
2551 \toks@={}
2552 \@whilenum \count1<\@coadvisor \do{%
2553 \ifcase\count1
2554 \toks@=\expandafter{\csname CoppeCoadvisorName:\the\count1%
2555 \expandafter\endcsname\expandafter\space%
2556 \csname CoppeCoadvisorSurname:\the\count1\endcsname\\}
2557 \else
2558 \toks@=\expandafter\expandafter\expandafter{%
2559 \expandafter\the\expandafter\toks@%
2560 \expandafter&\expandafter\space%
2561 \csname CoppeCoadvisorName:\the\count1\expandafter\endcsname%
2562 \expandafter\space\csname CoppeCoadvisorSurname:\the\count1\endcsname\\}
2563 }%
2564 \fi
2565 \advance\count1 by 1}
2566 \the\toks@
2567 \end{tabularx}%
2568 \fi
2569 \fi}

```

`abstract (env.)` This is a specialization of the abstract in the article standard class.

```

2570 \newenvironment{abstract}{%
2571 \cleardoublepage
2572 \thispagestyle{plain}
2573 \abstract@toptext\par
2574 \vspace*{8.6mm}
2575 \begin{center}
2576 \sloppy\nohyphens{\MakeUppercase\local@title}\par
2577 \vspace*{13.2mm}
2578 \@authname\ \@authsurn \par
2579 \vspace*{7mm}
2580 \local@monthname/\coppe@docyear
2581 \end{center}\par
2582 \vspace*{1.5cm}

```

```

2583 \noindent%
2584 \begin{tabularx}{\textwidth}[b]{@{}l@{ }>{\raggedright\arraybackslash}X@{}}
2585   \local@advisorstring: &
2586   \count1=0
2587   \toks@={}
2588   \@whilenum \count1<\@advisor \do{%
2589     \ifcase\count1 % same as \ifnum0=\count1
2590       \toks@=\expandafter{\csname CoppeAdvisorName:\the\count1%
2591         \expandafter\endcsname\expandafter\space%
2592         \csname CoppeAdvisorSurname:\the\count1\endcsname\\}
2593     \else
2594       \toks@=\expandafter\expandafter\expandafter{%
2595         \expandafter\the\expandafter\toks@
2596         \expandafter&\expandafter\space
2597         \csname CoppeAdvisorName:\the\count1\expandafter\endcsname%
2598         \expandafter\space\csname CoppeAdvisorSurname:\the\count1\endcsname\\
2599       }%
2600     \fi
2601     \advance\count1 by 1}
2602   \the\toks@
2603 \end{tabularx}%
2604 \coppe@coadvisorrows{\local@coadvisorstring}\par
2605 \vspace*{2mm}
2606 \noindent\local@deptstring: \local@deptname\par
2607 \vspace*{7mm}}}%
2608 \coppe@printkeywords{\coppemainstring{keywordsname}}%
2609   {\CoppeKeyword:}{\value{keywords}}%
2610 \vspace*{\fill}}

```

foreignabstract (*env.*)

```

2611 \newenvironment{foreignabstract}{%
2612   \cleardoublepage
2613   \thispagestyle{plain}
2614   \begin{otherlanguage}{\coppe@foreignlang}
2615   \foreignabstract@toptext\par
2616   \vspace*{8.6mm}
2617   \begin{center}
2618     \sloppy\nohyphens{\MakeUppercase\foreign@title}\par
2619     \vspace*{13.2mm}
2620     \@authname\ \@authsurn \par
2621     \vspace*{7mm}
2622     \foreign@monthname/\coppe@docyear
2623   \end{center}\par
2624   \vspace*{1.5cm}
2625   \noindent%
2626   \begin{tabularx}{\textwidth}[b]{@{}l@{ }>{\raggedright\arraybackslash}X@{}}
2627     \foreign@advisorstring: &
2628     \count1=0
2629     \toks@={}
2630     \@whilenum \count1<\@advisor \do{%
2631       \ifcase\count1 % same as \ifnum0=\count1
2632         \toks@=\expandafter{\csname CoppeAdvisorName:\the\count1%
2633           \expandafter\endcsname\expandafter\space%
2634           \csname CoppeAdvisorSurname:\the\count1\endcsname\\}

```

```

2635 \else
2636 \toks@=\expandafter\expandafter\expandafter{%
2637 \expandafter\the\expandafter\toks@
2638 \expandafter&\expandafter\space
2639 \csname CoppeAdvisorName:\the\count1\expandafter\endcsname%
2640 \expandafter\space\csname CoppeAdvisorSurname:\the\count1\endcsname\\
2641 }%
2642 \fi
2643 \advance\count1 by 1}
2644 \the\toks@
2645 \end{tabularx}%
2646 \coppe@coadvisorrows{\foreign@coadvisorstring}\par
2647 \vspace*{2mm}
2648 \noindent\foreign@deptstring: \foreign@deptname\par
2649 \vspace*{7mm}}{%
2650 \ifnum\value{foreignkeywords}>0
2651 \coppe@printkeywords{\coppeforeignstring{keywordsname}}%
2652 {CoppeForeignKeyword:}\value{foreignkeywords}}%
2653 \else
2654 \coppe@printkeywords{\coppeforeignstring{keywordsname}}%
2655 {CoppeKeyword:}\value{keywords}}%
2656 \fi
2657 \end{otherlanguage}
2658 \vspace*{\fill}
2659 \global\let\@author\@empty
2660 \global\let\@date\@empty
2661 \global\let\foreign@title\@empty
2662 \global\let\foreign@title\relax
2663 \global\let\local@title\@empty
2664 \global\let\local@title\relax
2665 \global\let\author\relax
2666 \global\let\author\relax
2667 \global\let\date\relax}

```

brazilianabstract (*env.*) Optional third abstract for non-pt main languages: a Brazilian Portuguese abstract printed in addition to **abstract** (main language) and **foreignabstract** (English by default). Wraps its body in `\begin{otherlanguage}{brazilian}` so the babel typography (hyphenation, quotation marks) is correct regardless of the current main language. Uses the same `\local@...` Portuguese strings the **abstract** env uses today. Meaningful only when main is neither **brazilian** nor **english**; harmless otherwise (the document just won't include it).

```

2668 \newenvironment{brazilianabstract}{%
2669 \cleardoublepage
2670 \thispagestyle{plain}
2671 \begin{otherlanguage}{brazilian}
2672 \abstract@toptext\par
2673 \vspace*{8.6mm}
2674 \begin{center}
2675 \sloppy\nohyphens{\MakeUppercase\local@title}\par
2676 \vspace*{13.2mm}
2677 \@authname\ \@authsurn \par
2678 \vspace*{7mm}
2679 \local@monthname/\coppe@docyear

```

```

2680 \end{center}\par
2681 \vspace*{1.5cm}
2682 \noindent%
2683 \begin{tabularx}{\textwidth}[b]{@{}l@{ }>{\raggedright\arraybackslash}X@{}}
2684 \local@advisorstring: &
2685 \count1=0
2686 \toks@={}
2687 \@whilenum \count1<\@advisor \do{%
2688 \ifcase\count1
2689 \toks@=\expandafter{\csname CoppeAdvisorName:\the\count1%
2690 \expandafter\endcsname\expandafter\space%
2691 \csname CoppeAdvisorSurname:\the\count1\endcsname\\}
2692 \else
2693 \toks@=\expandafter\expandafter\expandafter{%
2694 \expandafter\the\expandafter\toks@
2695 \expandafter&\expandafter\space
2696 \csname CoppeAdvisorName:\the\count1\expandafter\endcsname%
2697 \expandafter\space\csname CoppeAdvisorSurname:\the\count1\endcsname\\
2698 }%
2699 \fi
2700 \advance\count1 by 1}
2701 \the\toks@
2702 \end{tabularx}%
2703 \coppe@coadvisorrows{\local@coadvisorstring}\par
2704 \vspace*{2mm}
2705 \noindent\local@deptstring: \local@deptname\par
2706 \vspace*{7mm}}{%
2707 % \coppestring takes ONE argument and resolves against the current language;
2708 % this runs before \end{otherlanguage}, so \language is already brazilian.
2709 \ifnum\value{braziliankeywords}>0
2710 \coppe@printkeywords{\coppestring{keywordsname}}%
2711 {CoppeBrazilianKeyword:}{\value{braziliankeywords}}%
2712 \else
2713 \coppe@printkeywords{\coppestring{keywordsname}}%
2714 {CoppeKeyword:}{\value{keywords}}%
2715 \fi
2716 \end{otherlanguage}
2717 \vspace*{\fill}}

```

\listoffigures

```

2718 \renewcommand\listoffigures{%
2719 \coppe@hasLof
2720 \if@twocolumn
2721 \@restonecoltrue\onecolumn
2722 \else
2723 \@restonecolfalse
2724 \fi
2725 \chapter*{\listfigurename}%
2726 \coppe@listtoc{\listfigurename}%
2727 \mkboth{\MakeUppercase\listfigurename}%
2728 {\MakeUppercase\listfigurename}%
2729 \@starttoc{lof}%
2730 \if@restonecol\twocolumn\fi
2731 }

```

\listoftables

```
2732 \renewcommand\listoftables{%
2733     \if@twocolumn
2734         \@restonecoltrue\onecolumn
2735     \else
2736         \@restonecolfalse
2737     \fi
2738     \chapter*{\listtablename}%
2739     \coppe@listtoc{\listtablename}%
2740     \@mkboth{%
2741         \MakeUppercase\listtablename}%
2742         {\MakeUppercase\listtablename}%
2743     \@starttoc{lot}%
2744     \if@restonecol\twocolumn\fi
2745 }
```

\printlosymbols

```
2746 \newcommand\printlosymbols{%
2747 \renewcommand\glossaryname{\listsymbolname}%
2748 \begingroup\@coppeglosspretrue
2749 \@input@{\jobname.los}\endgroup}
```

\makelosymbols

```
2750 \def\makelosymbols{%
2751     \newwrite\@losfile
2752     \immediate\openout\@losfile=\jobname.syx
2753     \newcommand\syml[3][\@bsphack\begingroup
2754     \ifstrepty{##1}{\def\@tempsymb1{##2=}}{\def\@tempsymb1{##1=}}}%
2755         \@sanitize%
2756         \@wrlos{\@tempsymb1}{##2}{##3}\typeout%
2757     {Writing index of symbols file \jobname.syx}%
2758     \let\makelosymbols\@empty%
2759 }%
2760 \@onlypreamble\makelosymbols

2761 \AtBeginDocument{%
2762 \ifpackageloaded{hyperref}{%
2763     \newcommand\@wrlos[3]{%
2764         \protected@write\@losfile{%
2765             {\string\indexentry{#1[#2] #3|hyperpage}{\thepage}}%
2766         \endgroup%
2767         \@esphack}}{%
2768     \newcommand\@wrlos[3]{%
2769         \protected@write\@losfile{%
2770             {\string\indexentry{#1[#2] #3}{\thepage}}%
2771         \endgroup%
2772         \@esphack}}}%

```

\printloabbreviations

```
2773 \newcommand\printloabbreviations{%
2774 \renewcommand\glossaryname{\listabbreviationname}%
2775 \begingroup\@coppeglosspretrue
2776 \@input@{\jobname.lab}\endgroup}
```

\makeloabbreviations

```

2777 \def\makeloabbreviations{%
2778   \newwrite\@labfile
2779   \immediate\openout\@labfile=\jobname.abx
2780   \newcommand\abbrev[3][\@bsphack\beginingroup
2781     \ifstrempy{##1}{\def\@tempsymb1{##2=}}{\def\@tempsymb1{##1=}}
2782     \@sanitize
2783     \@wrlab{\@tempsymb1}{##2}{##3}\typeout
2784     {Writing index of abbreviations file \jobname.abx}%
2785     \let\makeloabbreviations\@empty
2786 }
2787 \@onlypreamble\makeloabbreviations

```

```

2788 \AtBeginDocument{%
2789 \@ifpackageloaded{hyperref}{%
2790   \newcommand\@wrlab[3]{%
2791     \protected@write\@labfile{%
2792       {\string\indexentry{#1[#2] #3|hyperpage}{\thepage}}%
2793     \endgroup%
2794     \@esphack}}{%
2795   \newcommand\@wrlab[3]{%
2796     \protected@write\@labfile{%
2797       {\string\indexentry{#1[#2] #3}{\thepage}}%
2798     \endgroup%
2799     \@esphack}}}%

```

\newsigla Acronyms (siglas), manual2024 2.8 / R43. \newsigla{key}{SIGLA}{full form}
 \sigla declares one; \sigla{key} prints “full form (SIGLA)” on first use and registers it in the list of abbreviations and acronyms (through \abbrev, so \makeloabbreviations must be active), then just “SIGLA” on later uses. \sigla*{key} forces the full form. The first-use flag is set globally so it survives groups (floats, footnotes).

```

2800 \newcommand\newsigla[3]{%
2801   \@namedef{coppe@sigla@s@#1}{#2}%
2802   \@namedef{coppe@sigla@l@#1}{#3}%
2803 \newcommand\coppe@sigla@reg[1]{%
2804   \@ifundefined{abbrev}{}%
2805   {\abbrev{\@nameuse{coppe@sigla@s@#1}}{\@nameuse{coppe@sigla@l@#1}}}%
2806 \newcommand\coppe@sigla@full[1]{%
2807   \@nameuse{coppe@sigla@l@#1}\space(\@nameuse{coppe@sigla@s@#1})}
2808 \newcommand\coppe@sigla@undef[1]{%
2809   \PackageWarning{coppe}{Sigla '#1' nao declarada com \string\newsigla}%
2810   \textbf{??#1??}}
2811 \newcommand\coppe@sigla@first[1]{%
2812   \global\@namedef{coppe@sigla@done@#1}{\coppe@sigla@reg{#1}\coppe@sigla@full{#1}}
2813 \newcommand\coppe@sigla@norm[1]{%
2814   \@ifundefined{coppe@sigla@s@#1}{\coppe@sigla@undef{#1}}%
2815   {\@ifundefined{coppe@sigla@done@#1}{\coppe@sigla@first{#1}}%
2816     {\@nameuse{coppe@sigla@s@#1}}}%
2817 \newcommand\coppe@sigla@star[1]{%
2818   \@ifundefined{coppe@sigla@s@#1}{\coppe@sigla@undef{#1}}%
2819   {\global\@namedef{coppe@sigla@done@#1}{\coppe@sigla@full{#1}}}%
2820 \newcommand\sigla{\@ifstar\coppe@sigla@star\coppe@sigla@norm}

2821 %%% \AtBeginDocument{%

```

```

2822 %%%% \@ifpackageloaded{hyperref}{%
2823 %%%% \def\@wrlab#1#2{%
2824 %%%% \protected@write\@labfile{%
2825 %%%% {\string\indexentry{[#1] #2|hyperpage}{\thepage}}%
2826 %%%% \endgroup
2827 %%%% \@esphack}}{%
2828 %%%% \def\@wrlab#1#2{%
2829 %%%% \protected@write\@labfile{%
2830 %%%% {\string\indexentry{[#1] #2}{\arabic{page}}}%
2831 %%%% \endgroup
2832 %%%% \@esphack}}}

2833 % Catalog-label writers. Two language-related traps must be avoided here:
2834 % 1. The label NAMES contain ':' and '.', which babel makes active in
2835 % some languages (e.g. Brazilian's "shorthand" set). If the write
2836 % argument is tokenised while one of those is active, pdfTeX aborts
2837 % with "<inserted text> }\endwrite" because the shorthand expansion
2838 % injects unbalanced braces. \detokenize emits its argument as
2839 % catcode-12 characters regardless of the current activation, so the
2840 % label NAMES survive whatever language is current at writer time.
2841 % 2. Spanish babel redefines \@roman (and therefore \roman{...}) to
2842 % produce small-caps roman numerals via \es@scroman, whose expansion
2843 % contains an \uppercase group that is unbalanced inside \write. The
2844 % same crash signature appears. The fix is to bypass LaTeX's
2845 % \roman/\arabic counter wrappers entirely and use the raw e-TeX
2846 % primitives \romannumeral and \number directly on the \c@page count
2847 % register --- both are fully expandable, language-agnostic, and
2848 % always produce plain ASCII digits or lowercase roman letters.
2849 % \Roman{page} is left in place because babel-spanish only redefines
2850 % \@Roman under the (non-default) ucroman option.
2851 \AtBeginDocument{%
2852 \ltx@ifpackageloaded{hyperref}{%
2853 \def\coppe@bibEnd{%
2854 \immediate\write\@auxout{%
2855 \string\newlabel{\detokenize{bib:end}}%
2856 {}{\number\c@page}{\detokenize{page.}\number\c@page}{}}}%
2857 \def\coppe@bibBegin{%
2858 \immediate\write\@auxout{%
2859 \string\newlabel{\detokenize{bib:begin}}%
2860 {}{\number\c@page}{\detokenize{page.}\number\c@page}{}}}%
2861 \def\coppe@mainBegin{%
2862 \immediate\write\@auxout{%
2863 \string\newlabel{\detokenize{front:pageno}}%
2864 {}{\Roman{page}}{\detokenize{page.}\romannumeral\c@page}{}}}%
2865 \def\coppe@hasLof{%
2866 \immediate\write\@auxout{%
2867 \string\newlabel{\detokenize{cat:lofflag}}%
2868 {}{\detokenize{:~il.}}{\detokenize{page.}\romannumeral\c@page}{}}}%
2869 }{%
2870 \def\coppe@bibEnd{%
2871 \immediate\write\@auxout{%
2872 \string\newlabel{\detokenize{bib:end}}{\number\c@page}{}}}%
2873 \def\coppe@bibBegin{%
2874 \immediate\write\@auxout{%
2875 \string\newlabel{\detokenize{bib:begin}}{\number\c@page}{}}}%

```

```

2876 \def\coppe@mainBegin{%
2877 \immediate\write\@auxout{%
2878 \string\newlabel{\detokenize{front:pageno}}{\{\}\Roman{page}\}\}\}%
2879 \def\coppe@hasLof{%
2880 \immediate\write\@auxout{%
2881 \string\newlabel{\detokenize{cat:lofflag}}{\{\}\detokenize{:~il.};}\}\}\}%
2882 }%
2883 }
2884 % The reference list is now produced by biblatex's \printbibliography (see the
2885 % bibliography setup after babel, above). The former thebibliography
2886 % redefinition and its \bibindent are no longer used and have been removed; the
2887 % bib:begin/bib:end catalog labels are now written by \AtBeginBibliography /
2888 % \AtEndBibliography via \coppe@bibBegin and \coppe@bibEnd, defined just above.

```

longquote (*env.*)

```

2889 \newlength{\recuolongquote}
2890 \setlength{\recuolongquote}{4cm}
2891
2892 \newenvironment*{longquote}[1][default]{%
2893 \begin{list}{\}%
2894 \setlength{\leftmargin}{\recuolongquote}%
2895 \setlength{\rightmargin}{0pt}%
2896 \setlength{\itemindent}{0pt}%
2897 \setlength{\listparindent}{0pt}%
2898 \setlength{\parsep}{0pt}%
2899 \setlength{\topsep}{\baselineskip}%
2900 }%
2901 \item\relax
2902 \footnotesize
2903 \singlespacing
2904 \ifthenelse{\not\equal{#1}{default}}{%
2905 {\itshape\selectlanguage{#1}}%
2906 }%
2907 \ignorespaces
2908 }{%
2909 \end{list}%
2910 \ignorespacesafterend
2911 }

2912 \newenvironment{theglossary}{%
2913 \if@twocolumn%
2914 \@restonecoltrue\onecolumn%
2915 \else%
2916 \@restonecolfalse%
2917 \fi%
2918 \@mkboth{\MakeUppercase\glossaryname}%
2919 {\MakeUppercase\glossaryname}%
2920 \chapter*{\glossaryname}%
2921 \if@coppeglosspre
2922 \coppe@listtoc{\glossaryname}%
2923 \else
2924 \addcontentsline{toc}{chapter}{\glossaryname}%
2925 \fi
2926 \list{}

```



```

2927 {\setlength{\listparindent}{0in}%
2928 \setlength{\labelwidth}{1.0in}%
2929 \setlength{\leftmargin}{1.5in}%
2930 \setlength{\labelsep}{0.5in}%
2931 \setlength{\itemindent}{0in}}%
2932 \sloppy}%
2933 {\if@restonecol\twocolumn\fi%
2934 \endlist}
2935 %
2936 \renewenvironment{theindex}{%
2937 \if@twocolumn
2938 \@restonecolfalse
2939 \else
2940 \@restonecoltrue
2941 \fi
2942 \twocolumn[\@makeschapterhead{\indexname}]}%
2943 \@mkboth{\MakeUppercase\indexname}%
2944 {\MakeUppercase\indexname}%
2945 \thispagestyle{plain}\parindent\z@
2946 \addcontentsline{toc}{chapter}{\indexname}
2947 \parskip\z@ \@plus .3\p@\relax
2948 \columnseprule \z@
2949 \columnsep 35\p@
2950 \let\item\@idxitem}
2951 {\if@restonecol\onecolumn\else\clearpage\fi}
2952 \newcommand\listabbreviationname{\coppemainstring{listabbreviation}}
2953 \newcommand\listsymbolname{\coppemainstring{listsymbol}}
2954 \newcommand\glossaryname{\coppemainstring{glossary}}
2955 %
2956 \newcommand\local@advisorstring{Orientador}
2957 \newcommand\foreign@advisorstring{Advisor}
2958 % Fed by the string dispatcher instead of a hard-coded Portuguese literal.
2959 % Expansion is lazy (it happens inside \makefrontpage), so \coppe@mainlang is
2960 % already set by the time this is used.
2961 \ifthenelse{\boolean{maledoc}}{%
2962 \newcommand\local@approvedname{\coppemainstring{approvedmale}}%
2963 }{%
2964 \newcommand\local@approvedname{\coppemainstring{approvedfemale}}%
2965 }
2966 \ifthenelse{\boolean{maledoc}}{%
2967 \newcommand\local@approvedonname{\coppemainstring{approvedonmale}}%
2968 }{%
2969 \newcommand\local@approvedonname{\coppemainstring{approvedonfemale}}%
2970 }
2971 \newcommand\local@universityname{Universidade Federal do Rio de Janeiro}
2972 \newcommand\local@deptstring{Programa}
2973 \newcommand\foreign@deptstring{Department}
2974 \newcommand\local@cityname{Rio de Janeiro}
2975 \newcommand\local@statename{RJ}
2976 \newcommand\local@countryname{Brasil}
2977 %
2978 % 3.1.2.1.1(e) of the 2026 manual requires the natureza, the objetivo, the
2979 % name of the institution AND the area de concentracao in this block. The area
2980 % was missing; it is appended as its own sentence when \areaconcentracao was

```

```

2981 % given, and simply omitted otherwise.
2982 \newcommand\frontcover@maintext{
2983   \sloppy\nohyphens{\local@doctype\ de \@degreename\
2984   \ifthenelse{\boolean{maledoc}}{apresentado}{apresentada}
2985   ao Programa de P{\ ' o}s-gradua{\c c}{\~ a}o em \local@deptname,
2986   COPPE, da \local@universityname, como parte dos requisitos
2987   necess{\ ' a}rios {\ ' a} obten{\c c}{\~ a}o do t{\ ' i}tulo de
2988   \local@degname\ em \local@deptname.}%
2989   \ifthenelse{\equal{\coppe@areaconc}{}}{ }{%
2990     \par\vspace{1ex}%
2991     \sloppy\nohyphens{\ ' Area de concentra{\c c}{\~ a}o: \coppe@areaconc.}}%
2992 }
2993 %
2994 \newcommand\frontpage@maintext{
2995   \noindent {\MakeUppercase\local@doctype}
2996   \ifthenelse{\boolean{maledoc}}{SUBMETIDO}{SUBMETIDA}
2997   \sloppy\nohyphens{AO CORPO DOCENTE DO INSTITUTO ALBERTO LUIZ COIMBRA
2998   DE P{\ ' O}S-GRADUA{\c C}{\~ A}O E PESQUISA DE ENGENHARIA DA
2999   UNIVERSIDADE FEDERAL DO RIO DE JANEIRO COMO PARTE DOS REQUISITOS
3000   NECESS{\ ' A}RIOS PARA A OBTEN{\c C}{\~ A}O DO GRAU DE
3001   {\MakeUppercase\local@degname} EM CI{\^ E}NCIAS EM
3002   {\MakeUppercase\local@deptname.\par}}%
3003 }
3004 %
3005 \newcommand\frontpage@bottomtext{%
3006   \begin{center}
3007   {\MakeUppercase{\local@cityname, \local@statename\ -- \local@countryname}}\par
3008   {\MakeUppercase\local@monthname\ DE \coppe@docyear}
3009   \end{center}%
3010 }
3011 %
3012 \newcommand\abstract@toptext{%
3013   \noindent Resumo \ifthenelse{\boolean{maledoc}}{do}{da}
3014   \local@doctype\ \ifthenelse{\boolean{maledoc}}{apresentado}{apresentada}
3015   \sloppy\nohyphens{\ ' a} COPPE/UFRJ como parte dos requisitos
3016   necess{\ ' a}rios para a obten{\c c}{\~ a}o do grau de
3017   \local@degname\ em Ci{\^ e}ncias (\@degree)}
3018 }
3019 \newcommand\foreignabstract@toptext{%
3020   \noindent \sloppy\nohyphens{Abstract of \foreign@doctype\ presented to
3021   COPPE/UFRJ as a partial fulfillment of the requirements for the
3022   degree of \foreign@degname\ of Science (\@degree)}
3023 }
3024 %

```

Late-load packages: setspace, hyperref, biblatex

These three packages *must* be the last things the class loads, and in this exact order:

- **setspace** – supplies \singlespacing, \onehalfspacing and friends. It is loaded here so the line-spacing state it carries around is settled before hyperref patches all the cross-reference and TOC plumbing.

- `hyperref` – patches an enormous number of LaTeX internals (`\section`, `\label`, `\ref`, `\bibcite`, `list-of-*` commands, ...). Loading it as late as possible is the standard rule of thumb so it sees the final version of those macros, including the ones the `coppe` class redefines (e.g. `\@chapter`, `\listofquados`).
- `biblatex` – must come after `hyperref` so its citation links and back-references hook into `hyperref` correctly; otherwise the `\cite` commands print as plain (non-clickable) text.

```

3025 \RequirePackage{setspace}
3026 % Apply the default line spacing now that setspace is available. The flag was
3027 % set above by \DeclareOption{doublespacing}; absent the option, one-and-a-half.
3028 \if@coppedoublespace\doublespacing\else\onehalfspacing\fi
3029 % pdfx MUST precede hyperref -- it loads hyperref itself, with the options
3030 % PDF/A needs, and cannot fix up an already-loaded one. Level a-2b rather than
3031 % a-1b (the "ISO 19005-1" of Annex I): a-1b forbids transparency, which
3032 % tcolorbox and several graphics packages produce, so a-1b would fail on
3033 % documents that are otherwise perfectly fine. a-2b is ISO 19005-2, accepted by
3034 % Pantheon and by the SiBI's own instructions, which also list PDF/A-2b.
3035 \if@coppepdfa
3036   \RequirePackage[a-2b]{pdfx}
3037 % pdfTeX stamps /PTEX.Fullbanner into the document information dictionary.
3038 % It is not one of the keys ISO 19005-2 6.6.2.3.2 maps to an XMP property, and
3039 % no extension schema declares it, so it is a DocInfo entry with no XMP
3040 % counterpart -- a finding a PDF/A checker has every reason to raise, and one
3041 % that costs nothing to remove. -1 sets every suppression bit.
3042   \pdfsuppressptexinfo=-1
3043 % pdfx takes the XMP metadata from \jobname.xmpdata, which it reads when it is
3044 % LOADED -- before the preamble has run, so before \title, \author and
3045 % \keyword exist. The file is therefore WRITTEN by the class and picked up on
3046 % the NEXT run. The build already runs pdflatex three times for cross-
3047 % references, so the deposited PDF always carries current metadata; only a
3048 % first-ever compilation of a brand-new file emits pdfx's "no \jobname.xmpdata"
3049 % warning and lacks dc:title.
3050 %
3051 % The write happens at the END of the document, not at \begin{document}. This
3052 % class has always documented -- and every example follows it -- that \title,
3053 % \author, \department and \keyword go AFTER \begin{document}, next to
3054 % \maketitle. A write at \begin{document} therefore ran before any of them
3055 % existed and died with "Undefined control sequence" on \@authname, so the
3056 % pdfa option was unusable in exactly the style the manual teaches. It only
3057 % ever worked for a document that set its metadata in the preamble. At the end
3058 % everything is set whichever style the author chose, and \immediate still puts
3059 % the file on disk in time for the next run.
3060 %
3061 % The hook is enddocument/afteraux, and the stream is \@mainaux rather than one
3062 % of our own. TeX has sixteen output streams and example.tex was already using
3063 % all of them (toc, lof, lot, loa, lol, loq, los, idx, glo, aux, bcf, ...), so
3064 % \newwrite here failed with "No room for a new \write" on precisely the
3065 % document that most needs validating. enddocument/afteraux fires after LaTeX
3066 % has closed the .aux, which frees that stream for the one \immediate\openout
3067 % we need and costs the document nothing. Kernels older than 2020-10-01 have no
3068 % such hook; there the class falls back to \AtEndDocument with a stream of its

```

```

3069 % own, which is the pre-existing behaviour.
3070 %
3071 % Every field is guarded, and the guards are not cosmetic. \coppe@selecttitle
3072 % is \@nameuse{coppe@title@<lang>}, and \csname on a name that does not exist
3073 % yet MAKES it \relax: unexpandable, so \write puts the control sequence NAME
3074 % into the file instead of the title. The .xmpdata then carried the literal
3075 % \coppe@title@brazilian, pdfx read it back on the following run, and the
3076 % document stopped compiling at all -- a single bad run bricked the file until
3077 % someone deleted the .xmpdata by hand. A field that is not set is now simply
3078 % omitted, which pdfx handles, instead of poisoning the next run.
3079 %
3080 % The writes must be \immediate. \protected@write defers to the shipout, and
3081 % the stream is closed here and now, so deferred writes would land nowhere --
3082 % which is exactly what happened first time round: a 0-byte .xmpdata.
3083 % \let\protect\string inside the group gives the protection that
3084 % \protected@write would have given: accents and other robust commands are
3085 % written literally instead of being expanded. pdfx reads the .xmpdata as
3086 % LaTeX, so \'a arrives there intact.
3087 \ifdefined\AddToHook
3088   \let\coppe@xmpout\@mainaux
3089   \def\coppe@xmphook{\AddToHook{enddocument/afteraux}}%
3090 \else
3091   \newwrite\coppe@xmpout
3092   \let\coppe@xmphook\AtEndDocument
3093 \fi
3094 \coppe@xmphook{%
3095   \gdef\coppe@xmpkeys{%
3096     \count@=0
3097     \@whilenum\count@<\value{keywords}\do{%
3098       \ifnum\count@>0
3099         \g@addto@macro\coppe@xmpkeys{\string\sep\space}%
3100       \fi
3101       \protected@edef\coppe@tmp{\csname CoppeKeyword:\the\count@\endcsname}%
3102       \expandafter\g@addto@macro\expandafter\coppe@xmpkeys
3103       \expandafter{\coppe@tmp}%
3104       \advance\count@ by 1}%
3105   \immediate\openout\coppe@xmpout=\jobname.xmpdata
3106   \begingroup
3107     \let\protect\string
3108     \@ifundefined{coppe@title@\coppe@mainlang}{}%
3109       \immediate\write\coppe@xmpout{\string\Title{\coppe@selecttitle}}}%
3110     \@ifundefined{@authname}{}%
3111       \immediate\write\coppe@xmpout{%
3112         \string\Author{\@authname\space\@authsurn}}}%
3113     \immediate\write\coppe@xmpout{\string\Keywords{\coppe@xmpkeys}}%
3114     \ifthenelse{equal{\coppe@deptcode}{}}{%
3115       \immediate\write\coppe@xmpout{\string\Subject{\local@doctype\space de
3116         \@degree\name. Programa de \meta@deptname\space(\coppe@deptcode),
3117         COPPE/UFRJ}}}%
3118       \immediate\write\coppe@xmpout{\string\Publisher{\local@universityname}}}%
3119     \@ifundefined{coppe@docyear}{}%
3120       \immediate\write\coppe@xmpout{\string\Date{\coppe@docyear}}}%
3121   \endgroup
3122   \immediate\closeout\coppe@xmpout}%

```

```

3123 \fi
3124 \RequirePackage{hyperref}
3125 % Bibliography engine: the full coppe BibLaTeX/biber ABNT style, replacing the
3126 % former natbib+bibtex setup. The numbers class option selects the numeric
3127 % citation style; otherwise the author-date default is used. natbib=true keeps
3128 % \citet/\citep available for users migrating from natbib.
3129 \if@coppenumeric
3130   \RequirePackage[backend=biber,datamodel=coppe,%
3131     bibstyle=coppe-numeric,citestyle=coppe-numeric,natbib=true]{biblatex}
3132 \else
3133   \RequirePackage[backend=biber,datamodel=coppe,%
3134     bibstyle=coppe,citestyle=coppe,natbib=true]{biblatex}
3135 \fi
3136 % Reference-list heading "Referencias" (pt) / "References" (en) -- manual2024
3137 % 3.1.4.1 requires the post-textual list to be titled "Referencias", not
3138 % "Bibliografia" (babel's default \bibname). The title is rendered through
3139 % \coppestring so it follows the current main language in BOTH the chapter
3140 % heading and the .toc entry (biblatex's \bibstring proved unreliable across
3141 % the .toc write, and babel's \bibname caption override was equally fragile).
3142 \defbibheading{bibliography}{%
3143   \chapter*{\coppestring{references}}%
3144   \addcontentsline{toc}{chapter}{\coppestring{references}}
3145 % The catalog page range (bib:begin/bib:end) was written by the old
3146 % thebibliography redefinition, which biblatex's \printbibliography does not
3147 % use. biblatex offers \AtBeginBibliography but no end counterpart, so the
3148 % begin label uses that hook and the end label is emitted by wrapping
3149 % \printbibliography to write it just after the reference list.
3150 \AtBeginBibliography{\coppe@bibBegin}
3151 \let\coppe@orig@printbibliography\printbibliography
3152 \renewcommand\printbibliography[1][]{%
3153   \coppe@orig@printbibliography[#1]%
3154   \coppe@bibEnd}
3155 \</class>
3156 \<*glossary>
3157 actual '='
3158 quote '!'
3159 level '>'
3160 %%% delim_0  ", p. "
3161 delim_0 "\\dotfill "
3162 lethead_flag 0
3163 headings_flag 0
3164 preamble
3165 "\n\\begin{theglossary}\n  \\makeatletter"
3166 postamble
3167 "\n  \\end{theglossary}\n"
3168 \</glossary>
3169 \<*class>
3170 \newcommand{\annex}{
3171   \renewcommand{\appendixname}{\coppestring{annex}}%
3172   \appendix
3173   \@coppeannextrue
3174 % \appendix has already re-issued the centred chapter format, but it baked in
3175 % the \appendixname of that moment. Re-issue it now that the name is "Anexo".

```

```

3176 \coppe@appendixheadings
3177 }
3178 </class>

```

BibLaTeX style files

Generated from this .dtx by coppe.ins (one docstrip module each).

```

3179 <*dbx>
3180 \ProvidesFile{coppe.dbx}[2026/05/28 v4.0 CoppeTeX ABNT datamodel]
3181 % Custom entry types beyond biblatex's standard datamodel (games, TV, maps).
3182 % map = cartographic document (NBR 6023 4.2.12); biblatex has no native @map.
3183 % The remaining ABNT "exotic" types (legislation, jurisdiction, legal, patent,
3184 % standard, music, audio, image, artwork, video, dataset, letter, periodical,
3185 % performance) are already in biblatex's standard datamodel.
3186 \DeclareDatamodelEntrytypes{game,videogame,boardgame,tvshow,map}
3187 % Custom fields.
3188 \DeclareDatamodelFields[type=field,datatype=literal]{course} % "(... em <course>)"
3189 \DeclareDatamodelFields[type=field,datatype=literal]{coppedegree} % degree word
3190 \DeclareDatamodelFields[type=field,datatype=literal]{platform} % videogame platform
3191 \DeclareDatamodelFields[type=field,datatype=literal]{scale} % map scale (escala)
3192 \DeclareDatamodelFields[type=list,datatype=literal]{artist} % game/illustration art
3193 \DeclareDatamodelFields[type=list,datatype=literal]{director} % movie/tvshow/video
3194 \DeclareDatamodelFields[type=list,datatype=literal]{producer}
3195 \DeclareDatamodelFields[type=field,datatype=literal]{caderno} % newspaper section
3196 \DeclareDatamodelFields[type=field,datatype=literal]{relator} % jurisprudence rapporteur
3197 \DeclareDatamodelEntryfields{course,coppedegree}
3198 \DeclareDatamodelEntryfields{game,videogame,boardgame,software,movie,tvshow,video}%
3199 {artist,platform,director,producer,version}
3200 \DeclareDatamodelEntryfields[map]{scale}
3201 \DeclareDatamodelEntryfields[article]{caderno}
3202 \DeclareDatamodelEntryfields[jurisdiction]{relator}
3203 </dbx>
3204 <*bbx>
3205 \ProvidesFile{coppe.bbx}[2026/05/28 v4.0 CoppeTeX ABNT bibliography style]
3206 % The reference list is single-spaced (ABNT 2.4) via \singlespacing in
3207 % \AtBeginBibliography. coppe.cls already loads setspace, but the style must work
3208 % standalone too (e.g. \usepackage[bibstyle=coppe]{biblatex} in a plain article),
3209 % so load setspace here as well (idempotent).
3210 \RequirePackage{setspace}
3211 % ABNT reference list keeps the date at the end (imprenta), same for author-date
3212 % and numeric modes, so we derive from the standard style, not authoryear.
3213 \RequireBibliographyStyle{standard}
3214
3215 % Note: standard style never dashes repeated authors, which already matches the
3216 % post-2020 ABNT rule (repeat the entry in full), so no 'dashed' option is needed.
3217 \ExecuteBibliographyOptions{%
3218 maxbibnames=99, % ABNT 4.3.1.3: list all authors by default
3219 giveninits=false,
3220 uniquename=false,
3221 uniquelist=false,
3222 labeldateparts=true, % needed for author-date extradate (1999a/1999b)
3223 sorting=nyt,
3224 }

```

```

3225
3226 % Language strings (Disponível em / Acesso em / In, etc.)
3227 \DeclareLanguageMapping{brazilian}{brazilian-coppe}
3228 \DeclareLanguageMapping{english}{english-coppe}
3229
3230 % --- Names: all reversed "FAMILY, Given", separated by ";" in the list ---
3231 \DeclareNameAlias{author}{family-given}
3232 \DeclareNameAlias{editor}{family-given}
3233 \DeclareNameAlias{translator}{family-given}
3234 \DeclareNameAlias{default}{family-given}
3235 \DeclareDelimFormat[bib,biblist]{multinamedelim}{\addsemicolon\space}
3236 \DeclareDelimFormat[bib,biblist]{finalnamedelim}{\addsemicolon\space}
3237
3238 % ABNT (NBR 6023 8.1.1.4): a collection entered under the name of its
3239 % organizer prints the kind of responsibility in parentheses after the
3240 % name, abbreviated, lowercase and in the singular even for several
3241 % organizers -- e.g. "SOUSA, A. M. de; OLIVEIRA, L. da C. (org.)". The
3242 % standard style prints ", Org." (comma, no parentheses) when the editor
3243 % takes the author slot, so override editor+others to wrap the role string
3244 % in "(...)". The host position ("In: NAME (org.)") already parenthesises
3245 % the role via coppe:bookeditor below. The role word itself is the
3246 % lowercase, singular "org." (and "ed.", "cur.", ...) set in the language
3247 % packs (brazilian-coppe.lbx et al.).
3248 \renewbibmacro*{editor+others}{%
3249   \ifboolexpr{ test \ifuseeditor and not test {\ifnameundef{editor}} }
3250     {\printnames{editor}%
3251       \setunit*{\addspace}%
3252       \printtext{\mkbibparens{\usebibmacro{editor+othersstrg}}}%
3253       \clearname{editor}}
3254   {}}
3255 % Same parenthesised role for drivers that take the editor through the plain
3256 % editor macro instead of editor+others (e.g. @manual).
3257 \renewbibmacro*{editor}{%
3258   \ifboolexpr{ test \ifuseeditor and not test {\ifnameundef{editor}} }
3259     {\printnames{editor}%
3260       \setunit*{\addspace}%
3261       \printtext{\mkbibparens{\usebibmacro{editorstrg}}}%
3262       \clearname{editor}}
3263   {}}
3264
3265 % Uppercase the surname (family + prefix) only inside the bibliography list;
3266 % citations keep normal caps (post-2020 rule).
3267 % ABNT date (NBR 6023 §4.3.5.5.2): "dia mês. ano" -- without the brazilian "de"
3268 % between the parts (e.g. "4 fev. 1992", "abr. 1996"). #1=year, #2=month, #3=day
3269 % field names; the month stays abbreviated via \mkbibmonth.
3270 \protected\def\coppe@abntdate#1#2#3{%
3271   \iffieldundef{#3}{\stripzeros{\thefield{#3}}\nobreakspace}%
3272   \iffieldundef{#2}{\mkbibmonth{\thefield{#2}}\iffieldundef{#1}{\space}}%
3273   \iffieldundef{#1}{\stripzeros{\thefield{#1}}}}
3274 \AtBeginBibliography{%
3275   \singlespacing % entries single-spaced internally (2.4)
3276   \setlength{\bibitemsep}{\baselineskip}% one blank line between entries (2.4)
3277   \renewcommand*{\mkbibnamefamily}[1]{\MakeUppercase{#1}}%
3278   \renewcommand*{\mkbibnameprefix}[1]{\MakeUppercase{#1}}%

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3279 \let\mkbibdatelong\coppe@abntdate      % ABNT date format (drop the "de")
3280 \let\mkbibdateshort\coppe@abntdate
3281 }
3282
3283 % --- Titles: bold for whole-document types; plain for parts; entry-by-title
3284 %      (no author/editor) -> not bold, first word in CAPS (ABNT 4.2). ---
3285 % \coppeUCtitle uppercases the first SIGNIFICANT word of the title (ABNT 4.2).
3286 % If the title starts with an article (o/a/os/as/um/uma/uns/umas, the/an), the
3287 % article AND the next word are uppercased; otherwise just the first word. The
3288 % delimited macros split off words; \coppeUCkeep strips the trailing-space
3289 % sentinel so no spurious space is left before the following punctuation.
3290 \protected\def\coppeUCtitle#1{\coppeUCtitleA#1 \coppeUCstop}
3291 \protected\def\coppeUCtitleA#1 #2\coppeUCstop{%
3292   \coppeUCifart{#1}
3293   {\MakeUppercase{#1}\ifblank{#2}{ }\{\space\coppeUCtitleB#2\coppeUCstop}}%
3294   {\MakeUppercase{#1}\ifblank{#2}{ }\{\space\coppeUCkeep#2\coppeUCstop}}
3295 \protected\def\coppeUCtitleB#1 #2\coppeUCstop{%
3296   \MakeUppercase{#1}\ifblank{#2}{ }\{\space\coppeUCkeep#2\coppeUCstop}}
3297 \protected\def\coppeUCkeep#1 \coppeUCstop{#1}
3298 % \coppeUCifart{word}{if-article}{if-not}: case-insensitive membership test.
3299 \protected\def\coppeUCifart#1{%
3300   \begingroup
3301   \lowercase{\def\coppeUC@w{#1}}%
3302   \edef\coppeUC@w{\coppeUC@w,%}
3303   \expandafter\in@\expandafter{\coppeUC@w}{,a,o,as,os,um,uma,uns,umas,the,an,%}
3304   \ifin@\aftergroup\@firstoftwo\else\aftergroup\@secondoftwo\fi
3305   \endgroup}
3306 \DeclareFieldFormat{title}{%
3307   \ifboolexpr{ test {\ifnameundef{labelname}} }
3308     {#1}%
3309     {\mkbibbold{#1}}}
3310 % First word of the title in CAPS for entry-by-title (ABNT 4.2). The titlecase
3311 % format receives the RAW title text (the title format above only sees the
3312 % field-printing commands), so \coppeUCtitle can split off and uppercase the
3313 % first word here.
3314 \DeclareFieldFormat{titlecase}{%
3315   \ifboolexpr{ test {\ifnameundef{labelname}} }
3316     {\coppeUCtitle{#1}}%
3317     {#1}}
3318 \DeclareFieldFormat[article,incollection,inbook,inproceedings,supperiodical,review]{title}{%
3319   \DeclareFieldFormat[thesis]{title}{\mkbibbold{#1}}
3320 % biblatex's standard style quotes patent (and unpublished) titles; ABNT bolds
3321 % the title of a whole document, so override those back to bold.
3322 \DeclareFieldFormat[patent]{title}{\mkbibbold{#1}}
3323 % Legal documents (NBR 6023 4.2.6): the law's identification/ementa is NOT
3324 % highlighted; the entry leads with the jurisdiction (entity author, CAPS).
3325 \DeclareFieldFormat[legislation,jurisdiction,legal]{title}{#1}
3326 \DeclareFieldFormat[journaltitle]{\mkbibbold{#1}}
3327 \DeclareFieldFormat[booktitle]{\mkbibbold{#1}}
3328 \DeclareFieldFormat[maintitle]{\mkbibbold{#1}}
3329
3330 % --- Pages: ABNT uses "p." (also for ranges); theses use "f." for pagetotal ---
3331 \DeclareFieldFormat{pagetotal}{#1\addnbspace p\adddot}
3332 \DeclareFieldFormat[thesis]{pagetotal}{#1\addnbspace f\adddot}

```



```

3333 \DeclareFieldFormat{pages}{p\adddot\addnbspace#1}
3334
3335 % --- Periodical volume / number prefixes ---
3336 \DeclareFieldFormat{article}{volume}{v\adddot\addnbspace#1}
3337 \DeclareFieldFormat{article}{number}{n\adddot\addnbspace#1}
3338
3339 % --- "In:" before host title (incollection/inproceedings/inbook) ---
3340 \renewbibmacro*{in:}{\printtext{\bibstring{in}\addcolon\space}}
3341
3342 % --- Online: "Disponível em: <url>." + "Acesso em: <date>." via lbx strings ---
3343 \DeclareFieldFormat{url}{\bibstring{availablefrom}\addcolon\space\url{#1}}
3344
3345 % --- ABNT article driver: no "In: "; Journal, Local, v., n., p., date, DOI/URL.
3346 % Newspapers (entrysubtype=newspaper, set by the @artigodejournal source map)
3347 % differ: the newspaper name (journaltitle) and city, then either
3348 % "<date>. <caderno>, p. <pages>" (when a caderno/section is given), or
3349 % "p. <pages>, <date>" (page before date, when none),
3350 % per NBR 6023 §4.2.3.6. ---
3351 \DeclareBibliographyDriver{article}{%
3352 \usebibmacro{bibindex}%
3353 \usebibmacro{begentry}%
3354 \usebibmacro{author/translator+others}%
3355 \setunit{\labelnamepunct}\newblock
3356 \usebibmacro{title}%
3357 \newunit\newblock
3358 \printfield{journaltitle}%
3359 \setunit{\addcomma\space}\printlist{location}%
3360 \iffieldequalstr{entrysubtype}{newspaper}
3361 {\iffieldundef{caderno}
3362 {\setunit{\addcomma\space}\printfield{pages}%
3363 \setunit{\addcomma\space}\printdate}%
3364 {\setunit{\addcomma\space}\printdate%
3365 \setunit{\adddot\space}\printfield{caderno}%
3366 \setunit{\addcomma\space}\printfield{pages}}}%
3367 {\setunit{\addcomma\space}\printfield{volume}%
3368 \setunit{\addcomma\space}\printfield{number}%
3369 \setunit{\addcomma\space}\printfield{pages}%
3370 \setunit{\addcomma\space}\printdate}%
3371 \newunit\newblock
3372 \printfield{doi}%
3373 \newunit\newblock
3374 \usebibmacro{url+urldate}%
3375 \usebibmacro{finentry}}
3376
3377 % --- incollection: ABNT puts the host organizer before the booktitle, as
3378 % "AUTOR. Parte. In: ORGANIZADOR (Org.). Booktitle. Local: Editora, ano,
3379 % p. X-Y." The organizer role string is "Org./Orgs." (see the .lbx). ---
3380 \newbibmacro*{coppe:bookeditor}{%
3381 \ifnameundef{editor}
3382 {}
3383 {\printnames{editor}%
3384 \setunit*\addspace\printtext{\mkbibparens{\usebibmacro{editorstrg}}}%
3385 \clearname{editor}%
3386 \setunit{\adddot\space}}}

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```

3387 \newbibmacro*{coppe:partdriver}{%
3388   \usebibmacro{bibindex}\usebibmacro{begentry}%
3389   \usebibmacro{author/translator+others}%
3390   \setunit{\labelnamepunct}\newblock
3391   \usebibmacro{title}%
3392   \newunit\newblock
3393   \usebibmacro{in:}%
3394   \usebibmacro{coppe:bookeditor}%
3395   \printfield{booktitle}%
3396   \newunit\newblock
3397   \printfield{edition}%
3398   \newunit\newblock
3399   \printlist{location}%
3400   \setunit*{\addcolon\space}\printlist{publisher}%
3401   \setunit*{\addcomma\space}\printdate%
3402   \newunit\newblock
3403   \printfield{pages}%
3404   \setunit{\addcomma\space}\printfield{chapter}%
3405   \newunit\newblock
3406   \usebibmacro{url+urldate}%
3407   \usebibmacro{finentry}}
3408 \DeclareBibliographyDriver{incollection}{\usebibmacro{coppe:partdriver}}
3409 \DeclareBibliographyDriver{inbook}{\usebibmacro{coppe:partdriver}}
3410
3411 % --- type field: resolve a known bibliography string (babel-aware), else literal ---
3412 \DeclareFieldFormat{type}{\ifbibstring{#1}{\bibstring{#1}}{#1}}
3413
3414 % --- Thesis: "<type> (em <course>)" using the custom course field (ABNT). ---
3415 \DeclareFieldFormat{course}{\mkbibparens{\coppestring{in:}\addspace#1}}
3416 \DeclareBibliographyDriver{thesis}{%
3417   \usebibmacro{bibindex}\usebibmacro{begentry}%
3418   \usebibmacro{author}%
3419   \setunit{\labelnamepunct}\newblock
3420   \usebibmacro{title}%
3421   \newunit\newblock
3422   \printfield{type}\setunit*{\addspace}\printfield{course}%
3423   \newunit\newblock
3424   \printlist{institution}%
3425   \setunit*{\addcomma\space}\printlist{location}%
3426   \setunit*{\addcomma\space}\printdate%
3427   \newunit\newblock
3428   \printfield{pagetotal}%
3429   \newunit\newblock
3430   \printfield{note}%
3431   \newunit\newblock
3432   \usebibmacro{url+urldate}%
3433   \usebibmacro{finentry}}
3434 % --- Games / audiovisual: render the custom fields (artist, platform,
3435 %       director, producer, version) the generic misc driver ignores. ---
3436 \newbibmacro*{coppe:media}{%
3437   \iflistundef{artist}{\setunit{\addperiod\space}%
3438     \printtext{\coppestring{art}\addcolon\space}\printlist{artist}}%
3439   \iffieldundef{platform}{\setunit{\addperiod\space}%
3440     \printtext{\coppestring{platform}\addcolon\space}\printfield{platform}}%

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```

3441 \iflistundef{director}{\setunit{\addperiod\space}%
3442 \printtext{\coppestring{directedby}\addcolon\space}\printlist{director}}%
3443 \iflistundef{producer}{\setunit{\addperiod\space}%
3444 \printtext{\coppestring{producedby}\addcolon\space}\printlist{producer}}%
3445 \newbibmacro*{coppe:mediadriv}{%
3446 \usebibmacro{bibindex}\usebibmacro{begentry}%
3447 \usebibmacro{author/translator+others}%
3448 \setunit{\labelnamepunct}\newblock
3449 \usebibmacro{title}%
3450 \newunit\newblock
3451 \usebibmacro{coppe:media}%
3452 \newunit\newblock
3453 \printlist{publisher}%
3454 \setunit*{\addcomma\space}\printlist{location}%
3455 \setunit*{\addcomma\space}\printdate%
3456 \newunit\newblock
3457 \printfield{version}%
3458 \newunit\newblock
3459 \printfield{note}%
3460 \newunit\newblock
3461 \usebibmacro{url+urldate}%
3462 \usebibmacro{finentry}}
3463 \DeclareBibliographyDriver{game}{\usebibmacro{coppe:mediadriv}}
3464 \DeclareBibliographyDriver{videogame}{\usebibmacro{coppe:mediadriv}}
3465 \DeclareBibliographyDriver{boardgame}{\usebibmacro{coppe:mediadriv}}
3466 \DeclareBibliographyDriver{movie}{\usebibmacro{coppe:mediadriv}}
3467 \DeclareBibliographyDriver{tvshow}{\usebibmacro{coppe:mediadriv}}
3468 % Cartographic document (map, NBR 6023 4.2.12): like a book (author/title/
3469 % imprenta) plus the mandatory scale ("Escala 1:N").
3470 \DeclareFieldFormat{scale}{\bibstring{coppescale}\addnbspace#1}
3471 \DeclareBibliographyDriver{map}{%
3472 \usebibmacro{bibindex}\usebibmacro{begentry}%
3473 \usebibmacro{author/translator+others}%
3474 \setunit{\labelnamepunct}\newblock
3475 \usebibmacro{title}%
3476 \newunit\newblock
3477 \printlist{location}%
3478 \setunit*{\addcolon\space}\printlist{publisher}%
3479 \setunit*{\addcomma\space}\printdate%
3480 \newunit\newblock
3481 \printfield{scale}%
3482 \newunit\newblock
3483 \usebibmacro{url+urldate}%
3484 \usebibmacro{finentry}}
3485 % Patent (NBR 6023 4.2.5): inventor. \textbf{title}. Depositante: holder. type
3486 % number. date. The holder (depositante) is printed in normal case, not the
3487 % ALL-CAPS family treatment the bibliography gives the leading author.
3488 \newbibmacro*{coppe:depositor}{%
3489 \ifnameundef{holder}
3490 {}
3491 {\printtext{\bibstring{depositor}\addcolon\space}%
3492 \begingroup
3493 \def\mkbibnamefamily##1{##1}%
3494 \def\mkbibnameprefix##1{##1}%

```

```

3495         \printnames{holder}%
3496     \endgroup}}
3497 \DeclareBibliographyDriver{patent}{%
3498     \usebibmacro{bibindex}%
3499     \usebibmacro{begentry}%
3500     \usebibmacro{author/translator+others}%
3501     \setunit{\labelnamepunct}\newblock
3502     \usebibmacro{title}%
3503     \newunit\newblock
3504     \usebibmacro{coppe:depositor}%
3505     \newunit\newblock
3506     \printfield{type}%
3507     \setunit*{\addspace}\printfield{number}%
3508     \newunit\newblock
3509     \printdate%
3510     \newunit\newblock
3511     \usebibmacro{url+urldate}%
3512     \usebibmacro{finentry}}
3513 % Exotic ABNT types that biblatex's standard style leaves without a driver:
3514 % give each a baseline (book-like for published docs: norma/partitura; misc for
3515 % the rest) so they print. Per-type ABNT refinements can follow later.
3516 \DeclareBibliographyDriver{standard}{\usedriver{}{book}}
3517 \DeclareBibliographyDriver{music}{\usedriver{}{book}}
3518 % Legal documents (NBR 6023 4.2.6-4.2.8): jurisdiction/entity (CAPS author),
3519 % the law id + ementa (plain, not bold), then location and date.
3520 \newbibmacro*{coppe:legaldriver}{%
3521     \usebibmacro{bibindex}\usebibmacro{begentry}%
3522     \usebibmacro{author/translator+others}%
3523     \setunit{\labelnamepunct}\newblock
3524     \usebibmacro{title}%
3525     \newunit\newblock
3526     % Jurisprudência: "Relator: <nome>." after the decision title (NBR 6023).
3527     \iffieldundef{relator}{\printtext{Relator\addcolon\space}\printfield{relator}\newunit\ne
3528     \printlist{location}%
3529     \setunit{\addcomma\space}\printdate%
3530     \newunit\newblock
3531     \usebibmacro{url+urldate}%
3532     \usebibmacro{finentry}}
3533 \DeclareBibliographyDriver{legislation}{\usebibmacro{coppe:legaldriver}}
3534 \DeclareBibliographyDriver{jurisdiction}{\usebibmacro{coppe:legaldriver}}
3535 \DeclareBibliographyDriver{legal}{\usebibmacro{coppe:legaldriver}}
3536 \DeclareBibliographyDriver{audio}{\usedriver{}{misc}}
3537 \DeclareBibliographyDriver{video}{\usedriver{}{misc}}
3538 \DeclareBibliographyDriver{software}{\usedriver{}{misc}}
3539 \DeclareBibliographyDriver{image}{\usedriver{}{misc}}
3540 \DeclareBibliographyDriver{artwork}{\usedriver{}{misc}}
3541 \DeclareBibliographyDriver{performance}{\usedriver{}{misc}}
3542 \DeclareBibliographyDriver{dataset}{\usedriver{}{misc}}
3543
3544 % --- biber source map: Portuguese synonyms (entry types + fields) + thesis family ---
3545 \DeclareSourcemap{%
3546     \maps[datatype=bibtex]{%
3547         % entry-type PT synonyms -> canonical English
3548         \map{\step[tourcesource=livro,typetarget=book]}

```

```

3549 \map{\step[tourcesource=artigo,typetarget=article]}
3550 \map{\step[tourcesource=artigodejornal,typetarget=article,final]\step[fieldset=entrysubtyp
3551 \map{\step[tourcesource=partedelivro,typetarget=incollection]}
3552 \map{\step[tourcesource=emlivro,typetarget=inbook]}
3553 \map{\step[tourcesource=verbeta,typetarget=inreference]}
3554 \map{\step[tourcesource=anaais,typetarget=proceedings]}
3555 \map{\step[tourcesource=trabalhodeevento,typetarget=inproceedings]}
3556 \map{\step[tourcesource=periodico,typetarget=periodical]}
3557 \map{\step[tourcesource=relatorio,typetarget=report]}
3558 \map{\step[tourcesource=relatoriotecnico,typetarget=report]}
3559 \map{\step[tourcesource>manual,typetarget>manual]}
3560 \map{\step[tourcesource=correspondencia,typetarget=letter]}
3561 \map{\step[tourcesource=patente,typetarget=patent]}
3562 \map{\step[tourcesource=diverso,typetarget=misc]}
3563 \map{\step[tourcesource=jogo,typetarget=game]}
3564 \map{\step[tourcesource=jogoeletronico,typetarget=videogame]}
3565 \map{\step[tourcesource=jogodetabuleiro,typetarget=boardgame]}
3566 \map{\step[tourcesource=filme,typetarget=movie]}
3567 \map{\step[tourcesource=serietv,typetarget=tvshow]}
3568 \map{\step[tourcesource=programadettv,typetarget=tvshow]}
3569 % Exotic ABNT types (NBR 6023 4.2.5-4.2.14); targets are biblatex-native
3570 % except map (declared in coppe.dbx).
3571 \map{\step[tourcesource=legislacao,typetarget=legislation]}
3572 \map{\step[tourcesource=atonormativo,typetarget=legislation]}
3573 \map{\step[tourcesource=jurisprudencia,typetarget=jurisdiction]}
3574 \map{\step[tourcesource=documentojuridico,typetarget=legal]}
3575 \map{\step[tourcesource=documentocivil,typetarget=legal]}
3576 \map{\step[tourcesource=norma,typetarget=standard]}
3577 \map{\step[tourcesource=partitura,typetarget=music]}
3578 \map{\step[tourcesource=iconografico,typetarget=image]}
3579 \map{\step[tourcesource=obradearte,typetarget=artwork]}
3580 \map{\step[tourcesource=tridimensional,typetarget=artwork]}
3581 \map{\step[tourcesource=cartografico,typetarget=map]}
3582 \map{\step[tourcesource=mapa,typetarget=map]}
3583 \map{\step[tourcesource=audiovisual,typetarget=video]}
3584 \map{\step[tourcesource=conjuntodedados,typetarget=dataset]}
3585 \map{\step[tourcesource=fasciculo,typetarget=supperperiodical]}
3586 \map{\step[tourcesource=trabalhoapresentado,typetarget=unpublished]}
3587 % thesis family: retype to thesis and preset the (babel-aware) type word
3588 \map{\step[tourcesource=tese,typetarget=thesis]}
3589 \map{\step[tourcesource=dissertacao,typetarget=thesis]}
3590 \map{\step[tourcesource=masterdissertation,typetarget=thesis,final]\step[fieldset=type,fiel
3591 \map{\step[tourcesource=mscdissertation,typetarget=thesis,final]\step[fieldset=type,fiel
3592 \map{\step[tourcesource=dissertacaomestrado,typetarget=thesis,final]\step[fieldset=type,fi
3593 \map{\step[tourcesource=dscthesis,typetarget=thesis,final]\step[fieldset=type,fieldvalue=d
3594 \map{\step[tourcesource=tesedoutorado,typetarget=thesis,final]\step[fieldset=type,fieldval
3595 \map{\step[tourcesource=phdthesis,typetarget=thesis,final]\step[fieldset=type,fieldvalue=p
3596 \map{\step[tourcesource=tesephdt,typetarget=thesis,final]\step[fieldset=type,fieldvalue=phd
3597 \map{\step[tourcesource=tcc,typetarget=thesis,final]\step[fieldset=type,fieldvalue=tcc]}
3598 % field PT synonyms (unaccented) -> canonical English
3599 \map{\step[fieldsource=periodico,fieldtarget=journaltitle]}
3600 \map{\step[fieldsource=revista,fieldtarget=journaltitle]}
3601 \map{\step[fieldsource=jornal,fieldtarget=journaltitle]}
3602 \map{\step[fieldsource=titulo,fieldtarget=title]}

```

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3603 \map{\step[fieldsource=subtitulo,fieldtarget=subtitle]}
3604 \map{\step[fieldsource=autor,fieldtarget=author]}
3605 \map{\step[fieldsource=editora,fieldtarget=publisher]}
3606 \map{\step[fieldsource=local,fieldtarget=location]}
3607 \map{\step[fieldsource=ano,fieldtarget=year]}
3608 \map{\step[fieldsource=data,fieldtarget=date]}
3609 \map{\step[fieldsource=edicao,fieldtarget=edition]}
3610 \map{\step[fieldsource=paginas,fieldtarget=pages]}
3611 \map{\step[fieldsource=numero,fieldtarget=number]}
3612 \map{\step[fieldsource=tradutor,fieldtarget=translator]}
3613 \map{\step[fieldsource=organizador,fieldtarget=editor]}
3614 \map{\step[fieldsource=instituicao,fieldtarget=institution]}
3615 \map{\step[fieldsource=tipo,fieldtarget=type]}
3616 \map{\step[fieldsource=nota,fieldtarget=note]}
3617 \map{\step[fieldsource=dataacesso,fieldtarget=urldate]}
3618 \map{\step[fieldsource=curso,fieldtarget=course]}
3619 \map{\step[fieldsource=grau,fieldtarget=coppedegree]}
3620 \map{\step[fieldsource=artista,fieldtarget=artist]}
3621 \map{\step[fieldsource=plataforma,fieldtarget=platform]}
3622 \map{\step[fieldsource=director,fieldtarget=director]}
3623 \map{\step[fieldsource=produtor,fieldtarget=producer]}
3624 \map{\step[fieldsource=versao,fieldtarget=version]}
3625 \map{\step[fieldsource=escala,fieldtarget=scale]}
3626 }%
3627 }
3628
3629 % --- ABNT list layout: flush left, single spacing, blank line between entries ---
3630 \setlength{\bibhang}{0pt}
3631 \setlength{\bibitemsep}{\baselineskip}
3632
3633 </bbx>
3634 <*cbx>
3635 \ProvidesFile{coppe.cbx}[2026/05/28 v4.0 CoppeTeX ABNT author-date citations]
3636 % Author-date is the CoppeTeX 4.0 default. authoryear-comp gives compressed
3637 % author-year cites with normal-caps names (post-2020 rule) out of the box.
3638 \RequireCitationStyle{authoryear-comp}
3639
3640 % --- NBR 10520:2023 author separator -----
3641 % Parenthetical citations separate authors with ";" -> (Silva; Souza, 2020).
3642 % Narrative citations (\textcite, natbib \citet) keep comma + "e/and" ->
3643 % Silva, Souza e Oliveira (2020). The reference list keeps its own ";" via
3644 % coppe.bbx ([bib,biblist]); these contextless formats cover the cite side.
3645 \DeclareDelimFormat{multinamedelim}{\addsemicolon\space}
3646 \DeclareDelimFormat{finalnamedelim}{\addsemicolon\space}
3647 \DeclareDelimFormat{textcite}{multinamedelim}{\addcomma\space}
3648 \DeclareDelimFormat{textcite}{finalnamedelim}{\addspace\bibstring{and}\space}
3649
3650 % --- apud: citation of a citation (ABNT NBR 6023 §4.1.2.2 / NBR 10520) -----
3651 % ABNT lists only the work actually CONSULTED. So only the consulted work (#4, a
3652 % .bib key) is cited and enters the reference list; the ORIGINAL work (#2 author,
3653 % #3 year) was not consulted and is given as plain text -- it is never a cite key,
3654 % so it is never added to the bibliography.
3655 % \citetapud[<page>]{<orig author>}{<orig year>}{<consulted key>}
3656 % -> Orig Author (year) apud Consulted (yr[, page]) [narrative]

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```

3657 % \citepapud[<page>]{<orig author>}{<orig year>}{<consulted key>}
3658 % -> (Orig Author, year apud CONSULTED, yr[, page]) [parenthetical]
3659 % #1 = optional postnote (page/loc) on the consulted work; "apud" is latin -> italic.
3660 \DeclareRobustCommand{\citetapud}[4][]{%
3661 #2\space(#3)\space\mkbibemph{apud}\space\textcite[#1]{#4}}
3662 \DeclareRobustCommand{\citepapud}[4][]{%
3663 \mkbibparens{#2\addcomma\space#3\space\mkbibemph{apud}\space
3664 \citeauthor{#4}\addcomma\space\citeyear{#4}}%
3665 \ifblank{#1}{ }\{\addcomma\space\mkpageprefix[pagination]{#1}}}}
3666 \</cbx>
3667 \*lbxbr>
3668 \ProvidesFile{brazilian-coppe.lbx}[2026/05/28 v4.0 CoppeTeX brazilian strings]
3669 \InheritBibliographyExtras{brazilian}
3670 \NewBibliographyString{availablefrom,mscdiss,dscthesi s,tcc,depositor,coppescale}
3671 \DeclareBibliographyStrings{%
3672 inherit = {brazilian},
3673 availablefrom = {{Dispon\'\i vel em}{Dispon\'\i vel em}},
3674 urlseen = {{Acesso em}{Acesso em}},
3675 in = {{In}{In}},
3676 editor = {{organizador}{org\addot}},
3677 editors = {{organizadores}{org\addot}},
3678 depositor = {{Depositante}{Depositante}},
3679 coppescale = {{Escala}{Escala}},
3680 mscdiss = {{Disserta\c{c}\~{a}o de Mestrado}{Disserta\c{c}\~{a}o de Mestrado}},
3681 dscthesi s = {{Tese de Doutorado}{Tese de Doutorado}},
3682 phdthesi s = {{Tese de Doutorado}{Tese de Doutorado}},
3683 tcc = {{Trabalho de Conclus\~{a}o de Curso}{Trabalho de Conclus\~{a}o de Curso}}
3684 }
3685 \</lbxbr>
3686 \*lbxen>
3687 \ProvidesFile{english-coppe.lbx}[2026/05/28 v4.0 CoppeTeX english strings]
3688 \InheritBibliographyExtras{english}
3689 \NewBibliographyString{availablefrom,mscdiss,dscthesi s,tcc,depositor,coppescale}
3690 \DeclareBibliographyStrings{%
3691 inherit = {english},
3692 availablefrom = {{Available from}{Available from}},
3693 urlseen = {{Access on}{Access on}},
3694 in = {{In}{In}},
3695 editor = {{editor}{ed\addot}},
3696 editors = {{editors}{ed\addot}},
3697 depositor = {{Depositor}{Depositor}},
3698 coppescale = {{Scale}{Scale}},
3699 mscdiss = {{M\addot Sc\addot Dissertation}{M\addot Sc\addot Diss\addot}},
3700 dscthesi s = {{D\addot Sc\addot Thesis}{D\addot Sc\addot Thesis}},
3701 phdthesi s = {{Ph\addot D\addot Thesis}{Ph\addot D\addot Thesis}},
3702 tcc = {{Undergraduate Thesis}{Undergraduate Thesis}},
3703 }
3704 \</lbxen>
3705 \*lbxes>
3706 \ProvidesFile{spanish-coppe.lbx}[2026/05/28 v4.0 CoppeTeX spanish strings]
3707 \InheritBibliographyExtras{spanish}
3708 \NewBibliographyString{availablefrom,mscdiss,dscthesi s,tcc,depositor,coppescale}
3709 \DeclareBibliographyStrings{%
3710 inherit = {spanish},

```

```

3711 availablefrom = {{Disponible en}{Disponible en}},
3712 urlseen       = {{Consultado el}{Consultado el}},
3713 in            = {{En}{En}},
3714 editor        = {{editor}{ed\adddot}},
3715 editors       = {{editores}{ed\adddot}},
3716 depositor     = {{Solicitante}{Solicitante}},
3717 coppescale    = {{Escala}{Escala}},
3718 msdiss        = {{Tesis de Maestr'\{i\}a}{Tesis de Maestr'\{i\}a}},
3719 dscthesi      = {{Tesis Doctoral}{Tesis Doctoral}},
3720 phdthesis     = {{Tesis Doctoral}{Tesis Doctoral}},
3721 tcc           = {{Trabajo de Fin de Carrera}{Trabajo de Fin de Carrera}},
3722 }
3723 </lbox>
3724 <*lboxfr>
3725 \ProvidesFile{french-coppe.lbx}[2026/05/28 v4.0 CoppeTeX french strings]
3726 \InheritBibliographyExtras{french}
3727 \NewBibliographyString{availablefrom,msdiss,dscthesi,tcc,depositor,coppescale}
3728 \DeclareBibliographyStrings{%
3729   inherit      = {french},
3730   availablefrom = {{Disponible \{a\}}{Disponible \{a\}}},
3731   urlseen      = {{Consult\{e\} le}{Consult\{e\} le}},
3732   in           = {{Dans}{Dans}},
3733   editor       = {{\{e\}diteur}{\{e\}d\adddot}},
3734   editors      = {{\{e\}diteurs}{\{e\}d\adddot}},
3735   depositor    = {{D\{e\}posant}{D\{e\}posant}},
3736   coppescale   = {{\{e\}chelle}{\{e\}chelle}},
3737   msdiss       = {{M\{e\}moire de Master}{M\{e\}moire de Master}},
3738   dscthesi     = {{Th\{e\}se de Doctorat}{Th\{e\}se de Doctorat}},
3739   phdthesis    = {{Th\{e\}se de Doctorat}{Th\{e\}se de Doctorat}},
3740   tcc          = {{M\{e\}moire de fin d'\{e\}tudes}{M\{e\}moire de fin d'\{e\}tudes}},
3741 }
3742 </lboxfr>
3743 <*lboxit>
3744 \ProvidesFile{italian-coppe.lbx}[2026/05/28 v4.0 CoppeTeX italian strings]
3745 \InheritBibliographyExtras{italian}
3746 \NewBibliographyString{availablefrom,msdiss,dscthesi,tcc,depositor,coppescale}
3747 \DeclareBibliographyStrings{%
3748   inherit      = {italian},
3749   availablefrom = {{Disponibile da}{Disponibile da}},
3750   urlseen      = {{Consultato il}{Consultato il}},
3751   in           = {{In}{In}},
3752   editor       = {{curatore}{cur\adddot}},
3753   editors      = {{curatori}{cur\adddot}},
3754   depositor    = {{Depositante}{Depositante}},
3755   coppescale   = {{Scala}{Scala}},
3756   msdiss       = {{Tesi di Laurea Magistrale}{Tesi di Laurea Magistrale}},
3757   dscthesi     = {{Tesi di Dottorato}{Tesi di Dottorato}},
3758   phdthesis    = {{Tesi di Dottorato}{Tesi di Dottorato}},
3759   tcc          = {{Tesi di Laurea Triennale}{Tesi di Laurea Triennale}},
3760 }
3761 </lboxit>
3762 <*langes>
3763 %% coppe-lang-spanish.def -- CoppeTeX 4.x Spanish class-string pack.
3764 %% Loaded by coppe.cls when \documentclass[spanish]{coppe} (or any other

```



```

3765 %% spanish slot) is active. Populates the dispatcher table for the
3766 %% spanish language and registers the biblatex language mapping.
3767 %% Institutional names (universityname/cityname/statename/countryname)
3768 %% are deliberately kept Portuguese: the COPPE/UFRJ cover and folha de
3769 %% rosto are a Brazilian institutional template that does not translate.
3770 \ProvidesFile{coppe-lang-spanish.def}[2026/05/28 v4.0 CoppeTeX spanish class strings]
3771 \copperdefstring{spanish}{appendix}      {Ap\'}endice}
3772 \copperdefstring{spanish}{annex}         {Anexo}
3773 \copperdefstring{spanish}{frame}         {Cuadro}
3774 \copperdefstring{spanish}{listframe}     {Lista de Cuadros}
3775 \copperdefstring{spanish}{source}        {Fuente}
3776 \copperdefstring{spanish}{program}       {Programa}
3777 \copperdefstring{spanish}{listprogram}   {Lista de Programas}
3778 \copperdefstring{spanish}{references}    {Referencias}
3779 \copperdefstring{spanish}{in:}          {en}
3780 \copperdefstring{spanish}{art}           {Arte}
3781 \copperdefstring{spanish}{platform}      {Plataforma}
3782 \copperdefstring{spanish}{directedby}    {Direcci\'on}
3783 \copperdefstring{spanish}{producedby}    {Producci\'on}
3784 \copperdefstring{spanish}{listabbreviation}{Lista de Abreviaturas y Siglas}
3785 \copperdefstring{spanish}{listsymbol}    {Lista de S\'imbolos}
3786 \copperdefstring{spanish}{glossary}      {Glosario}
3787 \copperdefstring{spanish}{advisor}       {Director}
3788 \copperdefstring{spanish}{advisors}      {Directores}
3789 \copperdefstring{spanish}{dept}          {Programa}
3790 \copperdefstring{spanish}{approvedmale}  {Aprobado por}
3791 \copperdefstring{spanish}{approvedfemale}{Aprobada por}
3792 \copperdefstring{spanish}{approvedonmale}{Aprobado el}
3793 \copperdefstring{spanish}{approvedonfemale}{Aprobada el}
3794 \copperdefstring{spanish}{keywordsname} {Palabras clave}
3795 \copperdefstring{spanish}{volumename}   {Volumen}
3796 \copperdefstring{spanish}{coadvisor}    {Codirector}
3797 \copperdefstring{spanish}{coadvisors}   {Codirectores}
3798 \copperdefstring{spanish}{ofname}       {de}
3799 \copperdefstring{spanish}{universityname}{Universidade Federal do Rio de Janeiro}
3800 \copperdefstring{spanish}{cityname}     {Rio de Janeiro}
3801 \copperdefstring{spanish}{statename}    {RJ}
3802 \copperdefstring{spanish}{countryname} {Brasil}
3803 \copperdefstring{spanish}{monthname1}   {enero}
3804 \copperdefstring{spanish}{monthname2}   {febrero}
3805 \copperdefstring{spanish}{monthname3}   {marzo}
3806 \copperdefstring{spanish}{monthname4}   {abril}
3807 \copperdefstring{spanish}{monthname5}   {mayo}
3808 \copperdefstring{spanish}{monthname6}   {junio}
3809 \copperdefstring{spanish}{monthname7}   {julio}
3810 \copperdefstring{spanish}{monthname8}   {agosto}
3811 \copperdefstring{spanish}{monthname9}   {septiembre}
3812 \copperdefstring{spanish}{monthname10} {octubre}
3813 \copperdefstring{spanish}{monthname11} {noviembre}
3814 \copperdefstring{spanish}{monthname12} {diciembre}
3815 \copperdefstring{spanish}{algorithm2eopt}{spanish}
3816 \copperdefstring{spanish}{babelname}   {spanish}
3817 %% biblatex is loaded by coppe.cls AFTER this .def file is read; defer the
3818 %% language-mapping registration until \begin{document} so biblatex's

```

```

3819 %% \DeclareLanguageMapping is available.
3820 \AtBeginDocument{\DeclareLanguageMapping{spanish}{spanish-coppe}}
3821 \langedes
3822 \langfr
3823 %% coppe-lang-french.def -- CoppeTeX 4.x French class-string pack.
3824 %% See coppe-lang-spanish.def for the structure. Institutional names
3825 %% remain Portuguese; only document-content strings translate.
3826 \ProvidesFile{coppe-lang-french.def}[2026/05/28 v4.0 CoppeTeX french class strings]
3827 \copperdefstring{french}{appendix} {Annexe}
3828 \copperdefstring{french}{annex} {Annexe}
3829 \copperdefstring{french}{frame} {Encadr\'e}
3830 \copperdefstring{french}{listframe} {Liste des encadr\'es}
3831 \copperdefstring{french}{source} {Source}
3832 \copperdefstring{french}{program} {Programme}
3833 \copperdefstring{french}{listprogram} {Liste des programmes}
3834 \copperdefstring{french}{references} {R\'ef\'erences}
3835 \copperdefstring{french}{in:} {dans}
3836 \copperdefstring{french}{art} {Illustration}
3837 \copperdefstring{french}{platform} {Plateforme}
3838 \copperdefstring{french}{directedby} {R\'ealis\'e par}
3839 \copperdefstring{french}{producedby} {Produit par}
3840 \copperdefstring{french}{listabbreviation} {Liste des abr\'eviations et acronymes}
3841 \copperdefstring{french}{listsymbol} {Liste des symboles}
3842 \copperdefstring{french}{glossary} {Glossaire}
3843 \copperdefstring{french}{advisor} {Directeur}
3844 \copperdefstring{french}{advisors} {Directeurs}
3845 \copperdefstring{french}{dept} {Programme}
3846 \copperdefstring{french}{approvedmale} {Approuv\'e par}
3847 \copperdefstring{french}{approvedfemale} {Approuv\'ee par}
3848 \copperdefstring{french}{approvedonmale} {Approuv\'e le}
3849 \copperdefstring{french}{approvedonfemale} {Approuv\'ee le}
3850 \copperdefstring{french}{keywordsname} {Mots-cl\'es}
3851 \copperdefstring{french}{volumename} {Volume}
3852 \copperdefstring{french}{coadvisor} {Co-directeur}
3853 \copperdefstring{french}{coadvisors} {Co-directeurs}
3854 \copperdefstring{french}{ofname} {de}
3855 \copperdefstring{french}{universityname} {Universidade Federal do Rio de Janeiro}
3856 \copperdefstring{french}{cityname} {Rio de Janeiro}
3857 \copperdefstring{french}{statename} {RJ}
3858 \copperdefstring{french}{countryname} {Brasil}
3859 \copperdefstring{french}{monthname1} {janvier}
3860 \copperdefstring{french}{monthname2} {f\'evrier}
3861 \copperdefstring{french}{monthname3} {mars}
3862 \copperdefstring{french}{monthname4} {avril}
3863 \copperdefstring{french}{monthname5} {mai}
3864 \copperdefstring{french}{monthname6} {juin}
3865 \copperdefstring{french}{monthname7} {juillet}
3866 \copperdefstring{french}{monthname8} {ao\~ut}
3867 \copperdefstring{french}{monthname9} {septembre}
3868 \copperdefstring{french}{monthname10} {octobre}
3869 \copperdefstring{french}{monthname11} {novembre}
3870 \copperdefstring{french}{monthname12} {d\'ecembre}
3871 \copperdefstring{french}{algorithm2eopt} {french}
3872 \copperdefstring{french}{babelname} {french}

```

```

3873 \AtBeginDocument{\DeclareLanguageMapping{french}{french-coppe}}
3874 \</langfr>
3875 \<langit>
3876 %% coppe-lang-italian.def -- CoppeTeX 4.x Italian class-string pack.
3877 %% See coppe-lang-spanish.def for the structure. The algorithm2e
3878 %% language option for Italian is spelled "italiano" -- that's the
3879 %% spelling algorithm2e ships, even though babel calls the language
3880 %% "italian".
3881 \ProvidesFile{coppe-lang-italian.def}[2026/05/28 v4.0 CoppeTeX italian class strings]
3882 \copperdefstring{italian}{appendix}      {Appendice}
3883 \copperdefstring{italian}{annex}         {Allegato}
3884 \copperdefstring{italian}{frame}         {Riquadro}
3885 \copperdefstring{italian}{listframe}     {Elenco dei riquadri}
3886 \copperdefstring{italian}{source}        {Fonte}
3887 \copperdefstring{italian}{program}       {Programma}
3888 \copperdefstring{italian}{listprogram}   {Elenco dei programmi}
3889 \copperdefstring{italian}{references}    {Riferimenti}
3890 \copperdefstring{italian}{in:}           {in}
3891 \copperdefstring{italian}{art}           {Illustrazione}
3892 \copperdefstring{italian}{platform}      {Piattaforma}
3893 \copperdefstring{italian}{directedby}    {Regia di}
3894 \copperdefstring{italian}{producedby}    {Prodotto da}
3895 \copperdefstring{italian}{listabbreviation}{Elenco di abbreviazioni e sigle}
3896 \copperdefstring{italian}{listsymbol}    {Elenco dei simboli}
3897 \copperdefstring{italian}{glossary}      {Glossario}
3898 \copperdefstring{italian}{advisor}       {Relatore}
3899 \copperdefstring{italian}{advisors}      {Relatori}
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3902 \copperdefstring{italian}{approvedfemale}{Approvata da}
3903 \copperdefstring{italian}{approvedonmale}{Approvato il}
3904 \copperdefstring{italian}{approvedonfemale}{Approvata il}
3905 \copperdefstring{italian}{keywordsname}  {Parole chiave}
3906 \copperdefstring{italian}{volumename}   {Volume}
3907 \copperdefstring{italian}{coadvisor}     {Correlatore}
3908 \copperdefstring{italian}{coadvisors}    {Correlatori}
3909 \copperdefstring{italian}{ofname}        {di}
3910 \copperdefstring{italian}{universityname}{Universidade Federal do Rio de Janeiro}
3911 \copperdefstring{italian}{cityname}      {Rio de Janeiro}
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3917 \copperdefstring{italian}{monthname4}    {aprile}
3918 \copperdefstring{italian}{monthname5}    {maggio}
3919 \copperdefstring{italian}{monthname6}    {giugno}
3920 \copperdefstring{italian}{monthname7}    {luglio}
3921 \copperdefstring{italian}{monthname8}    {agosto}
3922 \copperdefstring{italian}{monthname9}    {settembre}
3923 \copperdefstring{italian}{monthname10}   {ottobre}
3924 \copperdefstring{italian}{monthname11}   {novembre}
3925 \copperdefstring{italian}{monthname12}   {dicembre}
3926 \copperdefstring{italian}{algorithm2eopt}{italiano}

```

```

3927 \copperdefstring{italian}{babelname}      {italian}
3928 \AtBeginDocument{\DeclareLanguageMapping{italian}{italian-coppe}}
3929 \end{langit}
3930 \numbbx
3931 \ProvidesFile{coppe-numeric.bbx}[2026/05/28 v4.0 CoppeTeX ABNT numeric bibliography]
3932 % Numeric reference list for the ABNT numeric system (NBR 10520). It reuses the
3933 % full coppe ABNT style (drivers, CAPS surnames, bold titles, "In:", p./f.,
3934 % the PT->EN source map and the language mappings) and only adds the [n] list
3935 % label and the numeric sorting on top.
3936 \RequireBibliographyStyle{coppe}
3937
3938 % ABNT numeric system lists references in the order they are first cited
3939 % (NBR 10520), i.e. unsorted. Override with sorting=nty for an alphabetical
3940 % numbered list. 'labelnumber' generates the [n] field.
3941 \ExecuteBibliographyOptions{labelnumber,sorting=none}
3942
3943 % Bracketed labels, both in the list and for any shorthand entries.
3944 \DeclareFieldFormat{labelnumberwidth}{\mkbibbrackets{#1}}
3945 \DeclareFieldFormat{shorthandwidth}{\mkbibbrackets{#1}}
3946
3947 % Reference list driven by the [n] label (copied from biblalex's numeric.bbx),
3948 % keeping the ABNT blank line between entries via \bibitemsep set in coppe.bbx.
3949 \defbibenvironment{bibliography}
3950 {
3951   \list
3952     {\printtext[labelnumberwidth]{%
3953       \printfield{labelprefix}%
3954       \printfield{labelnumber}}}
3955     {\setlength{\labelwidth}{\labelnumberwidth}%
3956       \setlength{\leftmargin}{\labelwidth}%
3957       \setlength{\labelsep}{\biblabelsep}%
3958       \addtolength{\leftmargin}{\labelsep}%
3959       \setlength{\itemsep}{\bibitemsep}%
3960       \setlength{\parsep}{\bibparsep}}%
3961     {\endlist}
3962     {\item}
3963
3964 \numcbx
3965 \numcbx
3966 \ProvidesFile{coppe-numeric.cbx}[2026/05/28 v4.0 CoppeTeX ABNT numeric citations]
3967 % Numeric citation system (ABNT NBR 10520): in-text references are bracketed
3968 % numbers [1], compressed and sorted within a single cite group. This is the
3969 % opt-in alternative to the author-date default (coppe.cbx); it pairs with
3970 % bibstyle=coppe-numeric. With the biblalex natbib option, \citep -> [1] and
3971 % \citet -> Author [1].
3972 \RequireCitationStyle{numeric-comp}
3973 \numcbx

```

10.2 The bibliographic databases

The two BibLaTeX/biber databases shipped with the class — `example.bib` (the sample database cited by the example monograph) and `coppe.bib` (the references of this manual) — are generated from this source too, so the whole distribution still has a single editable origin. They live inside an `\iffalse` block below: the

documentation driver never typesets them, and only `coppe.ins` extracts them, through the `examplebib` and `coppebib` docstrip targets. Edit the entries here, never the generated `.bib` files.

10.3 The generated companion files

Everything the distribution ships is extracted from this file. The class, the BibLaTeX styles, the language packs, the databases and the `.ist` were already here; the demonstration documents, the `latexmkrc` and the regression tests joined them, so that a single `pdflatex coppe.ins` rebuilds the whole tree and there is no second place to edit. Each one sits in an `\iffalse` block: the documentation driver never typesets them, and only `coppe.ins` extracts them.

The tests live in `src/tests/` rather than a sibling `tests/`. That is not taste: `TeX` refuses `\openout` on a path containing `..`, so a `.ins` run in `src/` can write into a subdirectory and never into a parent. Putting the suite under `src/` is what lets it be generated at all.

What is NOT generated from here, and why: `tools/*.ps1` and `src/tests/run-tests.ps1` are the build harness, not artefacts of the class; `NORMA_COPPE_2026.tex` and `futuremanual2026.tex` are documents about the norm that happen to live in this repository, with their own life cycle.

10.3.1 `example_pt.tex` -- pt-main demo

`example_pt.tex` – CoppeTeX 4.0 multilingual demo, main language: brazilian. Short demonstration document (cover + folha + abstracts + 1 chapter) used by the CPGP submission package. The full pt-main example with bibliography and lists is the historical `example.tex`.

10.3.2 `example_en.tex` -- en-main demo

`example_en.tex` – CoppeTeX 4.0 multilingual demo, main language: english.

10.3.3 `example_es.tex` -- es-main demo

`example_es.tex` – CoppeTeX 4.0 multilingual demo, main language: spanish. Requires the language pack files `coppe-lang-spanish.def` and `spanish-coppe.lbx` to be installed alongside `coppe.cls`.

10.3.4 `example_fr.tex` -- fr-main demo

`example_fr.tex` – CoppeTeX 4.0 multilingual demo, main language: french. Requires the language pack files `coppe-lang-french.def` and `french-coppe.lbx` to be installed alongside `coppe.cls`.

10.3.5 `example_it.tex` -- it-main demo

`example_it.tex` – CoppeTeX 4.0 multilingual demo, main language: italian. Requires the language pack files `coppe-lang-italian.def` and `italian-coppe.lbx` to be installed alongside `coppe.cls`.

10.3.6 example_pdfa.tex -- the full example as PDF/A

example_pdfa.tex – o example.tex completo, compilado com a opção ‘pdfa’.

Existe para dar ao veraPDF o documento mais exigente que a classe produz: o exemplo inteiro, com bibliografia, listas, figuras, tabelas, algoritmos e as caixas do tcolorbox – ou seja, transparência, que é justamente o que o a-1b proibiria e o a-2b admite. Validar só um documento curto não diria nada sobre o que vai de fato para o depósito.

\PassOptionsToClass tem de vir ANTES do \documentclass que o example.tex traz dentro de si; é o único jeito de acrescentar uma opção de classe a um arquivo gerado pelo docstrip sem duplicá-lo aqui e deixar as duas cópias divergirem com o tempo.

10.3.7 covers_5languages.tex -- the five covers side by side

covers_5languages.tex – One-page visual demonstration of the CoppeTeX 4.0 multilingual model: the five built-in main languages side by side, all sharing the same Portuguese institutional template, differing only in the title (and the body language, not visible on the cover).

This file uses the article class deliberately – it is a meta-document about the coppe class, not a thesis. It expects the five cover PNGs cover_pt.png, cover_en.png, cover_es.png, cover_fr.png, cover_it.png to live in the same directory (generated by pdftoppm from the example_<lang>.pdf demos).

10.3.8 latexmkrc -- latexmk/Overleaf build configuration

latexmkrc – CoppeTeX build configuration (works locally and on Overleaf).

Run with: latexmk example (or latexmk futuremanual2026, your thesis...) latexmk reads this file automatically when it is in the build directory. On Overleaf, place this file at the project root; Overleaf runs latexmk.

Engine: pdfLaTeX by default (fast). For LuaLaTeX (recommended, full Unicode) set \$pdf_mode = 4 below, or use the "% !TeX program = lualatex" magic comment / the Overleaf "Menu > Compiler" setting.

10.3.9 tests/test_brazilian_one_advisor.tex

```
% % test_brazilian_one_advisor.tex – regression guard. % % Historical bug:
when only ONE \advisor was given, the cover/folha-de-rosto % rendering used to
fall over (the original \maketitle loop assumed at % least two advisors). This test
pins the single-advisor path: it must % compile cleanly AND the cover label must
read "Orientador" (singular), % not "Orientadores". % % Scope: just enough
thesis to render \maketitle and the front pages. % No body chapters – they are
covered by test_brazilian_two_advisors.tex. % % Run via ../tests/run-tests.ps1.
%
```

10.3.10 tests/test_brazilian_two_advisors.tex

```
% % test_brazilian_two_advisors.tex – regression test for the coppe class. %
Brazilian main language, two advisors (the common case; the cover label % should
print "Orientadores"). % % Run via ../tests/run-tests.ps1, which sets TEXIN-
PUTS so this file finds % coppe.cls and example.bib in ../src/. EXTEND THIS
```

FILE whenever a new % bug is found in the two-advisor / common path: keep every existing case % working and add a minimal repro for the new one. % % Covered features (as of 2026/05/29, CoppeTeX v4.0, nlinguas branch): % * metadata: \title, \foreigntitle, \author, \advisor (x2), \examiner (x3), % \department, \date, \keyword (x3) % * \maketitle (cover + folha de rosto + ficha catalografica) % * \frontmatter: \dedication, agradecimentos chapter*, abstract, foreignabstract % * lists: \listoffigures, \listoftables, \listofquadros, \listofprogramas, % \printloabbreviations, \printlosymbols, \tableofcontents % * \mainmatter chapters covering: figures with \source/\fonte, booktabs % table, quadro float, subcaption subfigures, longtable, amsmath + % amssymb math, algorithm2e algorithm, listings Python block, longquote, % footnotes (singlespacing+filete via setspace), \sigla / \sigla*, % \syml, \abbrev, \citet, \citep, \autoref cross-references % * back matter: \printbibliography, \appendix (x2), \annex % % Compile under pdfLaTeX. Under LuaLaTeX/XeLaTeX, the class skips amssymb % and the end-of-document ClassWarning will fire (test it by adding % \usepackage{unicode-math} + \set-mathfont{Latin Modern Math} below). %

10.3.11 tests/test_brazilian_three_advisors.tex

% % test_brazilian_three_advisors.tex – upper-bound regression test. % % Three \advisor calls is the most the COPPE/UFRJ cover is expected to % typeset reasonably; the class must still render the cover and folha de % rosto cleanly, with the plural label "Orientadores". % % Scope: just enough thesis to render \maketitle and the front pages. % No body chapters – they are covered by test_brazilian_two_advisors.tex. % % Run via ../tests/run-tests.ps1. %

10.3.12 tests/test_assinaturas.tex

% % test_assinaturas.tex – regression test for the coppe class. % Exercises the pre-textual pages introduced/reshaped for the SiBI manual, % 9th revised edition (2026): % % * class option ‘assinaturas’ – folha de aprovacao with one signature % rule per board member instead of the compact list (Annex D) % * \examiner and \advisor with the OPTIONAL institution argument, and the % titulacao argument that pre-4.1 discarded silently % * folha adicional with the Coleta CAPES fields (3.1.2.1.2, Annex H): % \tipoproducao, \projetovinculado, \nomeprojeto, \areaconcentracao, % \agenciafomento % * \linhapesquisa and \areaconcentracao on the folha de rosto % * \dataaprovacao on the folha de aprovacao % % Deliberately mixes calls WITH and WITHOUT the optional institution, so a % regression in the optional-argument handling shows up here. % % Run via ../tests/run-tests.ps1. Compile under pdfLaTeX. %

10.3.13 tests/test_assinaturas_7.tex

% % test_assinaturas_7.tex – regression test for the coppe class. % % The folha de aprovacao under the ‘assinaturas’ option gives every board % member a signature rule of its own (Annex D). The class had only ever been % exercised up to five members, and a doctoral board at the COPPE routinely % reaches seven, so this test pins the tall end of the range: two advisors % plus five examiners, one of them with a long name and a long institution, % which is the combination most likely to wrap a line and push the sheet over. % % What "passing" means here

is narrower than "pdflatex exited 0": the folha % de aprovacao must still fit on ONE page. tools/check-aprovacao.py measures % that from the rendered PDF; run-tests.ps1 only sees the exit code, so a % silent overflow would show up as an Overfull \vbox in the log rather than % as a failure. Both are checked. % % Run via ../tests/run-tests.ps1. Compile under pdfLaTeX. %

10.3.14 tests/test_coorientador.tex

Guards the `coorientador` class option: the coadvisors have to show up on all three abstract pages, under the label of the page's own language, and the advisors have to stay exactly where they were. Two coadvisors on purpose, so the plural form of the label is exercised as well.

10.3.15 tests/test_pdfa.tex

% % test_pdfa.tex – regression test for the coppe class. % Exercises the pre-textual pages introduced/reshaped for the SiBI manual, % 9th revised edition (2026): % % * class option ‘assinaturas’ – folha de aprovacao with one signature % rule per board member instead of the compact list (Annex D) % * \examiner and \advisor with the OPTIONAL institution argument, and the % titulacao argument that pre-4.1 discarded silently % * folha adicional with the Coleta CAPES fields (3.1.2.1.2, Annex H): % \tipoproducao, \projetovinculado, \nomeprojeto, \areaconcentracao, % \agenciafomento % * \linhapesquisa and \areaconcentracao on the folha de rosto % * \dataaprovacao on the folha de aprovacao % % Deliberately mixes calls WITH and WITHOUT the optional institution, so a % regression in the optional-argument handling shows up here. % % Run via ../tests/run-tests.ps1. Compile under pdfLaTeX. %

Acknowledgments

Thanks to all COPPE_{TEX} users who have reported their experience with this class. We also thank to professor Fernando Lizarralde and Heiko Oberdiek for their helpful comments. The authors would like to thank the National Council for Scientific and Technological Development (CNPq) of Brazil.

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<code>\XeLaTeX</code>		998	<code>\xspace</code>		867

Change History

v0.0		v3.1	
General: Creation Date.	1	<code>\department</code> : Included a sort key in <code>symb</code>	50
v0.1		v3.2	
General: Documentation:		General: Fixed version problem between <code>cls</code> and <code>dtx</code>	1
bibliography fixed, title translation.	1	v3.3	
Sourceforge submission.	1	General: Extend <code>abrev</code> to work like <code>symb</code> and accept a sorting key .	1
v0.2		v3.4	
General: Unification of the code for the list of symbols and abbreviations.	1	General: Some examples for figures, tables, longtables, etc. .	1
v0.3		v3.4.1	
General: Added 'draft' option. . . .	1	General: Fix english option to consider brazilian a second language, describe it in document	48
<code>\maketitle</code> : Added number of examiners test.	59	v3.5	
Generalization.	59	General: New command <code>annex</code> . .	96
v0.4		v3.5.1	
General: Beta documentation. . . .	1	General: New command <code>annex</code> now supports brazilian or english .	96
v0.5		v4.0	
<code>abstract</code> : Changed from macro to environment.	66	General: Catalog-label writers (<code>coppe@mainBegin</code> , <code>coppe@bibBegin</code> , <code>coppe@bibEnd</code> , <code>coppe@hasLof</code>) made language-agnostic: <code>detokenize</code> wrap on label names protects ':' and '.' from babel shorthand activation, and the raw <code>etex</code> primitives <code>romannumeral/number</code> replace LaTeX's <code>roman/arabic</code> to bypass <code>spanish-babel</code> 's <code>small-caps @roman</code> redefinition that would otherwise crash <code>pdftex</code> with "Unbalanced write command" on every <code>es-main</code> document.	1
<code>\backmatter</code> : Added <code>mainmatter</code> pages counter.	55	Class-wide string dispatcher (<code>copperdefstring</code> , <code>coppestring</code> , <code>coppemainstring</code> , <code>coppeforeignstring</code>) replaces scattered <code>iflanguage</code> and <code>if</code> english branches; <code>pt/en</code> outputs remain byte-identical. .	1
<code>foreignabstract</code> : Changed from macro to environment.	67	Multilingual model: three fixed slots (<code>main</code> , <code>foreign</code> , <code>optional</code> <code>third</code>); five built-in language options (<code>brazilian</code> , <code>english</code> ,	
v1.0			
General: First COPPETeX release. .	1		
v2.0			
General: COPPETeX release 2.0. . .	1		
v2.1			
General: COPPETeX release 2.1:			
Matching the new rules.	1		
<code>\advisor</code> : Advisors, co-advisors, co-co-advisors, etc., all of them are simply considered advisors. .	51		
v2.2			
General: Matching new guidelines, including new logo.	1		
v2.2.2			
General: Fixed some text constants in <code>.bib</code> and documented it here. .	1		
v3.0			
General: Added support for monographs in English.	48		
New approval page layout. . . .	61		
<code>\department</code> : Added new course on Nanotechnology.	50		
<code>\examiner</code> : Examiners expansion without degree.	52		

spanish, french, italian);	usecoppelanguage,
plug-in mechanism for any	brazilianabstract environment. . . 1
other Babel language via	
coppe-lang-lang.def plus	\coppetexfinalpage: Initial
lang-coppe.lbx. 1	coppetexfinalpage colophon
New user API: titlein,	page. 56